

# Chemical monitoring reporting guidance: 2026 data collection

European Food Safety Authority (EFSA)

Alexios Zormpas, Emanuela Marchese\*, Paula Medina Pastor, Marta Vericat Ferrer\*\*,  
Alicia Gutiérrez Linares, Stefania Salvatore, Daniela Brocca and Valentina Bocca

\*Net Service S.p.A., Italy (EFSA framework contractor)

\*\*MAPS S.p.A., Italy (EFSA framework contractor)

## Abstract

This document should be used for reporting to EFSA samples analysed for chemicals. It provides guidance on how to use the Standard Sample Description (SSD2) data model for submission to the EU of analytical results of food and feed samples taken during control activities carried out to monitor residues of pesticides and veterinary medicinal products, contaminants, food additives, and food flavourings. This document does not replace – but complements and updates some aspects of – the general EFSA Guidance on Standard Sample Description (SSD2) and Guidance on Data Exchange (GDE2). It is meant to provide guidance on the specific technical and legislative requirements as well as clarity on data quality validation for chemical monitoring (ChemMon) data at national and EU levels.

© European Food Safety Authority, 2026

**Key words:** SSD2, food samples, pesticide residues, food additives, food flavourings, veterinary medicinal product residues, contaminants, animal feed

**Question number:** EFSA-Q-2025-00483

**Correspondence:** [Ask a Question](#)



**Acknowledgements:** EFSA wishes to thank the national members of the EFSA scientific Network on Chemical Monitoring data collection and the following EFSA staff for the support provided to this output: Vaia Mitoula, Giuseppe Antonio Triacchini, Maria Pia Fabrile, Luca Belmonte, Elisa Fasanelli, Rubén Fuertes, Giandomenico Furnari, Luca Pasinato, José Ángel Gómez Ruiz, Zsuzsanna Horvath, and Alexandra Tard.

**Suggested citation:** EFSA (European Food Safety Authority), Zormpas A, Marchese E, Medina Pastor P, Vericat Ferrer M, Salvatore S, Gutiérrez Linares A, Brocca D and Bocca V, 2026 Chemical monitoring reporting guidance: 2026 data collection. EFSA Supporting publication 2026:EN-9911. 158 pp. doi: 10.2903/sp.efsa.2026.EN-9911

**ISSN:** 2397-8325

© European Food Safety Authority, 2026

Reproduction is authorised provided the source is acknowledged.

## Table of contents

|                                                                                                                                                                                          |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Abstract.....                                                                                                                                                                            | 1  |
| Table of contents .....                                                                                                                                                                  | 3  |
| List of tables.....                                                                                                                                                                      | 4  |
| 1 Introduction .....                                                                                                                                                                     | 5  |
| 1.1 Background.....                                                                                                                                                                      | 5  |
| 2 SSD2 data model elements .....                                                                                                                                                         | 17 |
| B.01 Sampling programme identification code (progId) (mandatory) .....                                                                                                                   | 17 |
| B.02 Programme legal reference (progLegalRef) (mandatory) .....                                                                                                                          | 18 |
| B.03 Sampling strategy (sampStrategy) (mandatory) .....                                                                                                                                  | 20 |
| B.04 Programme type (progType) (mandatory) .....                                                                                                                                         | 21 |
| B.05 Sampling method (sampMethod) (optional) .....                                                                                                                                       | 27 |
| B.06 Sampler (sampler) (mandatory).....                                                                                                                                                  | 28 |
| B.07 Sampling point (sampPoint) (mandatory) .....                                                                                                                                        | 28 |
| C.01 Sampling event identification code (sampEventId) (optional).....                                                                                                                    | 29 |
| C.02 Sampling unit type (sampUnitType) (optional) .....                                                                                                                                  | 29 |
| C.03 Sampling unit size (sampUnitSize) and C.04 Sampling unit size unit<br>(sampUnitSizeUnit) (optional).....                                                                            | 30 |
| D.01 Sample taken identification code (sampId) (mandatory).....                                                                                                                          | 30 |
| D.03 Country of sampling (sampCountry) (mandatory).....                                                                                                                                  | 31 |
| D.06 Year of sampling (sampY), D.07 Month of sampling (sampM), D.08 Day of sampling<br>(sampD) (mandatory) .....                                                                         | 32 |
| D.11 Original sample identifier (sampInfo.origSampId) (optional) .....                                                                                                                   | 32 |
| E.02 Coded description of the matrix of the sample taken (sampMatCode) (mandatory).....                                                                                                  | 32 |
| E.03 Text description of the matrix of the sample taken (sampMatText) (optional) .....                                                                                                   | 56 |
| F.01 Sample analysed identification code (sampAnId), G.01 Coded description of the<br>matrix analysed (anMatCode), and G.02 Text description of the matrix analysed<br>(anMatText) ..... | 56 |
| E.04 Country of origin on the sample taken (origCountry) (mandatory).....                                                                                                                | 56 |
| E.06 Area of origin for fisheries or aquaculture activities code of the sample taken<br>(origFishAreaCode) (optional) .....                                                              | 58 |
| F.03 Year of analysis (analysisY) (mandatory) .....                                                                                                                                      | 59 |
| H.01 Sample analysed portion sequence (anPortSeq) (optional) .....                                                                                                                       | 59 |
| J.01 Identification code of the laboratory (labId) (mandatory) .....                                                                                                                     | 60 |
| J.02 Laboratory accreditation (labAccred) (mandatory) .....                                                                                                                              | 60 |
| J.03 Laboratory country (labCountry) (mandatory) .....                                                                                                                                   | 60 |
| K.01 Type of parameter (paramType) (optional) .....                                                                                                                                      | 60 |
| K.02 Coded description of parameter (paramCode) (mandatory) .....                                                                                                                        | 63 |
| L.01 Analytical method identification (anMethRefId) (mandatory).....                                                                                                                     | 64 |
| L.03 Analytical method type (anMethType), L.04 Analytical method code (anMethCode)<br>(mandatory) .....                                                                                  | 65 |
| M.01 Result identification code (resId) (mandatory) .....                                                                                                                                | 66 |
| M.02 Accreditation procedure for the analytical method (accredProc) (mandatory).....                                                                                                     | 66 |
| M.03 Result unit (resUnit) (mandatory) .....                                                                                                                                             | 67 |
| M.04 Result limit of detection (resLOD) (optional) .....                                                                                                                                 | 67 |
| M.05 Result limit of quantification (resLOQ) (optional) .....                                                                                                                            | 67 |
| M.08 CCalpha (CCalpha) and M.09 CCbeta (CCbeta) (optional) .....                                                                                                                         | 68 |
| M.10 Result value (resVal) (optional) .....                                                                                                                                              | 69 |
| M.11 Result value recovery rate (resValRec) and M.12 Result value corrected for recovery<br>(resValRecCorr) (optional).....                                                              | 70 |
| M.13 Expression of result percentage (exprResPerc) and M.14 Expression of result type<br>(exprResType) (optional) .....                                                                  | 71 |
| M.15 Result qualitative value (resQualValue) (optional) .....                                                                                                                            | 72 |
| M.16 Type of results (resType) (mandatory) .....                                                                                                                                         | 72 |



|                                                                                                                                                        |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| M.17 Result value uncertainty (resValUncert) (optional).....                                                                                           | 74  |
| M.20 resInfo.notSummed (optional).....                                                                                                                 | 74  |
| N.01 Limit for the result evaluation (evalLowLimit), N.03 Type of limit for the result evaluation (evalLimitType) (optional).....                      | 75  |
| N.04 Evaluation of the result (evalCode) (mandatory) .....                                                                                             | 76  |
| N.05 Action taken (actTakenCode), N.06.1 Conclusion of follow-up investigation (evalInfo.conclusion) and N.06.2 Comment (evalInfo.com) (optional)..... | 77  |
| N.06.3 Result assessment (evalInfo.resAsses) (optional) .....                                                                                          | 79  |
| N.06.4 Restriction or exception (evalInfo.restrictionException) (optional) .....                                                                       | 80  |
| 3 Specific requirements for baby foods' data submission .....                                                                                          | 87  |
| 4 Specific requirements for chemical contaminants' data submission .....                                                                               | 87  |
| 5 Algorithm to classify samples according to legislative matrix groups .....                                                                           | 99  |
| 5.1 Veterinary medicinal product residues .....                                                                                                        | 99  |
| 5.2 Pesticide residues .....                                                                                                                           | 99  |
| 6 Business Rules .....                                                                                                                                 | 99  |
| 7 Controlled terminology catalogues .....                                                                                                              | 146 |
| 8 Legal Limits database .....                                                                                                                          | 150 |
| 9 Reporting flags in the EFSA Scientific Data Warehouse .....                                                                                          | 150 |
| 10 Validation, National and Annual Reports .....                                                                                                       | 151 |
| 11 Data validation and acceptance .....                                                                                                                | 154 |
| References .....                                                                                                                                       | 156 |
| List of abbreviations .....                                                                                                                            | 157 |

## List of tables

|                                                                                                                                                                                                                                       |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Table 1: Main changes introduced with the 2025 Chemical Monitoring data collection ...                                                                                                                                                | 11  |
| Table 2: Combinations of codes to be used to describe the type of sampling programmes/programme legal reference/sampling strategy for pesticide residues, VMPPR, contaminants, food additives and food flavourings data domains. .... | 23  |
| Table 3: Reportable unspecific country codes for sampCountry and origCountry .....                                                                                                                                                    | 26  |
| Table 4: FoodEx2 main facet descriptions and their relevance for each data domain.....                                                                                                                                                | 34  |
| Table 5: paramType codes and correspondent interpretation .....                                                                                                                                                                       | 61  |
| Table 6: List of analysis hierarchies of PARAM catalogue and associated domain.....                                                                                                                                                   | 64  |
| Table 7: Full list of SSD2 elements, mapping to SSD1 and mandatory SSD2 elements..                                                                                                                                                    | 64  |
| Table 8: Specific reporting requirements for certain contaminants, food additives and food flavourings. ....                                                                                                                          | 71  |
| Table 9: Full list of general business rules (GBR) with troubleshooting tips .....                                                                                                                                                    | 83  |
| Table 10: Full list of chemical monitoring business rules with troubleshooting tips .....                                                                                                                                             | 115 |
| Table 11: Full list of catalogues and the hierarchies in use for chemical monitoring reporting .....                                                                                                                                  | 121 |
| Table 12: List of LEGREF values for the progLegalRef data element and domain attributes.....                                                                                                                                          | 148 |
| Table 13: Summary of reports .....                                                                                                                                                                                                    | 152 |



# 1 Introduction

## 1.1 Background

This guidance is intended for the use of data providers reporting datasets (files) to EFSA in SSD2 format that contain results for chemical parameters in relation to the occurrence of contaminants in food and feed, to food additives, to food flavourings, and to the monitoring of pesticide residues and veterinary medicinal product residue (VMPR) levels in food and feed.

Except for the statutory EU Annual Report domain (pesticide residues and VMPR) samples taken in any year can be transmitted to the EFSA Scientific Data Warehouse (sDWH) when the data provider has the data ready. Only samples taken in the specific calendar year will be included in each year's national and Annual Report<sup>1</sup>.

EFSA receives the results of testing for substances by laboratories in food and feed under:

- Regulation (EC) No 396/2005<sup>2</sup> on maximum residue levels of pesticides in or on food and feed of plant and animal origin; Regulation (EU) 2021/1355<sup>3</sup> on multiannual national control programmes for pesticide residues to be established by Member States and Regulation (EU) 2024/989<sup>4</sup> on the coordinated multiannual control programme of the Union for the year 2025;
- Regulation (EC) No 178/2002<sup>5</sup> (as amended) on general food law;
- Regulation (EC) No 2017/625<sup>6</sup> on official controls and associated delegated acts;
- Regulation (EU) 2023/915<sup>7</sup> on maximum levels for certain contaminants in food;
- Regulation (EC) No 1333/2008<sup>8</sup> for food additives;
- Regulation (EC) No 1334/2008<sup>9</sup> for food flavourings;

<sup>1</sup> National and Annual Reports are required under some legislations. The formats and content are defined and agreed upon through the collaboration of the Commission, Member States and EFSA. Terminology varies across legislation and includes the 'EU Summary Report' and 'EU Annual Report'. See Section 10 for further details.

<sup>2</sup> Regulation (EC) No 396/2005 of the European Parliament and of the Council of 23 February 2005 on maximum residue levels of pesticides in or on food and feed of plant and animal origin and amending Council Directive 91/414/EEC. OJ L 70, 16.3.2005, p. 1–16.

<sup>3</sup> Commission Implementing Regulation (EU) 2021/1355 of 12 August 2021 on multiannual national control programmes for pesticide residues, to be established by Member States. OJ L 291, 13.8.2021, p. 120–121.

<sup>4</sup> Commission Implementing Regulation (EU) 2024/989 of 2 April 2024 concerning a coordinated multiannual control programme of the Union for 2025, 2026 and 2027 to ensure compliance with maximum residue levels of pesticides and to assess the consumer exposure to pesticide residues in and on food of plant and animal origin and repealing Implementing Regulation (EU) 2023/731. OJ L, 2024/989, 3.4.2024.

<sup>5</sup> Regulation (EC) No 178/2002 of the European Parliament and of the Council of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety. OJ L 31, 1.2.2002, p. 1–24.

<sup>6</sup> Regulation (EU) 2017/625 of the European Parliament and of the Council of 15 March 2017 on official controls and other official activities performed to ensure the application of food and feed law, rules on animal health and welfare, plant health and plant protection products, amending Regulations (EC) No 999/2001, (EC) No 396/2005, (EC) No 1069/2009, (EC) No 1107/2009, (EU) No 1151/2012, (EU) No 652/2014, (EU) 2016/429 and (EU) 2016/2031 of the European Parliament and of the Council, Council Regulations (EC) No 1/2005 and (EC) No 1099/2009 and Council Directives 98/58/EC, 1999/74/EC, 2007/43/EC, 2008/119/EC and 2008/120/EC, and repealing Regulations (EC) No 854/2004 and (EC) No 882/2004 of the European Parliament and of the Council, Council Directives 89/608/EEC, 89/662/EEC, 90/425/EEC, 91/496/EEC, 96/23/EC, 96/93/EC and 97/78/EC and Council Decision 92/438/EEC (Official Controls Regulation). OJ L 95, 7.4.2017, p. 1–142.

<sup>7</sup> Commission Regulation (EU) 2023/915 of 25 April 2023 setting maximum levels for certain contaminants in food and repealing Regulation (EC) No 1881/2006. OJ L 119, 5.5.2023, p. 103–157.

<sup>8</sup> Regulation (EC) No 1333/2008 of the European Parliament and of the Council of 16 December 2008 on food additives. OJ L 354, 31.12.2008, p. 16–33.

<sup>9</sup> Regulation (EC) No 1334/2008 of the European Parliament and of the Council of 16 December 2008 on flavourings and certain food ingredients with flavouring properties for use in and on foods and amending Council



- Regulation (EU) 2022/1644<sup>10</sup> and Regulation (EU) 2022/1646<sup>11</sup> on the official controls of VMPP in live animals and animal products;
- Council Directive 2002/32/EC<sup>12</sup> on undesirable substances in animal feed;
- Regulation (EU) 2016/127<sup>13</sup>, Regulation (EU) 2016/128<sup>14</sup> and Directive 2006/125/EC<sup>15</sup> regarding baby food;
- Regulation (EU) 2019/1793<sup>16</sup> on official controls of certain goods from third countries and its reviewed annexes;
- Regulation (EU) 2019/1873<sup>17</sup> on intensified official controls on products of animal origin;
- Regulation (EU) 2022/931<sup>18</sup> and Regulation (EU) 2022/932<sup>19</sup> on the performance of official controls on the presence of contaminants in food.

Additionally, results on the presence of chemicals in food not covered by the above legislation can also be reported to EFSA.

In the frame of the chemical monitoring (ChemMon) data collection programme, occurrence data on plasticisers such as phthalates, structurally similar substances and replacement substances were collected by EFSA<sup>20</sup>. An ad hoc collection of results generated

Regulation (EEC) No 1601/91, Regulations (EC) No 2232/96 and (EC) No 110/2008 and Directive 2000/13/EC. OJ L 354, 31.12.2008, p. 34–50.

<sup>10</sup> Commission Delegated Regulation No (EU) 2022/1644 of 7 July 2022 supplementing Regulation (EU) 2017/625 of the European Parliament and of the Council with specific requirements for the performance of official controls on the use of pharmacologically active substances authorised as veterinary medicinal products or as feed additives and of prohibited or unauthorised pharmacologically active substances and residues thereof. OJ L 248, 26.9.2022, p. 3–17.

<sup>11</sup> Commission Implementing Regulation (EU) No 2022/1646 of 23 September 2022 on uniform practical arrangements for the performance of official controls as regards the use of pharmacologically active substances authorised as veterinary medicinal products or as feed additives and of prohibited or unauthorised pharmacologically active substances and residues thereof, on the specific content of multi-annual national control plans and specific arrangements for their preparation. OJ L 248, 26.9.2022, p. 32–45.

<sup>12</sup> Directive 2002/32/EC of the European Parliament and of the Council of 7 May 2002 on undesirable substances in animal feed – Council statement. OJ L 140, 30.5.2002, p. 10–22.

<sup>13</sup> Commission Delegated Regulation (EU) 2016/127 of 25 September 2015 supplementing Regulation (EU) No 609/2013 of the European Parliament and of the Council as regards the specific compositional and information requirements for infant formula and follow-on formula and as regards requirements on information relating to infant and young child feeding. OJ L 25, 2.2.2016, p. 1–29.

<sup>14</sup> Commission Delegated Regulation (EU) 2016/128 of 25 September 2015 supplementing Regulation (EU) No 609/2013 of the European Parliament and of the Council as regards the specific compositional and information requirements for food for special medical purposes. OJ L 25, 2.2.2016, p. 30.

<sup>15</sup> Commission Directive 2006/125/EC of 5 December 2006 on processed cereal-based foods and baby foods for infants and young children. OJ L 339, 6.12.2006, p. 16.

<sup>16</sup> Commission Implementing Regulation (EU) 2019/1793 of 22 October 2019 on the temporary increase of official controls and emergency measures governing the entry into the Union of certain goods from certain third countries implementing Regulations (EU) 2017/625 and (EC) No 178/2002 of the European Parliament and of the Council and repealing Commission Regulations (EC) No 669/2009, (EU) No 884/2014, (EU) 2015/175, (EU) 2017/186 and (EU) 2018/1660. OJ L 277, 29.10.2019, p. 89–129.

<sup>17</sup> Commission Implementing Regulation (EU) 2019/1873 of 7 November 2019 on the procedures at border control posts for a coordinated performance by competent authorities of intensified official controls on products of animal origin, germinal products, animal by-products and composite products. OJ L 289, 8.11.2019, pp. 50–54.

<sup>18</sup> Commission Delegated Regulation (EU) 2022/931 of 23 March 2022 supplementing Regulation (EU) 2017/625 of the European Parliament and of the Council by laying down rules for the performance of official controls as regards contaminants in food. OJ L 162, 17.6.2022, p. 7–12.

<sup>19</sup> Commission Implementing Regulation (EU) 2022/932 of 9 June 2022 on uniform practical arrangements for the performance of official controls as regards contaminants in food, on specific additional content of multi-annual national control plans and specific additional arrangements for their preparation. OJ L 162, 17.6.2022, p. 13–22.

<sup>20</sup> On 24/07/2020 EFSA received from the EU Commission (DG SANTE) the first part of a 2-part mandate on the re-evaluation of the risks to public health related to the presence of phthalates, structurally similar substances and replacement substances from food contact materials (FCM). As one of the tasks of part 1 of the mandate (Mandate Number M-2020-0183), a list of substances with potential use as a plasticiser has been identified. Out of those, the ones with an authorisation for use in FCM at the EU or national level have been prioritised (high, medium or low priority; EFSA CEP Panel, 2022.). Only the prioritised substances were subject to this data collection under the ChemMon data collection programme (EFSA CEP Panel, 2022. Identification and





in the context of experimental studies on Food Contact Materials (FCM) before their actual use, e.g. tests on migration of plasticisers from FCM using food/food simulants, or tests on concentration of plasticisers in FCM was issued separately in 2022 and was relaunched in 2023<sup>21</sup>. The data collection of such data under the Plasticisers\_FCM data collection is considered complete and is not expected to be repeated in the future.

On 23 March 2022 and 9 June 2022, Commission Delegated Regulation (EC) No 2022/931<sup>22</sup> and Commission Implementing Regulation (EC) No 2022/932<sup>23</sup> entered into force respectively, laying down rules for the performance of official controls as regards contaminants in food. These two Regulations aim to ensure continuity of the rules of Directive 96/23/EC on the content of the multiannual national control plans (MANCP) and its preparation, as well as the minimum frequency of official controls, as regards contaminants in food, within the framework of Regulation (EU) 2017/625. These Regulations apply as of 1 January 2023.

On 12 May 2023, Commission Recommendation (EC) No 2023/965<sup>24</sup> entered into force with a pilot data collection on the monitoring of data on the occurrence (use levels and analytical results) of food additives, in accordance with Regulation (EC) No 1333/2008, and food flavourings, as laid out in Regulation (EC) No 1334/2008.

Further, Implementing Regulation (EU) 2019/723<sup>25</sup> establishes the standard model form for the information and data to be included in the EC Annual Report on Official Controls (AROC) submitted by each Member State to the European Commission (EC). Table 1.4 of this form will be filled by the EC with the number of samples in status accepted in EFSA sDWH at the time agreed with the MANCP Network members<sup>26</sup>.

Results from Border Control Posts (BCPs) are to be reported to TRACES (Article 133 of Regulation (EU) No 2017/625). However, SANTE Unit E.2 and Unit E.4 require Member States (MS) to report information on these border control analyses also to EFSA via the SSD2 format due to different levels of detail compared with TRACES. More specifically, Member States are required for VMPPR (Article 9 of Regulation (EU) No 2022/1646), contaminants (Article 8 of Regulation (EU) No 2022/932) and pesticides domains (Article 31 of Regulation (EC) No 396/2005 and in SCoPAFF - Pesticide Residue meeting<sup>27</sup>), to report information on this border control analysis to EFSA via the SSD2 format. Future

---

prioritisation for risk assessment of phthalates, structurally similar substances and replacement substances potentially used as plasticisers in materials and articles intended to come into contact with food. EFSA Journal 2022;20(5):7231, 26 pp. <https://doi.org/10.2903/j.efsa.2022.7231>

<sup>21</sup> An ad-hoc 'data collection on plasticisers in and migrating from FCM' (Plasticisers\_FCM\_2022) for migration tests using food simulants and concentration into the materials was issued separately in 2022 and was relaunched in 2023. The concentration of plasticising agents in or migration from FCM collected in the ad hoc Plasticisers\_FCM\_2022 call, are intended to be established through testing under controlled laboratory conditions of contact with food simulants (or exceptionally with the food itself) before the actual use of the FCM.

<sup>22</sup> Commission Delegated Regulation (EU) 2022/931 of 23 March 2022 supplementing Regulation (EU) 2017/625 of the European Parliament and of the Council by laying down rules for the performance of official controls as regards contaminants in food. OJ L 162, 17.6.2022, p. 7–12

<sup>23</sup> Commission Implementing Regulation (EU) 2022/932 of 9 June 2022 on uniform practical arrangements for the performance of official controls as regards contaminants in food, on specific additional content of multi-annual national control plans and specific additional arrangements for their preparation. OJ L 162, 17.6.2022, p. 13–22

<sup>24</sup> Commission Recommendation (EU) 2023/965 of 12 May 2023 on the methodology for the monitoring of food additive and food flavouring intake. OJ L 129, 16.5.2023, p. 17–24.

<sup>25</sup> Commission Implementing Regulation (EU) 2019/723 of 2 May 2019 laying down rules for the application of Regulation (EU) 2017/625 of the European Parliament and of the Council as regards the standard model form to be used in the Annual Reports submitted by Member States. OJ L 124, 13.5.2019, p. 1–31.

<sup>26</sup> For further information related to this Network, please contact [SANTE-F-AROC@ec.europa.eu](mailto:SANTE-F-AROC@ec.europa.eu); [sante-consult-f4@ec.europa.eu](mailto:sante-consult-f4@ec.europa.eu)

<sup>27</sup> Agenda point A.17 of the 18-19 September 2023 SCoPAFF meeting:  
[https://food.ec.europa.eu/system/files/2023-10/sc\\_phyto\\_20230918\\_ppr\\_sum.pdf](https://food.ec.europa.eu/system/files/2023-10/sc_phyto_20230918_ppr_sum.pdf)



assessment of SSD2 and TRACES and compatibility format will be undertaken with the aim to assist Member States in avoiding double reporting.

In 2013, EFSA published a revision of the Standard Sample Description (EFSA, 2013) (SSD2), which provides the data specification for submitting laboratory results in samples from the food chain. This version incorporates FoodEx2 (EFSA, 2015), a food description system which allows detailed classification of complex food items and is compatible with the EU menu food consumption surveys (EFSA, 2014b). Data providers are requested to transmit all chemical monitoring data in the SSD2 format. This offers the opportunity to collect all chemical monitoring data through a single mechanism, reducing duplicate reporting and conflicting requirements. Some modifications have been made that may reduce the reporting burden for data providers and ensure that the data received are fit for purpose for compliance and exposure assessments and potentially re-usable for other scientific purposes. Input from the Chemical Monitoring Data Collection Network members has also been considered when preparing this document.

For newly adopted and amended regulations not applicable to data collected by EFSA in the reference year, the provisions of the older regulations remain in force.

The following documents should be consulted in conjunction with this document:

- Standard Sample Description ver. 2.0. (EFSA, 2013).
- Guidance on Data Exchange version 2.0 (EFSA, 2014a).
- The food classification and description system FoodEx2 (revision 2) (EFSA, 2015).
- Webinar: The FoodEx2 classification system and guidance on its harmonised use.<sup>28</sup>
- Catalogue Browser user guide (EFSA & Ioannidou, 2019b).
- EFSA FoodEx2 Interpreting and Checking Tool user guide (EFSA & Ioannidou, 2019c).
- Data Collection Framework (DCF) User Manual<sup>29</sup> (EFSA, DCF publication).

This document is designed to provide harmonised guidance on the implementation of the SSD2 and the GDE2 and to solve the issues listed below. It replaces all previous EFSA guidance on the reporting of chemical monitoring results from the contaminants, food additives, pesticide residues and VMPP domains. This guidance must be used in conjunction with SSD2 and GDE2 which provide details of the workflow, data validation and the EFSA Data Collection Framework principles. It also provides details where experience gained has resulted in modifications to the SSD2 and GDE2 Guidance. Where divergence exists between the SSD2 and GDE2 original guidance publications and this document (e.g. in field size or mandatory status), the details in this guidance must be followed:

- 1) *Multiple domain data collections.* The inclusion of data under Regulation (EU) 2022/1644 and Regulation (EU) 2022/1646 (repealing Council Directive 96/23/EC), which addresses residues of substances in samples of animal origin that are, pesticides and veterinary residues, has highlighted the issue of duplicate reporting and uncertainty regarding which data collection a sample should be submitted to. It was agreed to create a single SSD2 data collection for the results of laboratory analysis for chemical substances found in food, feed, animals, and plants. The results to be included in National and Annual Reports generated by EFSA will be selected using specific analysis hierarchies for each chemical monitoring domain (e.g. by specific Regulation or hazard group). The analysis hierarchies will be

<sup>28</sup> <https://www.efsa.europa.eu/it/events/event/180926>

<sup>29</sup> This document is available within the DCF for users to download.





applied to the PARAM and MTX catalogues plus other catalogues where necessary. A programme legal reference is also required to select data for inclusion in the Annual Reports for European monitoring programmes. This approach simplifies reporting from Member States and allows a more holistic assessment of specific hazards of concern for EFSA. This document no longer addresses the specific reporting requirements and analysis hierarchies for the area of FCM substances, which belonged to the chemical contaminant's domain in ChemMon2022 and ChemMon2023.

- 2) *Conflicts in mandatory SSD2 elements and Business Rules (BRs) for the same substance in different data collections.* Some of these requirements come specifically from legislation, e.g. the requirement to report CCalpha and CCbeta for VMPP results. However, in other cases they are a result of differing priorities or approaches to data quality when previous reporting guidance was specified. This document seeks to resolve conflicts wherever possible and provide a harmonised data model and data quality criteria in the form of business rules for all ChemMon results. Whenever substances are reported that fall into more than one domain, BR of all applicable domains will be applied. The SSD2 BRs for all chemicals are included and applied to the ChemMon data collection. This reporting guidance applies to all the data domains; where a data element is defined as optional for ChemMon transmissions in general, this does not preclude it from being a mandatory element for the collection of data related to a specific domain or substance and this would continue to be defined in the specific BR as also detailed in Table 8.
- 3) *Compound elements.* Some data providers have indicated that the creation of the strings required to report compound elements has resulted in additional technical overheads. The SSD2 XSD schema definition now allows:
  - the reporting of each data element listed in the SSD2 specification separately; or
  - as compound elements.
- 4) *Conversion of national values to EFSA catalogue terminology.* SSD2 requires the mapping to EFSA-coded terminologies and the FoodEx2 system requires a complex mapping to both base terms and facets to fully classify and describe the samples taken. Where there are differences in granularity or philosophy of terminologies used in data provider systems there is a risk of mis mapping with subsequent data quality issues. EFSA now provides a catalogue browser application (EFSA, online-b) which links to the latest version of the catalogue and web services to enable better conversion.
- 5) *Public access to document requests and Open Data.* In recent years there have been many requests for access to datasets, largely under the Regulation on Public Access to Documents<sup>30</sup>. EFSA, in collaboration with Member States, has moved to an Open by Default approach for data in the Scientific Data Warehouse (EFSA, 2019a). However, this<sup>31</sup> requires that sensitive information, e.g. personal data and commercially sensitive data, is protected. Therefore, the use of free text fields in

<sup>30</sup> Regulation (EC) No. 1049/2001 of the European Parliament and of the Council of 30 May 2001 regarding public access to European Parliament, Council and Commission documents <http://data.europa.eu/eli/reg/2001/1049/oj>.

<sup>31</sup> Paragraph 3.2.2 of 'Publication of scientific data from EU-coordinated monitoring programmes and surveys' (EFSA et al., 2019).



ChemMon data submissions have been reduced and only those where the content is clearly specified remain. Geographical identifiers below country level and unique identifiers which can be linked to public registers of food business operators are required only when necessary to support risk assessment. The Transparency Regulation (TR, Regulation (EU) 2019/1381<sup>32</sup>), which became applicable on 27/03/2021, has introduced new legal provisions on the publication of data and information supporting requests from the Commission for a scientific output which may also apply to data collected under the ChemMon data collections. According to Article 38(1)(c) of Regulation (EC) No 178/2002<sup>33</sup>, the proactive transparency requirements will be applicable to documents, studies and data submitted to EFSA to support application dossiers or mandates for scientific output received by EFSA on/after 27/03/2021. Similarly, pursuant to Article 38(1) (d) of the aforementioned Regulation, information on which EFSA's outputs are based shall be made proactively available together at the time of publication of the output. As indicated at the 4<sup>th</sup> meeting of the Network on chemical monitoring data collection<sup>34</sup>, Member States who submit data on behalf of natural or legal persons – such as industry – in relation to a post-TR mandate, have the right to submit confidentiality requests for certain data in accordance with the provisions of Articles 39–39e of Regulation (EC) No 178/2002, applicable sectoral legal acts and EFSA's Practical Arrangements on transparency and confidentiality<sup>35</sup>.

- 6) All resources linked to this document (structural metadata, catalogues, business rules, schema definitions) will be published in the EFSA Knowledge Junction<sup>36</sup> in a machine-readable format and human-readable where appropriate.

For the sake of clarity, throughout the document the general concept of 'residue' will be used to indicate residues coming from added substances (e.g. pesticides) and residues of substances present in the food unintentionally (e.g. environmental contaminants).

It is acknowledged that for national annual monitoring, any subsequent changes proposed (to elements, catalogues or BR) in October–November of year *X* can only be applied to results reported in year *X*+2 since the changes must be known before the sampling officers collect the samples in year *X*+1. However, this would not apply for changes aimed at implementing new or amended legal requirements that cannot be anticipated by EFSA. Network members must contribute to and participate in the process of revision on an ongoing basis with emphasis on suggestions being submitted during the last quarter of each year (September–November).

The major changes introduced for the 2026 Chemical Monitoring compared with the previous year's data collection are listed in Table 1.

<sup>32</sup> Regulation (EU) 2019/1381 of the European Parliament and of the Council of 20 June 2019 on the transparency and sustainability of the EU risk assessment in the food chain and amending Regulations (EC) No 178/2002, (EC) No 1829/2003, (EC) No 1831/2003, (EC) No 2065/2003, (EC) No 1935/2004, (EC) No 1331/2008, (EC) No 1107/2009, (EU) 2015/2283 and Directive 2001/18/EC, OJ L 231, 6.9.2019, p. 1–28.

<sup>33</sup> Regulation (EC) No 178/2002 of the European Parliament and of the Council of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety, OJ L31, 1.2.2002, p.1–48.

<sup>34</sup> Minutes of the meeting available here: <https://www.efsa.europa.eu/sites/default/files/2021-12/4th-efsa-scientific-network-chemical-monitoring-data-collection-minutes.pdf>

<sup>35</sup> [https://www.efsa.europa.eu/sites/default/files/corporate\\_publications/files/210111-PAs-transparency-and-confidentiality.pdf](https://www.efsa.europa.eu/sites/default/files/corporate_publications/files/210111-PAs-transparency-and-confidentiality.pdf)

<sup>36</sup> <https://zenodo.org/communities/efsa-kj/about>

Table 1: Main changes introduced with the 2026 Chemical Monitoring data collection

| Change                                                                                                           | Type of change | Reason for change                    | Details                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>New regulation (N422A)</b>                                                                                    | New addition   | Legal Requirement                    | Commission Implementing Regulation (EU) 2019/1873 on intensified official controls at border control posts on products of animal origin, germinal products, animal by-products and composite products has been included in Table 2 of this guidance.<br><br>This regulation has been flagged for contaminants, pesticides, VMPR, food additives and food flavourings.                         |
| <b>Addition of Sampling strategy ST20A in combination with K009A and K018A for pesticides (progLegRef N027A)</b> | Table 2        | Sampling requirement                 |                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Amendment of existing BR (CHEMON08)</b>                                                                       | Error          | Quality requirement                  | If the value in the data element 'Result value' (resVal) is equal to the value in the data element 'Result LOQ' (resLOQ), then the value in the data element 'Type of result' (resType) must be equal to 'Numerical Value' (Val)                                                                                                                                                              |
| <b>Merge of existing BRs CHEMON39_a and CHEMON39_b to CHEMON39</b>                                               | Error          | CHEMON39_a and CHEMON39_b are merged | If the result is reported for food additives (is_add=1), or food flavourings (is_flav=1), then the value in the data element 'Legislative classes' (sampMatCode.legis) must be used to report the additional product classification (food category code) based on Regulations (EC) No 1333/2008 and (EC) No 1334/2008 on food additives, and food flavourings, respectively, as last amended. |
| <b>Amendment of existing BR (CHEMON40)</b>                                                                       | Error          | Quality requirement                  | If the value in the data element 'Type of result' (resType) is 'Qualitative Value (Binary)' (BIN), or Non-Quantified Value (below LOQ)' (LOQ), or 'Non-Detected Value (less than LOD)' (LOD), then the data element 'Result value' (resVal) must be empty.                                                                                                                                    |
| <b>NEW BR CHEMON43_b</b>                                                                                         | Warning        | Quality requirement                  | If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1) then the value in the data element 'Sampling year' (sampY) should be lower than the submission year or equal to the submission year minus 1.<br><br>From warning to error message in 2027.<br><br>CHEMON43 and CHEMON43_b will be merged in 2027.                                                     |



|                                                                                |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------|--------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inactivation of existing BR (CHEMON47)</b>                                  | Inactivation | Quality requirement                               | Inactivation of business rule due to the BR CHEMON40 which is more complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Inactivation of existing BR (CHEMON49)</b>                                  | Inactivation | Quality requirement                               | CHEMON49 is included in the updated description of CHEMON50, therefore, not needed as a separate business rule.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Amendment of existing BR (CHEMON50)</b>                                     | Error        | Quality requirement                               | If the value in the data element 'Programme type' (progType) is equal to 'Official (National and EU) programme' (K018A) or to 'Official (EU) programme' (K009A), and the value in 'Programme legal reference' (progLegalRef) contains 'Regulation (EC) No 396/2005 (amended)' (N027A), then the value in 'Sampling strategy' (sampStrategy) can only be equal to 'Objective sampling' (ST10A) or 'Selective sampling' (ST20A).                                                                                                                                                                                                                                                |
| <b>Merge of existing BRs CHEMON79_a, CHEMON79_b and CHEMON79_c to CHEMON79</b> | Error        | CHEMON79_a, CHEMON79_b, and CHEMON79_c are merged | If the result is reported for contaminants (is_occ=1), food additives (is_add=1), or food flavourings (is_flav=1), then the value in 'Analytical Method' (anMethCode) should be different from 'Unknown' (F500A) or 'Unspecified' (F598A) or 'Classification not possible' (F001A).                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Amendment of existing BR (CHEMON84_a)</b>                                   | Error        | Quality requirement                               | If the result is reported for food additives (is_add=1), or food flavourings (is_flav=1), then the 'Expression of result type' (exprResType) must be reported.<br><br>From warning to error message in 2026.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Amendment of existing BR (CHEMON84_b)</b>                                   | Warning      | Quality requirement                               | If the result is reported for contaminants (is_occ=1) in food, then the 'Expression of result type' (exprResType) is highly recommended to be reported.<br><br>Business rule remains a warning message for contaminants.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Amendment of existing BR (CHEMON86)</b>                                     | Warning      | Quality requirement                               | If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1), then the 'physical-state' facet of the sample (F03) if not implicitly present in the 'description of the matrix of the sample' (sampMatCode) is highly recommended to be reported when the sample refers to the following food categories/sub-categories:<br>1 - Dairy products and analogues<br>6.3 - Breakfast cereals<br>12.5 - Soups and broths<br>12.6 - Sauces<br>13 - Foods intended for specific groups as defined by Regulation (EU) No 609/2013<br>14.1.2 - Fruit juices and vegetable juices<br>14.1.3 - Fruit nectars and vegetable nectars<br>14.1.4 - Flavoured drinks |



|                                              |         |                     |                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------|---------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              |         |                     | <p>14.1.5 - Coffee, tea, fruit infusions<br/>17 - Food supplements</p> <p>Updated business rule, which returns a warning message. From warning to error message in 2027 for the following food categories 12.5, 13, 14.1.4, 14.1.5, 17.</p>                                                                                               |
| <b>Amendment of existing BR (CHEMON88)</b>   | Warning | Quality requirement | <p>If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1), then 'restriction or exception' (evalInfo.restrictionException) is highly recommended to be reported.</p> <p>This business rule might become an error in 2027.</p>                                                                            |
| <b>Amendment of existing BR (CHEMON90_a)</b> | Warning | Quality requirement | <p>If the paramCode "RF-0102-001-PPP" 'Copper' is reported, then the use of facet 'Part-consumed-analysis' (F20) and/or 'Process' (F28) is needed.</p> <p>From warning to error message in 2027. Business rule remains a warning.</p>                                                                                                     |
| <b>Amendment of existing BR (CHEMON90_b)</b> | Error   | Quality requirement | <p>If the paramCode "RF-0102-001-PPP" 'Copper' is reported, then 'Result Value Uncertainty' (resValUncert) must be reported, if resType = VAL.</p> <p>From Warning to Error message in 2026.</p>                                                                                                                                          |
| <b>Amendment of existing BR (CHEMON94)</b>   | Error   | Quality requirement | <p>If the 'programme type' (progType) is 'Official (National) programme for Third Country Import' (K038A), then sampling point (sampPoint) can only be equal to 'Border Control Posts' (E010A).</p> <p>From warning to error message in 2026.</p>                                                                                         |
| <b>Amendment of existing BR (CHEMON95)</b>   | Warning | Quality requirement | <p>If the result is reported for PPP (is_pest=1) and 'Evaluation of the result' (evalCode) is equal to 'greater than maximum permissible quantities' (J003A), then a value in the data element 'country of origin', different from 'XX', 'AA', 'EU', 'XC', 'XD', 'XE' must be reported.</p> <p>It remains a warning in 2026.</p>          |
| <b>Amendment of existing BR (CHEMON96)</b>   | Error   | Quality requirement | <p>If the result is VMPP (is_vet=1) and 'programme type' (progType) is 'Official (National) programme' (K005A), then 'sampling strategy' (sampStrategy) can only be equal to 'Objective sampling' (ST10A), 'Selective sampling' (ST20A), 'Suspect sampling' (ST30A) or 'Other' (ST90A).</p> <p>From warning to error message in 2026.</p> |





|                                            |         |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------|---------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Amendment of existing BR (CHEMON97)</b> | Error   | Quality requirement | <p>If the result is reported for pesticides (is_pest=1), contaminants (is_occ=1), additives (is_add=1) or flavourings (is_flav=1) and programme type (progType) is 'Official (National) programme' (K005A), then sampling strategy (sampStrategy) can only be equal to 'Objective sampling' (ST10A), 'Selective sampling' (ST20A) or 'Suspect sampling' (ST30A).</p> <p>From warning to error message in 2026.</p>                                                                  |
| <b>Amendment of existing BR (CHEMON98)</b> | Error   | Quality requirement | <p>If the result is reported for contaminants (is_occ=1) and related to control plans, therefore, linked to Commission Delegated Regulation (EU) 2022/931 and Implementing Regulation (EU) 2022/932 (progLegRef N375A), then programme type (progType) can be only 'Official (National) programme' (K005A) or 'Official (National and EU) programme' (K018A) or 'Official (National) programme for Third Country Import' (K038A).</p> <p>From warning to error message in 2026.</p> |
| <b>New BR (CHEMON99)</b>                   | Warning | Quality requirement | <p>If programme type (progType) is 'Official (National) programme for Third Country Import' (K038A) or 'EU increased control programme on imported food' (K019A) or sampling point is 'Border Control Posts' (E010A), then country of origin (origCountry) cannot be equal to the sampling country (sampCountry).</p> <p>From warning to error message in 2027.</p>                                                                                                                 |
| <b>NEW BR (CHEMON100)</b>                  | Warning | Quality requirement | <p>If the result is reported for VMPr (is_vet=1), then 'evaluation of the result' (evalCode) can only be equal to 'less than or equal to maximum permissible quantities' (J002A), or 'greater than maximum permissible quantities' (J003A), or 'Result not evaluated' (J029A), or 'Compliant due to measurement uncertainty' (J031A), or 'detected' (J041A) or 'not detected' (J040A).</p> <p>From warning to error message in 2027.</p>                                            |
| <b>NEW BR (CHEMON101)</b>                  | Error   | Quality requirement | <p>If the value in 'Programme legal reference' (progLegalRef) is 'Commission Implementing Regulation (EU) No 2019/1873' (N422), then the value in 'Programme type' (progType) is equal to 'EU increased control programme on imported food' (K019A) and the value in 'Sampling strategy' (sampStrategy) can only be equal to 'Suspect sampling' (ST30A).</p> <p>Error message already in 2026.</p>                                                                                  |



|                           |         |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NEW BR (CHEMON102)</b> | Warning | Quality requirement | <p>If the result is reported for VMPR (is_vet=1), for the records with the same sampEventId, the elements repCountry, sampCountry, sampArea, origCountry, origArea, origFishAreaCode, origFishAreaText, procCountry, procArea should be constant.</p> <p>From warning to error message in 2027.</p>                                                                                                                                           |
| <b>NEW BR (CHEMON103)</b> | Warning | Quality requirement | <p>'Organic production' (A07SE) and 'Conventional (non-organic) production' (A0C6Y) should not be reported together in 'Production method' (sampMatCode.prod). Please note that in that specific case the sample will be considered as organic.</p> <p>From warning to error message in 2027.</p>                                                                                                                                             |
| <b>NEW BR (CHEMON104)</b> | Error   | Quality requirement | <p>If the value in 'Programme legal reference' (progLegalRef) is 'Commission Implementing Regulation (EU) No 2019/1873' (N422), then no other progLegalRef can be used.</p> <p>Error message already in 2026.</p>                                                                                                                                                                                                                             |
| <b>NEW BR (CHEMON105)</b> | Error   | Quality requirement | <p>If the value in 'Programme legal reference' (progLegalRef) is 'Commission Implementing Regulation (EU) No 2019/1793' (N317A), then no other progLegalRef can be used.</p> <p>Error message already in 2026.</p>                                                                                                                                                                                                                            |
| <b>NEW BR (CHEMON106)</b> | Warning | Quality requirement | <p>If the paramCode "RF-00000248-ADD" 'Potassium sorbate' is reported, then the paramText should be used to specify if the value is "Expressed as free acid" or "Expressed as salt".</p>                                                                                                                                                                                                                                                      |
| <b>NEW BR (CHEMON107)</b> | Warning | Quality requirement | <p>If one of the paramCode "RF-00000304-ADD" 'Sorbic acid', "RF-00000248-ADD" 'Potassium sorbate', "RF-00000062-ADD" 'Butylated hydroxytoluene (BHT)", "RF-00000370-TOX" 'Coumarin', "RF-00000333-TOX" 'Hydrocyanic acid' or "RF-00000334-TOX" 'Theobromine' is reported, and ResType is reported as "Qualitative Value (Binary)", and resQualValue is reported as 'NEG' then it is highly recommended to report the LOD.</p>                 |
| <b>NEW BR (CHEMON108)</b> | Warning | Quality requirement | <p>If the result is reported for additives (is_add=1) or flavourings (is_flav=1) then the following codes "Food colours" 'A047N', "Artificial food colour" 'A047Q', "Food additives other than flavours, colours and artificial sweeteners" 'A047R', "Food flavourings" 'A047A', "Natural food colour" 'A047P' and "Preparations for food flavouring" 'A0F3T' cannot be reported as sampMatCode.</p> <p>It might become an error in 2027.</p> |

|                             |         |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|---------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NEW BR (CHEMON109)</b>   | Warning | Quality requirement | If the reported sampMatCode contains an implicit additive/flavouring food legislative category (F33), then the explicit additive/flavouring F33 is not needed. It might become an error in 2027.                                                                                                                                                                                                                                                           |
| <b>NEW BR (LL_01_FA_FF)</b> | Warning | Quality requirement | If the value in 'Result type' (resType) is 'Numerical value' (VAL), and 'Type of limit for the result evaluation' (evalLimitType) is 'Maximum Permitted Level (MPL)' (W016A) or is not reported and the value in 'Result value' (resVal) is greater than the MPL, then the value in 'Evaluation of the result' (evalCode) should be different from 'Less than or equal to maximum permissible quantities' (J002A) or 'Compliant' (J037A)                   |
| <b>NEW BR (LL_02_FA_FF)</b> | Warning | Quality requirement | If the value in 'Result type' (resType) is 'Numerical value' (VAL), and the value in 'Type of limit for the result evaluation' (evalLimitType) is 'Maximum Permitted Level (MPL)' (W016A) or is not reported and the value in 'Result value' (resVal) is less than or equal to the MPL, then the value in 'Evaluation of the result' (evalCode) should be different from 'Greater than maximum permissible quantities' (J003A) and 'Non-compliant' (J038A) |
| <b>NEW BR (LL_03_FA_FF)</b> | Warning | Quality requirement | The substance (paramCode) reported should be authorised in this legislative category with this restriction/exception according to Regulation 1333/2008 and 1334/2008.                                                                                                                                                                                                                                                                                      |

## 2 SSD2 data model elements

The focus of this document is on all the mandatory and optional elements required for the preparation of charts and tables used in National Reports, EU Annual Reports, online reports of monitoring data and the elements needed to perform exposure assessments. SSD2 elements other than those described in this document can be reported at the discretion of the data provider. For food safety incidents or specific studies to investigate factors which may affect the levels of a hazard in tested food items, there may be a requirement to complete additional elements (e.g. production dates, hazard-specific product descriptions, information on production site hygiene classification/biosecurity level). These study-specific mandatory elements would be specified in the ad hoc study guidance or reporting instructions issued during the specification and planning phase and before the collection of any samples for these targeted data collections.

Specific elements of the SSD2 data model, including their references, are detailed below where implementation for chemical monitoring requires clarification. The references are taken from the original SSD2 guidance:

- Element code (e.g. B.01)
- Element label (e.g. sampling programme identification code)
- Element name (e.g. ProgID)

At the end of the present Guidance, Table 7 provides the full list of the SSD2 data elements along with e.g. their applicability and the concerned controlled terminology (catalogues).

### B.01 Sampling programme identification code (progId) (mandatory)

Reporting countries should use this field to specify their own codes for national sampling programmes or projects under which the sample was taken. All samples analysed under a programme for a specific purpose or objective should be grouped under this code. One or more of these codes can be used to group samples where there is a requirement to compare actual samples taken against national sampling plans or in cases where ad hoc studies have been performed to address a food safety issue.

This element is not used in national or EU Annual Reports issued by EFSA. It is requested and considered useful mainly for the data providers and/or national organisations. For example, it allows EFSA and data providers to collaboratively identify samples taken for the same purpose; past experience has shown that during the data validation step if a specific progId was reported for a subset of samples, this made it easier for EFSA and national data providers to retrieve and check whether those results were correctly coded. Further, where specific risk-based monitoring schemas are defined under the implementing acts as per the requirements of Regulation (EU) No 625/2017, the progId would make it possible to, for example, identify the samples under these schemas. EFSA has provided this element in the MicroStrategy validation dashboard so that the details of all samples reported with a given progId can be filtered and visualised.

A country that does not use progId can simply provide a default value for all records, considering that there are no restrictions on the values used as long as they are within the size limit that is set to 100 characters.

| Example                                                                                                                                              | XML                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| A default Lithuanian national programme identifier to group all samples taken according to the national plans intended to fulfil the VMPP monitoring | <progId>LT_2019_VMPP</progId>             |
| Total Diet Study conducted in Slovakia in 2016                                                                                                       | <progId>SK_2016_TDS</progId>              |
| National pesticide monitoring programme in France in 2018                                                                                            | <progId>FR_2018_NPMP</progId>             |
| Ad hoc Italian national programme on fipronil residues in poultry products                                                                           | <progId>IT_2018_FIPRONIL_POULTRY</progId> |
| Portuguese sampling programme to identify all samples taken in 2020 to detect the presence of mineral oils                                           | <progId>PT_Mineral_oils_2020</progId>     |
| Finnish results on the occurrence of sweeteners in food                                                                                              | <progId>FI_2021_sweeteners</progId>       |
| Results to identify the VMPP Romanian National risk-based control plan for third country imports                                                     | <progId>RO_2023_VMPP_Plan3</progId>       |

## B.02 Programme legal reference (progLegalRef) (mandatory)

This mandatory data element is used to specify the legal framework under which the sample analysis was undertaken.

It is also required in a harmonised data collection to support the separation of analytical results into the relevant National and Annual Reports prepared by EFSA.

This data element is repeatable; this allows multiple values reporting in order to enable the identification of samples analysed for substances falling under more than one legal framework. In particular, results that should be included in more than one annual/national report or addressed in EFSA data analysis related to results from different residue domains, should be reported with multiple progLegalRef.

It is also possible to report the legal framework at the result level rather than at the sample level whenever the sample contains analysis performed for different purposes.

Codes to report the progLegalRef can be selected from the LEGREF catalogue using the *ChemMonLegRef* hierarchy. Values in this catalogue are marked as being applicable to the different domains by attribute in the catalogue. This attribute is used to select results to be included in National and Annual Reports, or to be used for exposure assessment and data analysis. It is visible through the catalogue browser and has been summarised, for convenience, in Table 12 of this document.

For the results reflecting the VMPP National Control Plans of substances in group A3b, which according to Regulation (EU) 2022/1644 are not authorised as veterinary medicinal products but that have a Maximum Residue Level (MRL) or default MRL of 0.01 mg/kg under the pesticide residue legislation, the following applies:

- If the validation is to be done according to the pesticide residue domain, the progLegalRef to be used is the combination of 'N371A\$N027A' for VMPP and pesticides, respectively. The combination of sampling strategy and programme type



should be the one that reflects the VMPR National Control Plan as reflected in Table 2.

- If the validation is done according to the VMPR domain only, the progLegalRef to be used is 'N371A', and therefore only reported to VMPR as reflected in the Table 2 sampling strategy and programme type combinations.

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                              | XML                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| For pesticide residues: code to be used for samples of food products defined in Annex I of Regulation (EC) No 396/2005 (processed and unprocessed products) taken as part of the EU-coordinated programme (EU MACP) defined in Article 29 and in Regulation (EU) 2021/1355 <sup>3</sup> . Both pieces of legislation are covered by the existing progLegalRef code=N027A.                                                                                            | <progLegalRef>N027A</progLegalRef>                                          |
| Baby food samples were collected under Directive 2006/125/EC, or Regulation (EC) 2016/127. Each paramCode reported will fall under its corresponding domain. If a given paramCode belongs to pesticide and contaminant domains, that result will be taken for the pesticide Annual Report output, as well as for contaminant covering baby food outputs.                                                                                                             | <progLegalRef>N028A</progLegalRef> or<br><progLegalRef>N318A</progLegalRef> |
| Samples taken under Regulation (EU) 2022/1646, the legislative framework for the control of VMPR and other residues in samples of animal origin. This is the only code to be selected (alone or in combination with other codes, e.g. in combination with code N027A for the pesticide residue monitoring activities) in order to flag that the sample was taken according to the three different plans intended to fulfil the legal requirements of the Regulation. | <progLegalRef>N371A</progLegalRef>                                          |
| Samples taken under Regulation (EC) 1333/2008 on food additives.                                                                                                                                                                                                                                                                                                                                                                                                     | <progLegalRef>N112A</progLegalRef>                                          |
| Samples where results were taken to assess compliance against the maximum levels for certain contaminants in foodstuffs in Regulation (EU) 2023/915.                                                                                                                                                                                                                                                                                                                 | <progLegalRef>N379A</progLegalRef>                                          |
| Samples where results were taken to assess compliance against Member States' control plans as regards contaminants in food, according to Regulations (EU) 2022/932 and 2022/931.                                                                                                                                                                                                                                                                                     | <progLegalRef>N375A</progLegalRef>                                          |
| Chemical elements tested for in animal organs that would be included in the VMPR Annual Reports and are relevant for reporting chemical contaminants.                                                                                                                                                                                                                                                                                                                | <progLegalRef>N371A\$N379A</progLegalRef>                                   |
| Food samples taken to simultaneously fulfil the legal requirements laid down in both Regulation 2022/1646 and Regulation 396/2005 (i.e. milk, eggs, honey), which should be considered in both the VMPR and in pesticide national/EU Annual Reports.                                                                                                                                                                                                                 | <progLegalRef>N371A\$N027A</progLegalRef>                                   |
| Suspect or targeted samples taken at border inspection under Regulation (EU) 2019/1793.                                                                                                                                                                                                                                                                                                                                                                              | <progLegalRef>N317A</progLegalRef>                                          |
| Samples taken under Regulation (EC) 1334/2008 on food flavourings.                                                                                                                                                                                                                                                                                                                                                                                                   | <progLegalRef>N113A</progLegalRef>                                          |

## B.03 Sampling strategy (sampStrategy) (mandatory)

This element allows the classification of samples according to the sampling methodology applied. It is important that samples taken in a targeted or suspect way are analysed separately from those taken on a random basis.

For VMPP National and Annual Reports, all results should be reported with sampling strategies 'Objective', 'Target sampling' or 'Suspect sampling'. The sampling strategy 'Other' should be used to transmit results that are not considered in the VMPP National Control Plans as set up by Regulation (EU) 2022/1646.

For PPP domain, samples falling under the EU MACP should be reported with sampling strategy 'objective sampling' (ST10A) or "Selective sampling" (ST20A).

Codes can be selected from the SAMPSTR catalogue.

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | XML                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| For samples that were taken as surveillance samples (random sampling), e.g. for the EU-coordinated pesticides monitoring programme or samples that were selected without specific targeting towards products or producers likely to be non-compliant or occurrence of chemical contaminants in food and feed (e.g. dioxins) or surveillance samples taken to fulfil the requirements of the VMPP national randomised surveillance plan for production in the Member States (Plan 2) (Objective sampling).                                                                                                                                                                                                                                                                                                                | <code>&lt;sampStrategy&gt;ST10A&lt;/sampStrategy&gt;</code> |
| Risk-based sampling designed to assess compliance with legislation (e.g. samples under national programmes with the aim of detecting unlawful use or controlling compliance against legal limits laid down in legislation). Either domestic or non-domestic products (Selective sampling/targeted sampling). This is the code to be used when reporting samples taken under national programmes under Regulation (EU) 2021/1355 for pesticide residues. This is also the code that is typically used to report samples taken to fulfil requirements of the VMPP national risk-based control plan for production in the Member States and for third-country imports as per Regulation (EC) No 2022/1646 (Plan 1 and Plan 3). The same code shall be used also for the official controls as regards contaminants in foods. | <code>&lt;sampStrategy&gt;ST20A&lt;/sampStrategy&gt;</code> |
| Risk-based sampling targeting specific producers repeatedly reporting non-compliance, e.g. to enforce the provision of samples taken after Rapid Alert System for Food and Feed (RASFF) notifications or follow-up enforcement samples (Suspect sampling). Samples taken for reasons of suspicion or due to enhanced surveillance, e.g. under emergency measures at import. These samples are not considered in the EFSA exposure assessments. Must be used when reporting samples taken under Regulation (EU) 2019/1793 or Regulation (EU) 2019/1873.                                                                                                                                                                                                                                                                   | <code>&lt;sampStrategy&gt;ST30A&lt;/sampStrategy&gt;</code> |

| Example                                                                                                                                                                                        | XML                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Samples analysed for VMPPR that were collected in the framework of monitoring programmes developed under national legislation, which go beyond the programme as per Regulation (EU) 2022/1646. | <sampStrategy>ST90A</sampStrategy> |

## B.04 Programme type (progType) (mandatory)

This element is used to distinguish samples taken as part of EU control programmes and other sampling programmes.

For PPP domain, samples falling under the EU MACP should be reported using PRGTYP 'K009A'. When reporting samples falling under the MANCP and taken in the EU market, please code the sample with PRGTYP 'K005A'. If there is a need to use the same sample for other domain programmes, PRGTYP 'K018A' can be used, as long as the sample strategy is used in accordance with Table 2. When samples are taken at border, if the sample falls under the EU increased control programme on imported food of non-animal origin (Regulation (EU) 2019/1793) use K019A. Otherwise, K038A for all other import controls.

Codes can be selected from the PRGTYP catalogue.

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | XML                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Samples taken that are part of a programme which is designed and coordinated at a European level, e.g. as part of an EU MACP as defined in Article 29 of Regulation (EC) No 396/2005.                                                                                                                                                                                                                                                                                                                                                                                                                             | <progType>K009A</progType> |
| Samples taken that are part of a programme which is designed and coordinated at a national level (e.g. Regulation (EU) 2021/1355, 2022/932).                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <progType>K005A</progType> |
| Samples taken that are part of both an EU MACP and national programme. For pesticides, this code should also be used for samples where the analytical scope is wider than the pesticide/crop combination listed in the EU MACP (i.e. more pesticides analysed in an EU MACP commodity). This is the code to be used when reporting VMPPR results under the national risk-based and randomised surveillance plans for production in the Member States (Plan 1 and Plan 2). For contaminants, this code should also be used when reporting results related to the control plan for food placed on the Union market. | <progType>K018A</progType> |
| This code is used to describe samples that were taken in the context of increased control programmes on imported food, e.g. taken under Regulation (EU) No 2019/1793.                                                                                                                                                                                                                                                                                                                                                                                                                                             | <progType>K019A</progType> |
| Samples taken under the national risk-based control plan for third-country imports for VMPPR (Plan 3) , but also pesticides if different from K019A. Samples taken under the national risk-based control plan for third-country imports for contaminants according to 2022/932 regulation.                                                                                                                                                                                                                                                                                                                        | <progType>K038A</progType> |
| Occurrence data produced in total diet study (TDS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <progType>K010A</progType> |



- The table below (Table 2) summarises the constraints for coding sampling strategy, programme type and programme legal reference when reporting data for the pesticides, VMPPR, contaminants, food additives and food flavourings domains. The constraints are implemented as BR. All other combinations are acceptable in the other domains.

Table 2: Combinations of codes to be used to describe the type of sampling programmes/programme legal reference/sampling strategy for pesticide residues, VMPPR, contaminants, food additives, and food flavourings data domains.

| progLegalRef                          | N027A                                                   |                                                     |                                                                        | N028A/ N318A                                                                |              | N371A                                              |              |                                 |                | N129A, N379A, N323A, e.t.c                                                   | N375A                                                     |                | N112A                  | N113A                                                   | N317A                                               | N422A                  |
|---------------------------------------|---------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------|----------------------------------------------------|--------------|---------------------------------|----------------|------------------------------------------------------------------------------|-----------------------------------------------------------|----------------|------------------------|---------------------------------------------------------|-----------------------------------------------------|------------------------|
| Programme Legal Reference             | (Regulation 396/2005)<br><br>(Regulation (EU) 2024/989) | (Regulation 396/2005)<br><br>(Regulation 2021/1355) |                                                                        | (Regulation 2016/125/EC)<br>Regulation 2016/128 &<br>Directive 2006/125/EC) |              | (Regulation 2022/1646 and<br>Regulation 2022/1644) |              |                                 |                | (Regulation 178/2002),<br>(Regulation 2023/915),<br>(Regulation EU 625/2017) | (Regulation 2022/932 and<br>2022/931)                     |                | (Regulation 1333/2008) | (Regulation 1334/2008)                                  | (Regulation 2019/1793)                              | (Regulation 2019/1873) |
| Type of sampling programme (progType) | Pesticide residues: EU-coordinated programme            | Pesticide residues: National programmes             |                                                                        | Pesticide residues: baby food                                               | VMPR - Plan1 | VMPR - Plan2                                       | VMPR - Plan3 | VMPR - Other Control activities | Contaminants   | Contaminants                                                                 |                                                           | Food additives | Food flavourings       | Increased control on imported food of non-animal origin | Increased control on imported food of animal origin |                        |
|                                       |                                                         |                                                     | Pesticide residues control at borders (animal and non - animal origin) |                                                                             |              |                                                    |              |                                 |                | Control plan for food placed on the Union market                             | Control plan for food of animal origin entering the Union |                |                        |                                                         |                                                     |                        |
| K005A                                 | Official (National) programme                           | ST10A                                               |                                                                        | ST10A                                                                       |              | ST10A                                              |              |                                 | ST10A          | ST10A                                                                        |                                                           | ST10A          | ST10A                  |                                                         |                                                     |                        |
|                                       |                                                         | ST20A                                               |                                                                        | ST20A                                                                       |              |                                                    |              | ST20A                           | ST20A          | ST20A                                                                        |                                                           | ST20A          |                        |                                                         |                                                     |                        |
|                                       |                                                         | ST30A                                               |                                                                        | ST30A                                                                       |              |                                                    |              | ST30A                           | ST30A          | ST30A                                                                        |                                                           | ST30A          |                        |                                                         |                                                     |                        |
|                                       |                                                         |                                                     |                                                                        |                                                                             |              |                                                    |              | ST90A                           |                |                                                                              |                                                           |                |                        |                                                         |                                                     |                        |
| K009A                                 | ST10A<br>ST20A                                          |                                                     |                                                                        | ST10A<br>ST20A                                                              |              |                                                    |              |                                 | ST10A<br>ST20A |                                                                              |                                                           | ST10A<br>ST20A | ST10A<br>ST20A         |                                                         |                                                     |                        |
| Official (EU) programme               |                                                         |                                                     |                                                                        |                                                                             |              |                                                    |              |                                 |                |                                                                              |                                                           |                |                        |                                                         |                                                     |                        |



| progLegalRef                                                          | N027A                                                     |                                               |                                                                                          | N028A/ N318A                                                                | N371A                                              |                    |                    |                                          | N129A,<br>N379A,<br>N323A, e.t.c                                                         | N375A                                                                 | N112A                                                                          | N113A                     | N317A                     | N422A                                                                  |                                                                 |
|-----------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------|--------------------|--------------------|------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------|---------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------|
| Programme<br>Legal<br>Reference                                       | (Regulation<br>396/2005)                                  | (Regulation<br>396/2005)                      |                                                                                          | (Regulation 2016/125/EC)<br>Regulation 2016/128 &<br>Directive 2006/125/EC) | (Regulation 2022/1646 and<br>Regulation 2022/1644) |                    |                    |                                          | (Regulation<br>178/2002),<br>(Regulation<br>2023/915),<br>(Regulation<br>EU<br>625/2017) | (Regulation<br>2022/932 and<br>2022/931)                              | (Regulation<br>1333/2008)                                                      | (Regulation<br>1334/2008) | (Regulation<br>2019/1793) | (Regulation<br>2019/1873)                                              |                                                                 |
|                                                                       | (Regulation<br>(EU)<br>2024/989)                          | (Regulation<br>2021/1355)                     |                                                                                          |                                                                             |                                                    |                    |                    |                                          |                                                                                          |                                                                       |                                                                                |                           |                           |                                                                        |                                                                 |
| Type of<br>sampling<br>programme<br>(progType)                        | Pesticide<br>residues:<br>EU-<br>coordinated<br>programme | Pesticide<br>residues: National<br>programmes |                                                                                          | Pesticide residues: baby<br>food                                            | VMPR<br>-<br>Plan1                                 | VMPR<br>-<br>Plan2 | VMPR<br>-<br>Plan3 | VMPR -<br>Other<br>Control<br>activities | Contaminants                                                                             | Contaminants                                                          |                                                                                | Food<br>additives         | Food<br>flavourings       | Increased<br>control on<br>imported<br>food of<br>non-animal<br>origin | Increased<br>control on<br>imported food<br>of animal<br>origin |
|                                                                       |                                                           |                                               | Pesticide<br>residues control<br>at<br>borders<br>(animal and non<br>- animal<br>origin) |                                                                             |                                                    |                    |                    |                                          |                                                                                          | Control<br>plan<br>for<br>food<br>placed<br>on the<br>Union<br>market | Control<br>plan for<br>food of<br>animal<br>origin<br>entering<br>the<br>Union |                           |                           |                                                                        |                                                                 |
| K018A                                                                 | ST10A                                                     | ST10A                                         |                                                                                          | ST10A                                                                       |                                                    | ST10A              |                    |                                          | ST10A                                                                                    | ST10A                                                                 |                                                                                | ST10A                     | ST10A                     |                                                                        |                                                                 |
| Official<br>(National and<br>EU)<br>programme                         | ST20A                                                     | ST20A                                         |                                                                                          | ST20A                                                                       | ST20A                                              |                    |                    | ST20A                                    | ST20A                                                                                    | ST20A                                                                 |                                                                                | ST20A                     |                           |                                                                        |                                                                 |
|                                                                       |                                                           |                                               |                                                                                          |                                                                             |                                                    |                    |                    | ST30A                                    |                                                                                          |                                                                       |                                                                                |                           |                           |                                                                        |                                                                 |
|                                                                       |                                                           |                                               |                                                                                          |                                                                             |                                                    |                    |                    | ST90A                                    |                                                                                          |                                                                       |                                                                                |                           |                           |                                                                        |                                                                 |
| K019A                                                                 |                                                           |                                               |                                                                                          |                                                                             |                                                    |                    |                    |                                          |                                                                                          |                                                                       |                                                                                |                           | ST30A                     | ST30A                                                                  |                                                                 |
| EU increased<br>control<br>programme<br>on imported<br>food           |                                                           |                                               |                                                                                          |                                                                             |                                                    |                    |                    |                                          |                                                                                          |                                                                       |                                                                                |                           |                           |                                                                        |                                                                 |
| K038A                                                                 |                                                           |                                               | ST10A                                                                                    | ST10A                                                                       |                                                    |                    |                    | ST10A                                    |                                                                                          |                                                                       | ST10A                                                                          | ST10A                     |                           |                                                                        |                                                                 |
| Official<br>(National)<br>programme<br>for Third<br>Country<br>Import |                                                           |                                               | ST20A                                                                                    | ST20A                                                                       |                                                    | ST20A              |                    | ST20A                                    |                                                                                          | ST20A                                                                 | ST20A                                                                          |                           |                           |                                                                        |                                                                 |
|                                                                       |                                                           |                                               | ST30A                                                                                    | ST30A                                                                       |                                                    |                    |                    | ST30A                                    |                                                                                          | ST30A                                                                 | ST30A                                                                          | ST30A                     |                           |                                                                        |                                                                 |
|                                                                       |                                                           |                                               |                                                                                          |                                                                             |                                                    |                    |                    | ST90A                                    |                                                                                          |                                                                       |                                                                                |                           |                           |                                                                        |                                                                 |





Please, consider the following for VMPP for Table 2 including the descriptions of the flags assigned to each plan:

- Plan 1 refers to the national risk-based control plan for production in the Member States. A sample is flagged as Plan 1 if programme type (progType) is equal to "Official (National) programme" (K005A) or "Official (National and EU) programme" (K018A) and sampling strategy (sampStrategy) is equal to "Selective sampling" (ST20A) and the sample is considered as unprocessed. An exception in the processing has been included for insects;
- Plan 2 refers to the national randomised surveillance plan for production in the Member States. A sample is flagged as Plan 2 if programme type (progType) is equal to "Official (National) programme" (K005A) or "Official (National and EU) programme" (K018A) and sampling strategy (sampStrategy) is equal to "Objective sampling" (ST10A) and the sample is considered as unprocessed. An exception in the processing has been included for insects;
- Plan 3 refers to the national risk-based control plan for third-country imports. A sample is flagged as Plan 3 if programme type (progType) is equal to "Official (National) programme for Third Country Import" (K038A), sampling strategy (sampStrategy) is equal to "Selective sampling" (ST20A) and sampling point (sampPoint) is equal to "Border Control Posts" (E010A);
- Other refers to any additional sample taken not to be considered towards the minimum sampling frequencies set up in Regulation 2022/1646, mainly suspect samples and any other sample taken beyond the mentioned regulation (with the use of ST90A). Any sample with a combination not covered by the three plans, including processed products with combination of Plan 1 or Plan 2, will not have a flag assigned and will automatically be categorised as Other.

In relation to contaminants' control plan results, and in line with Table 2, please take into consideration the descriptions of the flags assigned to results for:

- the control plan for food placed on the Union market, programme type (progType) is equal to "Official (National and EU) programme" (K018A) or "Official (National) programme" (K005A) and sampling strategy (sampStrategy) is equal to "Selective sampling" (ST20A);
- Samples taken under the national risk-based control plan for third-country imports for contaminants, programme type (progType) is equal to "Official (National) programme for Third Country Import" (K038A), sampling strategy (sampStrategy) is equal to "Selective sampling" (ST20A) and sampling point (sampPoint) is equal to "Border Control Posts" (E010A).

In relation to pesticide control plan results, and in line with Table 2, please take into consideration the descriptions of the flags assigned to results for:

- EU MACP refers to the randomised surveillance EU multi-coordinated annual control programme plan by EU Member States. A sample is flagged as EU MACP if programme type (progType) is equal to "Official (EU) programme" (K009A) or "Official (National and EU) programme" (K018A) and sampling strategy (sampStrategy) is equal to "Objective sampling" (ST10A) or "Selective sampling" (ST20A) and the sample is one of the 12 listed in Annex I – Part A or baby food listed in Annex II, recital 2 of the EU MACP Regulation (EU) 2024/989.



- MANCP refers to the national risk-based control plan under Regulation (EU) 2021/1355 in the EU Member States. A sample is flagged as MANCP if programme type (progType) is equal to:
  - "Official (National) programme" (K005A) and sampling strategy (sampStrategy) is equal to "Objective sampling" (ST10A), to "Selective sampling" (ST20A) or to 'Suspect sampling' (ST30A) or
  - "Official (National and EU) programme" (K018A) and sampling strategy (sampStrategy) is equal to "Objective sampling" (ST10A) and is not one of the 12 commodities listed in the EU MACP Regulation (EU) 2024/989 or "Selective sampling" (ST20A) or
  - 'Official (National) programme for Third Country Import' (K038A) and sampling strategy (sampStrategy) is equal to "Objective sampling" (ST10A), to "Selective sampling" (ST20A) or to 'Suspect sampling' (ST30A).
- EU increased control programme on imported food refers to risk-based control plan for third-country imports. A sample is flagged under this plan if programme type (progType) is equal to "EU increased control programme on imported food" (K019A), sampling strategy (sampStrategy) is equal to "Suspect sampling" (ST30A), sampling point (sampPoint) is equal to "Border Control Posts" (E010A) and the origCountry and sampMatCode are those listed in revised annexes of Regulation (EU) 2019/1793.

Based on Table 2, a number of examples are presented below displaying the proposed combinations of codes based on each scenario:

| Example                                                                                                                                           | Combination of codes                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Randomised control plan result for food placed on the Union market; for a sample related to EU MACP for PPP.                                      | <progLegalRef>N027A</progLegalRef><br>& <progType>K009A</progType><br>& <sampStrategy>ST10A</sampStrategy>        |
| Control plan result for food placed on the Union market; for a sample related to VMPPR control Plan 1.                                            | <progLegalRef>N371A</progLegalRef><br>& <progType>K018A</progType><br>& <sampStrategy>ST20A</sampStrategy>        |
| Control plan result for food placed on the Union market regarding Plan 2 of VMPPR and MANCP for PPP; for a unique sample related to both domains. | <progLegalRef>N027A\$N371A</progLegalRef><br>& <progType>K018A</progType><br>& <sampStrategy>ST10A</sampStrategy> |
| Control plan result for food placed on the Union market related to contaminants' control plan.                                                    | <progLegalRef>N375A</progLegalRef><br>& <progType>K018A</progType><br>& <sampStrategy>ST20A</sampStrategy>        |



| Example                                                                                                                                                                                                               | Combination of codes                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control plan result for food of animal origin entering the Union; for a sample related to Contaminants' control plan.                                                                                                 | <code>&lt;progLegalRef&gt;N375A&lt;/progLegalRef&gt;</code><br><code>&amp; &lt;progType&gt;K038A&lt;/progType&gt;</code><br><code>&amp; &lt;sampStrategy&gt;ST20A&lt;/sampStrategy&gt;</code><br><code>&amp; &lt;sampPoint&gt;E010A&lt;/sampPoint&gt;</code>               |
| Control plan result for food of animal origin entering the Union; for a sample related to VMPP control Plan 3, national plan of Pesticides and Contaminants' control plan (food of animal origin entering the Union). | <code>&lt;progLegalRef&gt;N371A\$N027A\$N375A&lt;/progLegalRef&gt;</code><br><code>&amp; &lt;progType&gt;K038A&lt;/progType&gt;</code><br><code>&amp; &lt;sampStrategy&gt;ST20A&lt;/sampStrategy&gt;</code><br><code>&amp; &lt;sampPoint&gt;E010A&lt;/sampPoint&gt;</code> |

## B.05 Sampling method (sampMethod) (optional)

This element is used to provide a reference to the legislation, protocol or other documentation describing the method of selecting samples from the food chain.

If reported, the sampling method codes can be selected from the SAMPMD catalogue.

| Example                                                                                                                                                                      | XML                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Samples taken for the control of dioxins, dioxin-like PCBs and non-dioxin-like PCBs in certain foodstuffs according to Commission Regulation (EU) 2017/644                   | <code>&lt;sampMethod&gt;N040A&lt;/sampMethod&gt;</code>                                                                  |
| For samples taken in the context of Regulation (EC) No 396/2005 and according to Directive 2002/63/EC                                                                        | <code>&lt;sampMethod&gt;N009A&lt;/sampMethod&gt;</code>                                                                  |
| For samples analysed for VMPP according to Regulation (EU) 2022/1644 and Regulation (EU) 2021/808                                                                            | <code>&lt;sampMethod&gt;N042A&lt;/sampMethod&gt;</code>                                                                  |
| Samples where no standardised sampling methodology has been defined                                                                                                          | <code>&lt;sampMethod&gt;N020A&lt;/sampMethod&gt;</code>                                                                  |
| Samples analysed for the presence of mercury in different samples of fish collected in different places (different batches) and put together before the analysis             | <code>&lt;sampMethod&gt;N031A&lt;/sampMethod&gt;</code>                                                                  |
| Samples analysed for the presence of furan, 2-methylfuran and 3-methylfuran in food according to part B of the Annex to Commission Regulation (EC) No 333/2007 <sup>37</sup> | <code>&lt;sampMethod&gt;N011A&lt;/sampMethod&gt;</code>                                                                  |
| Analytical results referring to pooled samples, can be reported either as 'pooled/batch' or 'pooled'                                                                         | <code>&lt;sampMethod&gt;N002A&lt;/sampMethod&gt;</code><br>or<br><code>&lt;sampMethod&gt;N031A&lt;/sampMethod&gt;</code> |

<sup>37</sup> Commission Regulation (EC) No 333/2007 of 28 March 2007 laying down the methods of sampling and analysis for the official control of the levels of lead, cadmium, mercury, inorganic tin, 3-MCPD and benzo(a)pyrene in foodstuffs. OJ L 88, 29.3.2007, p. 29–38.

If the analytical result refers to 'pooled' samples□, the code N002A or N031A ('pooled/batch' or 'pooled') has to be selected; in addition, in these latter cases, it is mandatory to report the two data elements 'sampUnitSize' and 'sampUnitSizeUnit' (see also paragraph on C03 and C04). A BR (CHEMON77) has been implemented to verify if these directions are respected. Reporting these two data elements becomes mandatory from 2025 in case of pooled samples.

## B.06 Sampler (sampler) (mandatory)

This element identifies the person or persons responsible for taking the sample.

Codes can be selected from the SAMPLR catalogue.

| Example                                                                                                                                                                                                                          | XML                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Samples taken in the context of an official control; this is the case for the samples taken in the context of Regulation 396/2005 and Regulation (EU) 2022/1646 for pesticides and VMPP residues or feed additives, respectively | <sampler>CX02A</sampler> |
| Samples taken by food business operators (FBOs)                                                                                                                                                                                  | <sampler>CX01A</sampler> |

## B.07 Sampling point (sampPoint) (mandatory)

This element describes the point in the food chain where the sample was taken.

Codes can be selected from the SAMPNT catalogue.

The catalogue used in this data element is based on a list of terminology developed by Eurostat. The list details the activities of establishments at different points in the food chain. The list of activities of the sampling points proposed is subdivided into three hierarchy levels, the first of which is intended to identify the main stages in the production/consumption of food.

|                      | Example                                                                                                                                  | XML                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Primary production   | Milk samples taken at a farm                                                                                                             | <sampPoint>E100A</sampPoint> |
| Manufacturing        | Milk samples taken at the level of the dairy industry before the bulk tanker has discharged                                              | <sampPoint>E301A</sampPoint> |
| Distribution         | Samples taken at wholesale and retail sale                                                                                               | <sampPoint>E520A</sampPoint> |
| Packaging            | Eggs taken in the collection/packing centres (if it is possible to identify the source farm)                                             | <sampPoint>E600A</sampPoint> |
| Border Control Posts | Samples classed as import samples; this code shall be selected for reporting import sampling for VMPP Plan 3 and contaminants monitoring | <sampPoint>E010A</sampPoint> |

When EFSA generates the VMPP reports the specific 'SAMPNT.vmppClasses' hierarchy may be applied, which classifies the sampPoint codes into three main classes: 'Slaughter', 'Farm' and 'Other'.

In case of sample foodstuffs purchased directly by consumers via online platforms, the sampling point will depend on the origin i.e. Supermarket, Farm.

## C.01 Sampling event identification code (sampEventId) (optional)

The sampEventId is the unique identifier that represents the sampling unit extracted at a certain time from the sampled population. This identifier can be reported when multiple samples are taken from a single sampling unit at a point in time. The sampling unit could be a batch, an animal, a flock, a herd or a holding. The sampling unit type can optionally be reported in C.02 sampUnitType and C.05 sampUnitIds.

If a value is not reported in sampEventId, this data element will automatically be substituted by EFSA with the sampId during the data submission process.

For pesticide residues, this element is not used when counting the number of samples for the creation of reports and can be left empty.

In the context of Regulation (EU) 2022/1646 control activities (VMPR residues), if two samples are e.g. taken from a single pig at slaughter (one sample of kidney and one of muscle) the two samples should be reported with the same sampEventId and in the VMPR national sampling plan will correspond to one pig (sampEventId=CY2017X001 in the example below). In the below table an example is provided in which two samples (please note the two different sampId) from the same animal (a pig) are correctly reported with regard to the same sampEventId and will be both classified in the VMPR Product Category 'Pigs'.

| sampEventId | sampId     | sampMatCode | sampMatText    | sampUnitType   | sampUnitId | paramCode       | resId     | evalCode | evalInfo.com                                               |
|-------------|------------|-------------|----------------|----------------|------------|-----------------|-----------|----------|------------------------------------------------------------|
| CY2017X001  | CY2021X1/K | A01YM       | Pig kidney     | G199A (animal) | IT123      | RF-00000581-VET | CY2021R01 | J002A    | Confirmatory test in kidney                                |
| CY2017X001  | CY2021X1/M | A01RG       | Pig fresh meat | G199A (animal) | IT123      | RF-00000581-VET | CY2021R02 | J002A    | Confirmatory test in muscle was below the MRL of 100 µg/kg |

A BR checks if two or more samples reported with different sampId but belonging to the same sample event (sampEventId) are referring to the same animal species.

## C.02 Sampling unit type (sampUnitType) (optional)

This element describes the sampling unit defined in the sampling method. It can be used to indicate whether the sample contains material from multiple individuals or lots. This is used in the example in Section C.01 to indicate that the two samples were taken from one pig.

Codes can be selected from the SAMPUNTYP catalogue.

| Example                                                                                     | XML                                                         |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Milk samples taken at the level of the dairy industry before the bulk tanker has discharged | <code>&lt;sampUnitType&gt;G202A&lt;/sampUnitType&gt;</code> |
| Single samples (e.g. one animal or one fruit) which are not representative of a lot/batch   | <code>&lt;sampUnitType&gt;G203A&lt;/sampUnitType&gt;</code> |

### C.03 Sampling unit size (sampUnitSize) and C.04 Sampling unit size unit (sampUnitSizeUnit) (optional)

The size of the sampling unit and its unit of measurement can be reported using the optional fields `sampUnitSize` and `sampUnitSizeUnit`, respectively.

The optional data elements `sampUnitSizeUnit` and `sampUnitSize` are used to report how a sample is created before the analysis, providing information on the unit of measurement (e.g. 'Unit', 'Litre', etc.) and the amount (a number) linked to the unit provided, respectively.

| Example                                                                                                                                                                    | XML                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| An individual rice sample made up of 300 grams that have been collected and analysed for mycotoxins should be reported with the <code>sampUnitSizeUnit</code> G148A='Gram' | <code>&lt;sampUnitSizeUnit&gt;G148A&lt;/sampUnitSizeUnit&gt;</code><br><code>&lt;sampUnitSize&gt;300&lt;/sampUnitSize&gt;</code> |

However, in the specific case of pooled samples (please also refer to Section B.05 on 'Sampling method'), the element `sampUnitSizeUnit` must be reported with the code G005A ('Unit'), while the element `sampUnitSize` must report the number of single samples pooled. The system returns an error message if this BR is not followed.

| Example                                                                                                                                                                                                        | XML                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| A pooled sample made up of five units of fish collected in different points and combined before the sample is analysed for mercury levels must be reported with the <code>sampUnitSizeUnit</code> G005A='Unit' | <code>&lt;sampUnitSizeUnit&gt;G005A&lt;/sampUnitSizeUnit&gt;</code><br><code>&lt;sampUnitSize&gt;5&lt;/sampUnitSize&gt;</code> |

### D.01 Sample taken identification code (sampId) (mandatory)

Each sample must be identified by a unique sample identification reference not longer than 100 characters.

Where multiple analytical results are reported for a sample (e.g. results for different residues analysed in the same sample using multiresidue methods and/or several single residue methods), the same `sampId` must be used for all the results. Business rule, GBR2, applies to ensure that.

The sample identification code is used to determine the overall status of the sample (e.g. compliant/non-compliant against the MRL) based on all the results reported for the substances/marker residues measured in the sample. The sample identification code is used to enforce the total number of samples taken by MS in accordance with the requirements set in different Regulations.





| Example                                         | XML                                  |
|-------------------------------------------------|--------------------------------------|
| Unique identifier for sample from 2021 in Italy | <sampId>IT_2021_AS000023456</sampId> |

## D.03 Country of sampling (sampCountry) (mandatory)

This element is reported using the ISO 3166-1-alpha-2 country codes for the country where the sample was taken. Codes can be selected from the COUNTRY catalogue.

Where samples are analysed in a different country than the country of sampling (e.g. where a lab is located outside of the sampling country) this element is needed to correctly connect the results to the sampling country.

For pesticide residues, the sampCountry and the reporting organisation country must be the same (CHEMON58). The pesticides Annual Report will include records with sampCountry being one of the EU countries, Norway or Iceland. Samples taken in the overseas territories of EU countries must be reported as sampCountry of the corresponding EU country, e.g. samples taken in Guadeloupe must be reported as sampCountry=France.

Country of origin cannot be unknown due to EU legislation where traceability of goods placed on the market is a must. Therefore, EFSA stresses the importance of collecting this information and submitting it to EFSA. However, if some data providers are not capable to have this information, the following table (Table 3) will be used to recode unspecific country codes. In the case of non-compliant results, codes listed in Table 3 cannot be used.

EFSA recommends that the country of processing of the sample taken (E.08) is provided when bulk samples are reported and the country of origin is not available.

Table 3: Reportable unspecific country codes for sampCountry and origCountry

| Code description                    | Code | Use for pesticide residues reporting (Annual Reports) |              |
|-------------------------------------|------|-------------------------------------------------------|--------------|
|                                     |      | sampCountry                                           | origCountry  |
| <b>EEA (European Economic Area)</b> | AA   | Excluded                                              | Unknown (XX) |
| <b>Non-EEA</b>                      | XC   | Excluded                                              | Unknown (XX) |
| <b>Non-domestic, import</b>         | XD   | Excluded                                              | Unknown (XX) |
| <b>Non-European Union</b>           | XE   | Excluded                                              | Unknown (XX) |
| <b>Unknown</b>                      | XX   | Excluded                                              | Unknown (XX) |

| Example                                        | XML                           |
|------------------------------------------------|-------------------------------|
| Sample taken in Greece                         | <sampCountry>GR</sampCountry> |
| Sample taken in Northern Ireland <sup>38</sup> | <sampCountry>XI</sampCountry> |

<sup>38</sup> In accordance with the Agreement on the withdrawal of the United Kingdom of Great Britain and Northern Ireland from the European Union and the European Atomic Energy Community, and in particular Article 5(4) of the Windsor Framework (see Joint Declaration No 1/2023 of the Union and the United Kingdom in the Joint Committee established by the Agreement on the withdrawal of the United Kingdom of Great Britain and Northern Ireland from the European Union and the European Atomic Energy Community of 24 March 2023, [OJ L 102, 17.4.2023, p.87](#)) in conjunction with section 24 of Annex 2 to that Framework, for the purposes of this Regulation, references to Member States include the United Kingdom in respect of Northern Ireland

## D.06 Year of sampling (sampY), D.07 Month of sampling (sampM), D.08 Day of sampling (sampD) (mandatory)

The complete date on when the sample was taken is mandatory. The information on the date of sampling is required to check the sample compliance against legal limits applicable at the time of sampling and to select results for inclusion in the Annual Reports.

Samples taken in any year can be transmitted to the EFSA sDWH when the data provider has the data ready. However, in line with the European Commission (EC) interpretation, each EU VMPPR and pesticides reports should only include samples taken in the specific calendar year, which runs from January to December, and submitted within agreed deadlines. In relation to samples for contaminants, food additives, and food flavourings, reporting of historical data is possible. Therefore, data providers can include samples taken from previous years. However, only samples in the specific calendar year (equal to the submission year minus 1) will be considered on the food additives/food flavourings report.

| Example                 | XML                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Friday 16 February 2019 | <pre>&lt;sampY&gt;2019&lt;/sampY&gt; &lt;sampM&gt;02&lt;/sampM&gt; &lt;sampD&gt;16&lt;/sampD&gt;</pre> |

## D.11 Original sample identifier (sampInfo.origSampId) (optional)

This element can be used to indicate that subsequent sampling and testing is linked to an original non-compliant or contaminated sample and offers the possibility to separate samples taken to support a specific investigation from routine monitoring samples.

In order to make explicit a connection between two samples where one is feed known to have been given to an animal and where the same animal is the source of a food sample, data providers must use sampInfo.origSampId to connect the two samples. This applies to insects as food and the substrates they are fed when both are analysed for chemicals. Similarly, an explicit connection is relevant for follow-up samples, which originate from a positive control sample.

## E.02 Coded description of the matrix of the sample taken (sampMatCode) (mandatory)

Samples in FoodEx2 must be encoded according to the guidance for the harmonised use of the FoodEx2 system and the quality control of the codes (EFSA, 2015). To describe the food or feed product or matrix sampled, the code reflecting the highest level of detail is normally used (e.g. select the code for tomatoes instead of the code for Solanaceae).

FoodEx2 codes can be selected from the MTX catalogue using the reporting hierarchy, which includes base terms for food, feed and non-food animal matrices. If the sample is not yet declared/intended for use as either food or feed, the sample must be reported as food. To generate the national/Annual Reports and classify samples according to legal limits or legislative groups, specific analysis hierarchies (see Table 11) are defined using the information provided in sampMatCode (Base terms and facets – both implicit and explicit).



FoodEx2 requires that a base term is always supplied. If the implicit facets are enough to characterise the matrix sampled (see example below), only a base term needs to be reported. Where the base term chosen has implicit facets, reporting of explicit facets should only be additional ones not already covered by the implicit facets.

### ***VMPR samples coding***

Note that for VMPR monitoring, two facets must always be present: the source (F01 facet) and the part nature (F02 facet) (except for the special case of feed and water samples coding and processed-composite food as detailed in the paragraphs below). In most cases, these are already pre-assigned in the codes present in the reporting hierarchy as implicit facets when the FoodEx2 browser is used for coding the matrix samples (an exception to this is, for example, the case of non-food base terms). If reported, the explicit facets supersede the implicit facet already pre-assigned in the FoodEx2 selected code when the EFSA categorisation of the matrices according to the VMPR legislation is applied to categorise the matrix in the EFSA national and annual EU reports. The F01 source facet code is not implicitly pre-assigned in the codes present in the reporting hierarchy referring to raw primary commodities (RPC) derivatives or 'Processed' products; in this case the F01 should be added by the data provider. For example, if the code A031X='Hen egg mixed whole, dried' or A02PJ='Milk powder' is selected, then an F01 code shall be reported (e.g. F01.A057Z='Gallus (chicken) (as animal)' for the hen dried egg samples or F01.A057E='Cattle (as animal)' for the dried milk).

### ***Wild animal VMPR samples coding***

To encode samples related to wild animals, the code A07RY='Wild, gathered or hunted' from facet F21 on the production method list must always be added to the base term by the data provider (explicit facet); this is particularly important to code wild game samples under the VMPR domain.

Regulation (EU) 2022/1644 and Regulation (EU) 2022/1646 no longer consider Wild Game as one of the VMPR Product Category except in the case of results of the national risk-based control plan for third countries. The samples will still be classified as Wild Game and will be included in the VMPR National Report. However, these samples will not be considered for the VMPR EU Annual Report except for results of the mentioned plan.

### ***Feed and water VMPR samples coding***

Samples of feed and water given to animals should be coded as described in this paragraph. When a feed sample is analysed, the sampMatCode must be selected from the feed section of the reporting hierarchy of the MTX catalogue; the textual description of each base term under the feed section contains the wording '(feed)' (e.g. for the code A05CR='Barley grain (feed)').

To report feed samples, the sampMatCode must always contain an implicit or explicit facet F23 *target-consumer* code to characterise the species for which the feed was intended. Thus, if the specific explicit facet F23 is not reported and the implicit facet F23 is the generic 'Animal feed' ('A07TV') the sample will be classified in the VMPR Product Category 'Other'. Instead, if an explicit facet F23 specific to an animal species is reported, then the sample will be classified in a specific VMPR Product Category other than the Category 'Other'.

If the peculiar cases for which the intended target-consumer species for the feed is not known or if the feed is intended and given to animals of different animal species, then the facet code F23 does not need to be selected and explicitly reported. In the latter cases, please only select a code among the first 12 feed categories that already implicitly contains the F23 code 'A07TV'='Feed sample'; the sample will always be classified in the VMPR Product Category 'Other'.

In addition, please be aware that if more facet F23 codes are explicitly reported (and refer to different animal species i.e. they are in conflict) in the final sampMatCode for the very same sample, then the analytical result reported for this sample will not be included in the VMPR reports created by EFSA because the sample will be classified in the VMPR Product Category 'Other'.

| Example of sampMatCode reported for feed                                             | Implicit and explicit facets code in the reported sampMatCode                                                                                                                                                                                     | Note                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A0BBB#F23.A07VC<br>Base term selected:<br>A0BBB='Barley, roasted (feed)'             | Implicit facets:<br>F02: Part-nature= 'Feed-related (as part nature)'<br>F23: Target-consumer='Animal feed'<br>Explicit facet code added to the base term:<br>F23: Target-consumer=A07VC='Pig feed'                                               | Correct coding; based on the explicit F23 facet code added to the base term ABBB this sample will be classified in the VMPR Product Category 'Pigs'                                                                                                                                                                                                       |
| A0BBB#F23.A07VC\$F23.A18EH<br>Base term selected:<br>A0BBB= 'Barley, roasted (feed)' | Implicit facets:<br>F02: Part-nature= 'Feed-related (as part nature)'<br>F23: Target-consumer='Animal feed'<br>Explicit facet codes added to the base term:<br>F23: Target-consumer=A07VC='Pig feed'<br>F23: Target-consumer=A18EH='Chicken feed' | This coding refers to a sample of feed that is fed to two different species (pigs and chicken), therefore it will be classified in the VMPR Product Category 'Other'                                                                                                                                                                                      |
| A0BTL#F23.A07VC<br>Base term selected:<br>A0BTL='Turkeys/Complete feed (feed)'       | Implicit facets:<br>F02: Part-nature= 'Feed-related (as part nature)'<br>F23: Target-consumer='Turkey feed'=A18EM<br>Explicit facet code added to the base term:<br>F23: Target-consumer=A07VC='Pig feed'                                         | This sampMatCode is incorrect, i.e. presumably due to a coding mistake; the F23 codes are contradictory as they refer at the same time to feed given to two different animal species (turkey and pigs). If the coding is not to be amended, this sample will be classified in the VMPR Product Category 'Other', not in the group 'Poultry' nor in 'Pigs' |

| Example of sampMatCode reported for feed                        | Implicit and explicit facets code in the reported sampMatCode                                               | Note                                                                                                                                                                                                              |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A0BBB<br>Base term selected:<br>A0BBB= 'Barley, roasted (feed)' | Implicit facets:<br>F02: Part-nature= 'Feed-related (as part nature)'<br>F23: Target-consumer='Animal feed' | Acceptable coding; however, this sample will be classified in the VMPR Product Category 'Other' because the only facet F23 is the implicit A07TV='Animal Feed' and it does not refer to a specific animal species |

The codes under the feed section are grouped into 14 categories (e.g. 'Cereal grains and products derived thereof (feed)'). If a feed base term is selected from one of the first 13 categories, which implicitly contain the facet F23 target consumer code A07TV='Animal feed' which does not refer specifically to the target consumer species, then an explicit facet code for F23 target consumer must be selected by the data provider. The facet code to be selected must be one of the codes listed under the facet target consumer code A07TX='Feed for animals' (e.g. code A07VE= 'Rabbit feed'). The implicit and/or explicit facet F23 target consumer will be used by EFSA to classify the sample results in one of the VMPR legislative matrix categories used in the VMPR reports (bovines, pigs, sheep & goats, horses, poultry, rabbit, farmed game, wild game, aquaculture, milk, eggs and honey).

If the last category of feed codes in the reporting hierarchy is selected (i.e. 'Compound feed (feed)'= A0BT0), then the code already implicitly contains a facet code F23 designating the target-consumer species and the data provider doesn't need to report an explicit facet F23 term since the analytical result will automatically be included in the VMPR Annual Report in the appropriate VMPR legislative matrix category. In some cases (e.g. A0BV0= 'Unspecified Complete Feed'), it is not possible to specify a species in the implicit facets so an explicit facet describing the target-consumer is needed if the record is intended for inclusion in an Annual Report (e.g. VMPR).

| Category in feed hierarchy | Feed example                                                         | Implicit facets                                                                                        | XML                                                                                                                    |
|----------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>#1</b>                  | 'Barley, roasted (feed)' used for breeding pigs                      | F02: Part-nature= 'Feed-related (as part nature)'<br>F23: Target-consumer='Animal feed'                | <sampMatCode>A0BBB#F23.A07VC</sampMatCode><br><br>F23.A07VC is needed to allocate the record to species 'pigs'         |
| <b>#14</b>                 | 'Complementary feed (incomplete diet) (feed)' used for breeding pigs | F02: Part-nature= 'Feed-related (as part nature)'<br>F23: Target-consumer='Animal feed' and 'Pig feed' | <sampMatCode>A0BV8</sampMatCode><br><br>Explicit F23 facet term is not needed to allocate the record to species 'pigs' |

It should be noted that for the very particular cases where the feed base term implicitly contains an F23 facet 'Sheep and goat feed' (A07VF), it is always necessary to add one of the four codes grouped under the A07VF code. In particular, for goat feed samples, either 'Kids reared for reproduction or meat production feed' (code A18EX) or 'Dairy/reproductive goat feed' (code A18EX) should be specified; for sheep feed samples, either 'Lambs reared for reproduction or meat production feed' (code A18ET) or 'Dairy/reproductive sheep feed' (code A18EV) should be specified.

When samples of water given to the farmed animals are to be coded, please select from the list of terms in 'Non-food matrices' from the reporting hierarchy one of the codes in the 'Environment' group. Since these terms do not contain an implicit facet for the target consumer species, a code for the F23 facet must be added.

| Water example                                        | XML                                        |
|------------------------------------------------------|--------------------------------------------|
| Water from a 'Watering place for animals' for cattle | <sampMatCode>A0F7N#F23.A07TY</sampMatCode> |

### Non-food matrices VMPR samples coding

When non-food matrices have been analysed (e.g. urine or retina or hair samples from pig), the default base term A0C60='Non-food animal-related matrices' should be used. In such cases, it is important to include an explicit F01 'Source' facet code to characterise the source animal species as well as an F02 'Part nature' to characterise the sample.

| Example           | MTX (foodEx2) code                                                                                                                                                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cow hair sample   | <sampMatCode>A0C60#F02.A0ESP\$F01.A057E</sampMatCode><br>Base term: Non-food animal-related matrices (A0C60)<br>Part-nature facet: Hair as part-nature (A0ESP)<br>Source facet: Cattle (as animal) (A057E)     |
| Cow retina sample | <sampMatCode>A0C60#F02.A0ESJ\$F01.A057E</sampMatCode><br>Base term: Non-food animal-related matrices (A0C60)<br>Part-nature facet: Retina (as part-nature) (A0ESJ)<br>Source facet: Cattle (as animal) (A057E) |

### New VMPR product categories: insects, reptiles and edible casings

Regulation (EU) 2022/1646 introduces three new VMPR product categories: insects, reptiles and casings.

The only four insects authorised as novel food in the EU for food production are *Tenebrio molitor*, *Locusta migratoria*, *Acheta domesticus* and *Alphitobius diaperinus*. Therefore, only those insects can be reported to EFSA. The aforementioned insects can be sampled and reported to EFSA as frozen/whole animal, dried and/or powdered.

For edible casings, the base term to be used is A0F1J. When including the source, the source commodity (F27 facet) must be manually inserted.





| Example                   | MTX (foodEx2) code                                                                                                                                                                                                                                                           |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Migratory locust powdered | <p>&lt;sampMatCode&gt;A16XE#F01.A16VP\$F28.A07LA&lt;/sampMatCode&gt;</p> <p>Base term: Edible insect adults (A16XE)<br/>Source facet: Migratory locust (as animal) (A16VP)<br/>Process facet: Grinding/milling/crushing (A07LA)</p>                                          |
| Snakes' meat              | <p>&lt;sampMatCode&gt;A02LC&lt;/sampMatCode&gt;</p> <p>This code includes the following implicit facets:<br/>Part-nature facet: Reptile meat (as part-nature) F02.A0EMJ<br/>Source facet: Snakes (as animal) F01.A04SS</p>                                                   |
| Casings from adult sheep  | <p>&lt;sampMatCode&gt; A0F1J#F01.A0CDD\$F27.A01ZQ&lt;/sampMatCode&gt;</p> <p>Base term: Edible casings (A0F1J)<br/>Source facet: Sheep over 1 year (as animal) (A0CDD)<br/>Source Commodities facet: Sheep edible offal, non-muscle, other than liver and kidney (A01ZQ)</p> |

### Processed product samples for VMPPR

Annex III of Regulation (EU) 2022/1646 introduces the possibility of reporting processed products from the main raw product categories for samples gathered under the national risk-based control plan for third-country import (Plan 3).

The reporting of these samples might differ depending on the type of term (derivative, composite food simple, etc). In order to allow an easy mapping of the samples to the main VMPPR legislative commodity groups, the facet F33 on legislative classes has been introduced. The codes to be used can be found under the parent term "Veterinary Medicinal Product Residues classes (Annex III – 2022/1646)", but only one code should be reported.

In order to code these samples, the rules of the catalogue must be followed. The addition of the facet F33 will automatically assign the sample towards one of the VMPPR legislative commodity group. In case of a mixed sample (e.g. mix pork and beef), the data provider should decide towards which VMPPR legislative commodity the sample should be counted by adding only one code for the facet F33.

| Example                   | MTX (foodEx2) code                                                                                                                                                                                                                                                              |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Canned meat mixed poultry | <p>&lt;sampMatCode&gt;A024C#F04.A04DT\$F04.A04ED\$F33.A1ANV&lt;/sampMatCode&gt;</p> <p>Base term: Canned meat (A024C)<br/>Ingredients facet = Chicken msm (A04DT) and Turkey msm (A04ED)<br/>Legislative classes facet = VR – Poultry (A1ANV)</p>                               |
| Whey protein              | <p>&lt;sampMatCode&gt;A02PS#F33.A1ANY&lt;/sampMatCode&gt;</p> <p>Base term: Whey protein (A02PS)<br/>Legislative classes facet = VR – Milk (A1ANY)</p>                                                                                                                          |
| Mix cow/sheep hard cheese | <p>&lt;sampMatCode&gt;A02ZH#F27.A02LV\$F27.A02MC\$F33.A1ANY&lt;/sampMatCode&gt;</p> <p>Base term: Extra hard cheese (parmesan, grana type) (A02ZH)<br/>Source-commodities facet = Cow milk (A02LV) and Sheep milk (A02MC)<br/>Legislative classes facet = VR – Milk (A1ANY)</p> |

| Example                        | MTX (foodEx2) code                                                                                                                                                                                                                                                                                 |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pork spreadable cooked sausage | <code>&lt;sampMatCode&gt;A025K#F04.A01RG\$F04.A01VA\$F04.A01XJ\$F33.A1ANS&lt;/sampMatCode&gt;</code><br>Base term: Spreadable cooked sausages (A025K)<br>Ingredients facet = Pig fresh meat (A01RG), Pig fat tissue (A01VA) and Pig liver (A01XJ)<br>Legislative classes facet = VR – Pigs (A1ANS) |
| Smoked salmon                  | <code>&lt;sampMatCode&gt;A02KF#F33.A1ANX&lt;/sampMatCode&gt;</code><br>Base term: Smoked salmon (A02KF)<br>Legislative classes facet = VR – Aquaculture (A1ANX)                                                                                                                                    |
| Egg powder                     | <code>&lt;sampMatCode&gt;A031X#F33.A1ANZ&lt;/sampMatCode&gt;</code><br>Base term: Hen egg mixed whole, dried (A031X)<br>Legislative classes facet = VR – Eggs (A1ANZ)                                                                                                                              |
| Chicken sausage                | <code>&lt;sampMatCode&gt;A024G#F33.A1ANV&lt;/sampMatCode&gt;</code><br>Base term: Fresh raw sausage (A024G)<br>Legislative classes facet = VR – Poultry (A1ANV)                                                                                                                                    |
| Beef sausage                   | <code>&lt;sampMatCode&gt;A024G#F33.A1ANP&lt;/sampMatCode&gt;</code><br>Base term: Fresh raw sausage (A024G)<br>Legislative classes facet = VR – Bovines (A1ANP)                                                                                                                                    |

### ***Food additives and food flavourings samples coding***

For additives and flavourings, the legislative food category (F33 facet) must always be present. In some cases, the F33 facet is already pre-assigned in the reporting hierarchy of the FoodEx2 browser. If it is not implicitly assigned, it must be added explicitly. The legislative category “All categories of foods” is not allowed to be reported.

The physical-state facet (F03) should be considered for samples belonging to the following legislative food categories/subcategories:

1 - Dairy products and analogues

6.3 - Breakfast cereals

12.5 – Soups and broths

12.6 - Sauces

13 - Foods intended for specific groups as defined by Regulation (EU) No 609/2013

14.1.2 - Fruit juices as defined by Directive 2001/112/EC and vegetable juices

14.1.3 - Fruit nectars as defined by Directive 2001/112/EC and vegetable nectars and similar products

14.1.4 - Flavoured drinks

14.1.5 - Coffee, coffee and chicory extracts, tea, herbal- and fruit-infusions; coffee substitutes, coffee mixes and mixes for ‘hot beverages’

17 - Food supplements as defined in Directive 2002/46/EC of the European Parliament and of the Council excluding food supplements for infants and young children



If the F03 facet is not implicitly assigned, it is highly recommended to be added. Moreover, if the sample is formulated for infants (<12 months), the target-consumer facet (F23) should be added to the code if it is not implicitly assigned, specifying this information.

| Example of sampMatCode reported for food additives and flavourings domains                                                     | Implicit and explicit facets code in the reported sampMatCode                                                                                                                                                                                                                                                                                                                                                                                  | Note                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Base term selected:</b><br/>A03MX = 'Wine, red'</p> <p>Correct code:<br/>A03MX</p>                                       | <p><b>Implicit facets:</b><br/> <b>F02:</b> Part-nature = 'Wine (as part-nature)'<br/> <b>F27:</b> Source-Commodities = 'Wine grapes'    <b>F28:</b> Process = 'Winemaking'<br/> <b>F10:</b> Qualitative-Info = 'Red'<br/> <b>F33:</b> Legislative classes = 'FA-14.2.2 Wine and other products defined by Regulation (EEC) No 1234/2007, and alcohol-free counterparts'</p>                                                                   |                                                                                                                                                                                                             |
| <p><b>Base term selected:</b><br/>A02RH = 'Soft-ripened cheese'</p> <p><b>Correct code:</b><br/>A02RH#F03.A06JA\$F33.A0C5F</p> | <p><b>Implicit facets:</b><br/> <b>F02:</b> Part-nature = 'Ripened cheese (as part-nature)'<br/> <b>F27:</b> Source-Commodities = 'Milk'<br/> <b>F28:</b> Process = 'Cheesemaking'<br/> <b>F28:</b> Process = 'Ripening'<br/> <b>F10:</b> Qualitative-Info = 'Soft'</p> <p><b>Explicit facets:</b><br/> <b>F03:</b> Physical-state = 'Slices, steaks or other flat cuts'<br/> <b>F33:</b> Legislative classes = 'FA-01.7.2 Ripened cheese'</p> | <p>Correct coding: based on the explicit F03 and F33 facet codes added to the base term A02RH this product will be identified as sliced and classified in the food additive legislative category 01.7.2</p> |



| Example of sampMatCode reported for food additives and flavourings domains                                                          | Implicit and explicit facets code in the reported sampMatCode                                                                                                                                                                                                                                                                                                                                              | Note                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Base term selected:</b><br/>A03YN = 'Omelette, plain'</p> <p><b>Correct code:</b><br/>A03YN# F33.A0C33</p>                    | <p><b>Implicit facets:</b><br/>F02: Part-nature = 'Egg based dishes (as part-nature)'<br/>F04: Ingredient = 'Whole eggs'</p> <p><b>Explicit facets:</b><br/><b>F33:</b> Legislative classes = 'FA- 10.2 Processed eggs and egg products'</p>                                                                                                                                                               | <p>Correct coding: based on the explicit F33 facet code added to the base term A03YN this product will be classified in the food additive legislative category 10.2</p>                                        |
| <p><b>Base term selected:</b><br/>A03QH = 'Infant formula, milk and soya-based, liquid'</p> <p><b>Correct code:</b><br/>A03QH</p>   | <p><b>Implicit facets:</b><br/><b>F02:</b> Part-nature = 'Infant formulae (as part-nature)'<br/><b>F04:</b> Ingredient = 'Milk'<br/><b>F04:</b> Ingredient = 'Soya proteins'<br/><b>F03:</b> Physical-state = 'Liquid'<br/><b>F23:</b> Target-consumer = 'Infant or toddler food'<br/><b>F33:</b> Legislative-classes = 'FA-13.1.1 Infant formulae as defined by Commission Directive 2006/141/EC (1)'</p> |                                                                                                                                                                                                                |
| <p><b>Base term selected:</b><br/>A03SL = 'Vitamin only supplements'</p> <p><b>Correct code:</b><br/>A03SL#F03.A1B1Y\$F33.A0C15</p> | <p><b>Implicit facets:</b><br/><b>F02:</b> Part-nature = 'Food supplements and similar preparations (as part-nature)'<br/><b>F04:</b> Ingredient = 'Vitamins'</p> <p><b>Explicit facets:</b><br/><b>F03:</b> Physical-state = 'Syrup-type'<br/><b>F33:</b> Legislative-classes = 'FA-17.2 Food supplements supplied in a liquid form, excluding food supplements for infants and young children'</p>       | <p>Correct coding: based on the explicit F03 and F33 facet codes added to the base term A03SL this product will be identified as syrup-type and classified in the food additive legislative category 17.2.</p> |

### ***Coding of copper samples under pesticides or contaminants***

When reporting copper, under Chemical monitoring data collection, data providers are requested to include any facet that describe sample preparation i.e. if done in accordance with contaminant or pesticide domain. Therefore, depending on the commodity the following facets will need to be included:

Citrus fruits, miscellaneous fruits with inedible peel, cucurbits with inedible peel and potatoes: contaminants without peel (F20.A07QF) – pesticide residues with peel (F20.A07QE)

Chestnuts contaminants without shell (F20.A07QD) – pesticide residues with shell (F20.A07QC)

Stone fruits contaminants without stone (F20.A07QK) – pesticide residues with stone (F20.A07QJ) (i.e. accounting for stone)

Sweet corn: contaminants kernels without cob (F20.A07RF) – pesticide residues kernels with cob (F20.A07RE)

Coffee beans: contaminants only roasted beans (F28.A07GY) – pesticide residues green beans

Products of animal origin: contaminants muscle with fat (F20.A0F4T) - pesticide residues muscle after removal of trimmable fat (F20.A0F4V)

Fruits, root and tuber vegetables, bulb vegetables, fruiting vegetables, Brassica vegetables, leafy vegetables, legume vegetables, stem vegetables and fungi: contaminants washed (F28.A07JG) – pesticide residues not washed (no code)

Kindly be informed that Tables 4 and 8 include a reference to Copper and specific reporting requirements of facets within the matrix code reported (sampMatCode).

### ***FoodEx2 mapping and general considerations***

For pesticide monitoring, where samples must be reported according to the legislative classes specified in pesticide legislation (corresponding to the EFSA MATRIX catalogue), a mapping from FoodEx2 to MATRIX is recorded in the FoodEx2 and can be seen in the catalogue browser. This mapping is automatically applied in the EFSA data processing and data providers only need to report a correct FoodEx2 code for each sample transmitted to ensure the legislative MATRIX grouping will be correct. Further in the validation dashboards, the mapping can be checked. EFSA provides a separate document to support data providers in choosing the base terms and facets that are used by EFSA to flag the EUCP samples. EFSA highly recommends checking that samples have been flagged as EUCP in accordance with their expectations before accepting the dataset(s) in the confirmation dashboards.

For VMPPR monitoring, where samples must be analysed according to the legislative classes specified in VMPPR legislation, mapping from FoodEx2 to legislative categories is automatically applied in the EFSA data processing and data providers only need to report



a correct FoodEx2 code for each sample transmitted to ensure the legislative grouping will be correct<sup>39</sup>.

For acrylamide reporting under contaminants (see Table 6) explicit facet F33 Legislative classes must be reported. Table 6 should be consulted for other specific, mandatory and recommended reporting requirements applicable to contaminants; the same as when a sample is included in other domains but analysed for such substances.

For food additives and flavourings, it is mandatory to report the legislative class facet code (F33). Mapping from FoodEx2 base terms to legislative categories is automatically applied in the EFSA catalogue as implicit facets in most of the cases; if the F33 facet is not implicitly assigned, it is necessary to be added.

After the 8<sup>th</sup> Chemical Monitoring Data Collection Network meeting, it was proposed that data providers should also report all available data from drinking water samples, as results in this area are currently lacking. EFSA encourages Competent Authorities and data providers to collect and submit available information for pesticides and contaminants in drinking water intended for human consumption (FoodEx2 code A03DK 'Drinking water' and its children).

<sup>39</sup> The VMPP mapping has been updated to accommodate the changes introduced from the entry in force of Regulation (EU) 2022/1644 and Regulation (EU) 2022/1646.



Table 4: FoodEx2 main facet descriptions and their relevance for each data domain

| Facet                  | VMPR                                                                                                                                                                                                                                                                                                                                                                  | Pesticides                                                     | Contaminants                                                                                  | Additives                                                                                                                                                                                                                                                                     | Flavourings                                                                                                                                                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>F01 source</b>      | Indicate the type of animal species sampled (e.g. pig samples). This can include the purpose of rearing; for example, whether the chickens are laying hens or broilers<br>Classification of samples as bovines, pigs, sheep, goats, horses, poultry, aquaculture, rabbit or game are based on this facet. It is important to select a facet at species level or lower |                                                                | Define the 'origin' of the raw commodity; usually, it is already assigned as 'implicit facet' | Plant, animal, organism or source of the raw agricultural commodity.<br>For fish and seafood samples the species must be specified.<br>For algae-based products the species of algae must be specified<br>Plant, animal, organism or source of the raw agricultural commodity | Plant, animal, organism or source of the raw agricultural commodity.<br>For fish and seafood samples the species must be specified.<br>For algae-based products the species of algae must be specified<br>Plant, animal, organism or source of the raw agricultural commodity |
| <b>F02 part-nature</b> | Indicate the 'part' or 'tissue' of the animal tested (e.g. the liver). Must be reported since the MRL legal limit applied is dependent on the target tissue                                                                                                                                                                                                           | Part sampled. For example, indicating fat samples from animals | Part sampled                                                                                  | Part sampled                                                                                                                                                                                                                                                                  | Part sampled                                                                                                                                                                                                                                                                  |



| Facet                         | VMPR                                                                                                                                   | Pesticides                                              | Contaminants                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Additives                                                                       | Flavourings                                                                     |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|                               | The first step in the EFSA procedure for the VMPR matrix classification for samples such as eggs, milk or honey is based on this facet |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                 |                                                                                 |
| <b>F03 physical state</b>     |                                                                                                                                        |                                                         | For reporting the physical state (e.g. solid, jelly, liquid) of the tested food                                                                                                                                                                                                                                                                                                                                                                              | For reporting the physical state (e.g. solid, jelly, liquid) of the tested food | For reporting the physical state (e.g. solid, jelly, liquid) of the tested food |
| <b>F04 ingredient</b>         |                                                                                                                                        | For reporting the ingredients of composite food samples | Recommended for the following products: 'Potato crisps', 'Pre-cooked French fries, potato products for home cooking', 'Breakfast cereals (excluding muesli and porridge)', 'Substitute coffee (dry)' and 'Baby foods, other than processed cereal-based foods' rice-based products algae-based foods for special nutritional uses compound products for infants and small children, including ready-made meals, diet supplements, herb mixes and spice mixes | Repeatable facet to be used to characterise composite foods                     | Repeatable facet to be used to characterise composite foods                     |
| <b>F06 surrounding medium</b> |                                                                                                                                        |                                                         | For reporting the surrounding medium (e.g. oil, fat) of the food                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                 |                                                                                 |

| Facet                         | VMPR | Pesticides                                                                                                                                                                       | Contaminants                                                                                                                                                                                                                                                                        | Additives | Flavourings |
|-------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|
| <b>F07 fat content</b>        |      |                                                                                                                                                                                  | For reporting the fat content of the food. To be reported when the expression of results is fat weight (B003A)                                                                                                                                                                      |           |             |
| <b>F10 qualitative-info</b>   |      | Where whole grain cereal (i.e. wheat, barley, oat and rye) is to be reported under the EUCP, flour integral/not refined (A06HR) can be reported along with the cereal-based term | This facet is recommended for some substances (PARAM codes) belonging to the group of plasticising agents (e.g. phthalates) for expressing when a product is not packed; see also Table 8 for specific requirements                                                                 |           |             |
| <b>F11 alcohol content</b>    |      |                                                                                                                                                                                  | For reporting the alcohol content of the food                                                                                                                                                                                                                                       |           |             |
| <b>F17 extend-of-cooking</b>  |      |                                                                                                                                                                                  | Heat treatment applied to food required for furans and acrylamide                                                                                                                                                                                                                   |           |             |
| <b>F18 packaging format</b>   |      |                                                                                                                                                                                  | Allows sampling officers to describe the shape of the container or wrapper that holds the marketed product. This facet is recommended for some substances (PARAM codes) belonging to the group of plasticising agents (e.g. phthalates); see also Table 8 for specific requirements |           |             |
| <b>F19 packaging material</b> |      |                                                                                                                                                                                  | Allows sampling officers to describe the material of the container or wrapper                                                                                                                                                                                                       |           |             |

| Facet                             | VMPR | Pesticides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Contaminants                                                                                                                                                                                            | Additives | Flavourings |
|-----------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|
| <b>F20 part-consumed-analysed</b> |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | that holds the marketed product.<br>This facet is crucial for some substances (PARAM codes), e.g. bisphenol, group of plasticising agents (e.g. phthalates); see also Table 8 for specific requirements |           |             |
|                                   |      | Where meat (as part nature) or its sub-codes are reported, then part-consumed-analysed is recommended to be used, to indicate the presence of fat:<br>A0F4V='Excluding visible fat' or<br>A0F4T='Including visible fat'. According to Regulation (EC) No 396/2005 the MRLs for 'Muscle' apply to "Meat after removal of trimmable fat".<br>Therefore, to report muscle samples the code A0F4V – 'Excluding visible fat' should be used.<br><br>When reporting copper results, the part to which the sample |                                                                                                                                                                                                         |           |             |



| Facet                        | VMPR                                                                                                             | Pesticides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Contaminants                                                                                                                                      | Additives | Flavourings |
|------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|
|                              |                                                                                                                  | <p>preparation applies needs to be reported</p> <p>For example, reporting a banana without (w/o) peel, as a matrix, should be A01LC#F20.A07QF, while banana with peel should be codified as A01LC#F20.A07QE</p>                                                                                                                                                                                                                                                                                                          |                                                                                                                                                   |           |             |
| <b>F21 production method</b> | <p>A07RY should be used to identify wild game. Classification of samples as wild game is based on this facet</p> | <p>Required to perform the data analysis regarding the residue situation for organic food compared with conventionally produced food.</p> <p>Organic production methods can be reported here using the term A07SE. If the production method is known to be non-organic, the term A0C6Y (conventional non-organic production) should be reported. This term has always been mapped to the former SSD1 value 'non-organic' and has been renamed for better clarity. Where the production method is unknown, this facet</p> | <p>Recommended.</p> <p>Required to perform the data analysis regarding the mycotoxin situation in organic food compared with non-organic food</p> |           |             |

| Facet                      | VMPR                                                                                                                                                        | Pesticides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Contaminants                                                                                                                                                                                                                     | Additives                                                                                                                                                                                                                                    | Flavourings                                                                                                                                                                                                                                  |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                                                                                                                                                             | <p>must not be reported. For the purposes of reporting, 'non-organic' and no facet F21 value reported, will be grouped as 'non-organic' production, since legislation only requires that organic production must be clearly declared.</p> <p>Intensity of production, when known, can also be reported in this facet using A0C6Q (intensive production) or A18FG (extensive (non-intensive)) production. A07RY (wild or gathered or hunted) A07RY should be used to identify wild game (e.g. to distinguish wild boar (farmed) from wild)</p> |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                              |
| <b>F23 target consumer</b> | This facet must be used for feed samples to indicate the species for whom the feed is intended if not already implicitly included in the base term selected | This facet must be used for feed samples to indicate the species for whom the feed is intended if not already implicitly included in the base term selected                                                                                                                                                                                                                                                                                                                                                                                   | <p>This facet must be used for feed samples to indicate the species for whom the feed is intended if not already implicitly included in the base term selected.</p> <p>Also, it can be used to indicate the age of the human</p> | This facet must be used for feed samples to indicate the species for whom the feed is intended if not already implicitly included in the base term selected. Also, it can be used to indicate the age of the human consumer/population (e.g. | This facet must be used for feed samples to indicate the species for whom the feed is intended if not already implicitly included in the base term selected. Also, it can be used to indicate the age of the human consumer/population (e.g. |



| Facet                         | VMPR                                           | Pesticides                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Contaminants                                                                                                                                                                                                                                                                  | Additives                                      | Flavourings                                    |
|-------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------|
|                               |                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | consumer/population (e.g. infants, toddlers, children, and adults)                                                                                                                                                                                                            | infants, toddlers, children, and adults)       | infants, toddlers, children, and adults)       |
| <b>F27 source-commodities</b> |                                                | Report the representative lead crop. It defines the origin of the derivatives for 'processed' food samples made up of one single food/ingredient (e.g. orange juice or wine).<br>This facet describes the raw primary commodity (RPC) from which an ingredient or derivative has been obtained. However, in some food groups, like cheese or fruit juice, products of the same nature as those from one raw source, but from mixed raw sources are encountered. | This facet describes the raw primary commodity (RPC) from which an ingredient or derivative has been obtained. However, in some food groups, like cheese or fruit juice, products of the same nature as those from one raw source, but from mixed raw sources are encountered |                                                |                                                |
| <b>F28 process</b>            | Required to distinguish processed food samples | Required to distinguish processed food samples. This distinction is important for the pesticide residue data as MRL compliance is checked/verified considering the results expressed for 'unprocessed' food samples. For processed products derived from                                                                                                                                                                                                        | Required to distinguish processed food samples. However, a more detailed classification should be used where possible                                                                                                                                                         | Required to distinguish processed food samples | Required to distinguish processed food samples |



| Facet                       | VMPR                                                                                                                                                                                                                                          | Pesticides                                                                                                                                                                                                                                                                             | Contaminants | Additives | Flavourings |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|-------------|
|                             |                                                                                                                                                                                                                                               | <p>raw agricultural products (as specified in Annex I of Regulation (EC) No 396/2005), the most specific code for processing must be selected</p> <p>When reporting copper results, the processing applied to the analysed sample needs to be reported (e.g. F28.A07GY 'roasting')</p> |              |           |             |
| <b>F31 animal-age-class</b> | <p>Although not mandatory, this information is very useful for the interpretation of the results, especially in cases of non-compliance due to residues belonging to the substance groups of anti-thyroid agents (A1b) and steroids (A1c)</p> |                                                                                                                                                                                                                                                                                        |              |           |             |



| Facet                        | VMPR                                                                                                                                                                                                                                   | Pesticides | Contaminants                                                                                                                                                                                               | Additives                                                                                                       | Flavourings                                                                                                       |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>F32 gender</b>            | Although not mandatory, this information is very useful for the interpretation of the results, especially in cases of non-compliance due to residues belonging to the substance groups of anti-thyroid agents (A1b) and steroids (A1c) |            |                                                                                                                                                                                                            |                                                                                                                 |                                                                                                                   |
| <b>F33 legislative class</b> | Required for processed products.                                                                                                                                                                                                       |            | Required for samples analysed for acrylamide to describe the acrylamide legislative classes in Commission Recommendation 2019/1888/EU <sup>40</sup> and Commission Regulation (EC) 2017/2158 <sup>41</sup> | Required to describe the sample legislative food category, according to Regulation (EC) 1333/2008 <sup>42</sup> | Recommended to describe the sample legislative food category according to Regulation (EC) 1334/2008 <sup>43</sup> |

<sup>40</sup> Commission Recommendation (EU) 2019/1888 of 7 November 2019 on the monitoring of the presence of acrylamide in certain foods. OJ L 290, 11.11.2019.

<sup>41</sup> Commission Regulation (EU) 2017/2158 of 20 November 2017 establishing mitigation measures and benchmark levels for the reduction of the presence of acrylamide in food Text with EEA relevance. OJ L 304, 21.11.2017.

<sup>42</sup> Regulation (EC) No 1333/2008 of the European Parliament and of the Council of 16 December 2008 on food additives. OJ L 354, 31.12.2008, p. 16–33.

<sup>43</sup> Regulation (EC) No 1334/2008 of the European Parliament and of the Council of 16 December 2008 on flavourings and certain food ingredients with flavouring properties for use in and on foods and amending Council Regulation (EEC) No 1601/91, Regulations (EC) No 2232/96 and (EC) No 110/2008 and Directive 2000/13/EC. OJ L 354 31.12.2008, p. 34

| Example                                                                                                                                                                                 | XML                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pig kidney</b><br>A01YM=Pig kidneys are composed of the following implicit facets where F01.A057=Pigs and F02.A069N=Kidney                                                           | <sampMatCode>A01YM</sampMatCode>                                                                                                                                                                                                                                                              |
| <b>Sheep for meat production – Blood serum</b><br>A0C60=Non-food animal-related matrices F01.A0CDE=Sheep for meat production and F02.A0CEY=Blood serum (as part nature)                 | <sampMatCode>A0C60#F01.A0CDE\$F02.A0CEY</sampMatCode><br><sampMatCode.building>A0C60</sampMatCode.building><br><sampMatCode.source>A0CDE</sampMatCode.source><br><sampMatCode.part>A0CEY</sampMatCode.part>                                                                                   |
| <b>Urine sample from a dairy cow</b><br>A0C60=Non-food animal-related matrices F01.A0C9L=Dairy cow and F02.A0CET=Urine (as part nature) F31.A0C8V=Young non-adult mammal (above 1 year) | <sampMatCode>A0C60#F01.A0C9L\$F02.A0CET\$F31.A0C8V</sampMatCode><br><sampMatCode.building> A0C60</sampMatCode.building><br><sampMatCode.source>A057F</sampMatCode.source><br><sampMatCode.part>A0F0Y</sampMatCode.part><br><sampMatCode.animal-age-class>A0F0Y</sampMatCode.animal-age-class> |
| <b>Royal jelly</b>                                                                                                                                                                      | <sampMatCode>A0CVG</sampMatCode>                                                                                                                                                                                                                                                              |
| <b>Processed cereal-based foods for infants and young children</b>                                                                                                                      | <sampMatCode>A03QX</sampMatCode>                                                                                                                                                                                                                                                              |
| <b>Infant formula</b>                                                                                                                                                                   | <sampMatCode>A0EQM</sampMatCode>                                                                                                                                                                                                                                                              |
| <b>Follow-on formula</b>                                                                                                                                                                | <sampMatCode>A0EQL</sampMatCode>                                                                                                                                                                                                                                                              |
| <b>Herbal infusions specific for infants and young children, dry</b>                                                                                                                    | <sampMatCode>A16GS</sampMatCode>                                                                                                                                                                                                                                                              |
| <b>Pasteurised eggs from organic farming</b><br>A031G=Hen eggs matrices F21 A07SE=Organic and F28. A07HV=Pasteurisation                                                                 | <sampMatCode>A031G#F21.A07SE\$F28.A07HV</sampMatCode><br><sampMatCode.building>A031G</sampMatCode.building><br><sampMatCode.prod>A07SE</sampMatCode.prod><br><sampMatCode.techno>A07HV</sampMatCode.techno>                                                                                   |
| <b>Potato crisp (category 2)</b><br>indicate whether batch fried or continuously fried<br>A011L=Potato crisps or sticks                                                                 | <sampMatCode>A011L#F17.A07MY</sampMatCode><br><sampMatCode.building>A011L</sampMatCode.building><br><sampMatCode.cookext>A07MY</sampMatCode.cookext>                                                                                                                                          |

| Example                                                                                                                                                                                                                                                                                                               | XML                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F17.A07MY=Outside light brown <sup>44</sup>                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Crispbread (category 6)</b><br>indicate whether the product is baked<br>A0CHT=Crispbread<br>F28.A07GX=Baking                                                                                                                                                                                                       | <pre>&lt;sampMatCode&gt;A0CHT#F28.A07GX&lt;/sampMatCode&gt; &lt;sampMatCode.building&gt;A0CHT&lt;/sampMatCode.building&gt; &lt;sampMatCode.techno&gt;A07GX&lt;/sampMatCode.techno&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Roasted coffee (category 7)</b><br>indicates the extent of cooking (light brown, brown, outside dark brown/slightly burned)<br>A03GL=Coffee beans, roasted<br>F17.A07MY=Outside light brown<br>F17.A07MZ=Outside brown<br>F17.A07NA=Outside dark brown/slightly burned                                             | <pre>&lt;sampMatCode&gt;A03GL#F17.A07MY&lt;/sampMatCode&gt; &lt;sampMatCode&gt;A03GL#F17.A07MZ&lt;/sampMatCode&gt; &lt;sampMatCode&gt;A03GL#F17.A07NA&lt;/sampMatCode&gt; &lt;sampMatCode.building&gt;A03GL&lt;/sampMatCode.building&gt; &lt;sampMatCode.cookext&gt;A07MY&lt;/sampMatCode.cookext&gt; &lt;sampMatCode.building&gt;A03GL&lt;/sampMatCode.building&gt; &lt;sampMatCode.cookext&gt;A07MZ&lt;/sampMatCode.cookext&gt; &lt;sampMatCode.building&gt;A03GL&lt;/sampMatCode.building&gt; &lt;sampMatCode.cookext&gt;A07NA&lt;/sampMatCode.cookext&gt;</pre> |
| <b>Chewing gum</b>                                                                                                                                                                                                                                                                                                    | <pre>&lt;sampMatCode&gt;A035M&lt;/sampMatCode&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Dry peas</b><br>(drying of peas is not considered processed if the moisture content is between 16 and 19%)                                                                                                                                                                                                         | <pre>&lt;sampMatCode&gt;A0DCD&lt;/sampMatCode&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Green peas</b><br>(fresh peas without pods are considered unprocessed)                                                                                                                                                                                                                                             | <pre>&lt;sampMatCode&gt;A012J&lt;/sampMatCode&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Dry camomile flowers</b><br>(herbal infusion, dried product considered unprocessed)                                                                                                                                                                                                                                | <pre>&lt;sampMatCode&gt;A0D9D#F28.A07KG &lt;/sampMatCode&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Bovine meat (muscle)</b> from cow or ox or bull (base term code A01QX), <b>without/excluding the visible fat</b> (food tested and reported in line with the sample description to which pesticide and VMPPR MRL apply). The facet code F20 A0F4V='Excluding visible fat' is explicitly added to the base term code | <pre>&lt;sampMatCode&gt;A01QX#F20.A0F4V&lt;/sampMatCode&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Deer fresh meat (muscle)</b> from <b>hunted</b> , not farmed, animals.                                                                                                                                                                                                                                             | <pre>&lt;sampMatCode&gt;A01SA#F21.A07RY&lt;/sampMatCode&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

<sup>44</sup> List of food categories and sub-categories and respective codes based on Commission Recommendation 2010/307/EU on the monitoring of acrylamide and Commission Recommendation 2013/647/EU (Annex, point C).

| Example                                                                                                                                                                                                                                                                                                                             | XML                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <p>The facet code F21=A07RY must be selected to indicate that the animal was 'wild'; if the animal instead was 'farmed' or 'raised', then the facet code A07RY should not be reported</p>                                                                                                                                           |                                                                              |
| <p><b>Feed sample of maize grains</b></p> <p>Code A07XG='Maize grain (feed)' intended for chicken/poultry<br/>Facet F23 target consumer code=A07VD='Poultry feed'<br/>OR one of its sub-codes (e.g. A18EH='Chicken feed')</p>                                                                                                       | <pre>&lt;sampMatCode&gt;A07XG#F23.A18EH&lt;/sampMatCode&gt;</pre>            |
| <p><b>Water given to farmed rabbit</b></p> <p>The full code will be A0F7L#F23.A07VE='Freshwater for animal farming' &amp; 'Rabbit feed'</p>                                                                                                                                                                                         | <pre>&lt;sampMatCode&gt;A0F7L#F23.A07VE&lt;/sampMatCode&gt;</pre>            |
| <p><b>Pizza taken in a takeaway shop, packed in a laminated pizza box</b></p> <p>The full code will be A03ZN#F18.A07NL\$F19.A07PN<br/>Where:<br/>A03ZN=Pizza and pizza-like dishes PACKAGING-FORMAT=Box<br/>PACKAGING-MATERIAL=Laminated paper-plastic foil<br/>(the takeaway should be reported in the sampPoint data element)</p> | <pre>&lt;sampMatCode&gt;A03ZN#F18.A07NL\$F19.A07PN&lt;/sampMatCode&gt;</pre> |
| <p><b>Pizza taken in a takeaway shop, not packed</b></p> <p>The full code will be<br/>A03ZN#F10.A18PX<br/>where:<br/>A03ZN=Pizza and pizza-like dishes QUALITATIVE-INFO=Not Packed<br/>(the takeaway should be reported in the sampPoint data element)</p>                                                                          | <pre>&lt;sampMatCode&gt;A03ZN#F10.A18PX&lt;/sampMatCode&gt;</pre>            |

| Example                                                                                                                                                                                                                                                                                                                             | XML                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p><b>Infant formula based on milk, heated in the microwave inside a plastic feeding bottle</b></p> <p>The full code will be<br/>A03QF#F28.A07HB\$F18.A07NM\$F19.A16RX<br/>where:<br/>A03QF=Infant formula, milk-based, liquid<br/>PROCESS=Microwave-cooking, PACKAGING-FORMAT=Bottle<br/>PACKAGING-MATERIAL=Polypropylene (PP)</p> | <p>&lt;sampMatCode&gt;A03QF#F28.A07HB\$F18.A07NM\$F19.A16RX&lt;/sampMatCode&gt;</p> |
| <p><b>Hemp seed flour</b></p> <p>The full code will be<br/>A0F0N#F27.A016B<br/>Built starting from the base term A0F0N =Dried nuts/seeds and related flours and powders<br/>and then added F27 code A016B=hemp seeds</p>                                                                                                            | <p>&lt;sampMatCode&gt;A0F0N#F27.A016B&lt;/sampMatCode&gt;</p>                       |
| <p><b>Bread made with hemp seed flour</b></p> <p>The full code will be<br/>A0BY0#F04.A016B\$F04.A0F0M<br/>Where:<br/>A0BY0=Leavened bread and similar<br/>F04.A016B=Hemp seeds<br/>F04.A0F0M=Nut/seeds paste/emulsion/paste</p>                                                                                                     | <p>&lt;sampMatCode&gt;A0BY0#F04.A016B\$F04.A0F0M&lt;/sampMatCode&gt;</p>            |



## E.03 Text description of the matrix of the sample taken (sampMatText) (optional)

After describing the matrix with the most detailed level of information available using FoodEx2 in the sampMatCode field, this free text data element can be completed to report a full textual description of the product sampled and to provide additional relevant information. This will provide a possibility of crosschecking for the codes reported and could highlight any data quality problem at the data analysis level.

In cases in which suitable codes to describe the item sampled cannot be found, it is recommended to ask EFSA for support or the addition of codes.

The description of the product sampled should not include the brand name which can be reported in the element E.10 (sampMatInfo).

## F.01 Sample analysed identification code (sampAnId), G.01 Coded description of the matrix analysed (anMatCode), and G.02 Text description of the matrix analysed (anMatText)

When left empty the three elements sampAnId, anMatCode and anMatText will be assumed to have respectively the same value as the 'sample taken identification code' (element D.01 'sampId'), 'Coded description of the matrix of the sample taken' (element E.02 'sampMatCode'), and 'Text description of the matrix of the sample taken' (element E.03 'sampMatText'), respectively. Therefore, there is no need to provide these data elements unless the nature of the sample analysed differs from the sample taken.

Each sample analysed must be identified by a unique sample identification reference (sampAnId) no longer than 100 characters.

When the analytical results corresponding to different parameters are reported for the same sample analysed, the unique sample (analysed) identification number (sampAnId) must be maintained for that sample analysed in all transmissions.

The data element 'Coded description of the matrix analysed' (anMatCode) reports the matrix sample analysed characteristics encoded using the FoodEx2 catalogue's codes, while the data element 'Text description of the matrix analysed' (anMatText) describes it as free text and could be reported on a voluntary basis if e.g. the sample analysed description (anMatCode) through the FoodEx2 codes are not sufficiently detailed.

## E.04 Country of origin on the sample taken (origCountry) (mandatory)

The country of origin must be completed for all samples in ISO 3166-1-alpha-2 format. Reporting countries are encouraged to identify the origin of the product, particularly for unprocessed (raw) food products and for cases where a non-compliant sample has been found.



Goods whose production involved more than one country will be deemed to originate in the country where they underwent their last, substantial, economically justified processing

or working in an undertaking equipped for that purpose and resulting in the manufacture of a new product or representing an important stage of manufacture<sup>45</sup>. Thus, a single country of origin must be reported as described in the labelling requirements of Regulation (EU) No 1169/2011.

Codes can be selected from the COUNTRY catalogue.

In cases in which the exact country of origin is unknown, but some information is available, Table 3 above summarises the available unspecific country codes which are the same as for D.03: sampCountry.

It should be noted that records with origCountry reported as an unspecific code ('XX', 'AA', 'EU', 'XC', 'XD', 'XE') will be grouped as unknown (code 'XX') for statutory reporting, e.g. the pesticides Annual Report.

For pesticide residue annual reporting, samples taken in the overseas territories of EU countries must continue to be reported as their true country of origin (origCountry), but these samples may be reclassified for the Annual Report formatting to be presented as part of the EU country and the EFSA algorithm will use the attribute 'moperRecodedCountry', visible in the COUNTRY catalogue, to reclassify them as domestic samples.

For VMPPR Plan 3, the country of origin should not be equal to an EU country including the country of the reporting organisation. The results for Plan 3 with an EU country as origin of the sample will then be rejected of the report.

| Example                                                                                                                                                                                                    | XML                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Rice from Thailand packed in Iceland                                                                                                                                                                       | <origCountry>TH</origCountry> |
| An individual country cannot be determined (unspecific codes such as XC, XD, XE can be selected). If needed at the reporting stage, e.g. for the pesticides Annual Report, these may be aggregated by EFSA | <origCountry>XE</origCountry> |

## E.06 Area of origin for fisheries or aquaculture activities code of the sample taken (origFishAreaCode) (optional)

For fish, seafood and other marine products the Food and Agriculture Organization fishing area should be reported.

Codes can be selected from the FAREA catalogue.

<sup>45</sup> Regulation (EU) No 1169/2011 of the European Parliament and of the Council of 25 October 2011 on the provision of food information to consumers, amending Regulations (EC) No 1924/2006 and (EC) No 1925/2006 of the European Parliament and of the Council, and repealing Commission Directive 87/250/EEC, Council Directive 90/496/EEC, Commission Directive 1999/10/EC, Directive 2000/13/EC of the European Parliament and of the Council, Commission Directives 2002/67/EC and 2008/5/EC and Commission Regulation (EC) No 608/2004.

| Example                                                                                                                                                                            | XML                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baltic herring caught in Skagerrak and Kattegat tested for brominated flame retardants (percentage fat should always be reported in this case, results reported for each congener) | <pre>&lt;sampMatCode&gt; &lt;origFishAreaCode&gt;M27IIIa&lt;/origFishAreaCode&gt; &lt;paramCode&gt; &lt;exprResPerc&gt;fatPerc=45&lt;/exprResPerc&gt; &lt;exprResType&gt;B003A&lt;/exprResType&gt;</pre> |

### F.03 Year of analysis (analysisY) (mandatory)

The year of analysis must be reported for all results.

| Example                 | XML                                                |
|-------------------------|----------------------------------------------------|
| Sample analysed in 2017 | <pre>&lt;analysisY&gt;2017&lt;/analysisY&gt;</pre> |

### H.01 Sample analysed portion sequence (anPortSeq) (optional)

This element contains a sequential number (e.g. 1, 2, 3) to be used when a laboratory sample is analysed for the same substance more than once. anPortSeq should only be used for repeated analysis of the same sample with the same method (i.e. under repeatable conditions) and only when legally required (e.g. aflatoxins in dried fruits where three laboratory samples belonging to the same original sample must be analysed according to Regulation (EU) 2023/2782). If different methods are used for the same sample and parameter, only the most accurate and reliable result should be reported (irrespective of the existence of a legal limit).

When reporting results which are subject to assessment against a legal limit, anPortSeq must not be used. Either:

- the result derived with the most accurate or reliable analysis must be reported; or
- when samples were analysed with equally accurate techniques, the mean value should be reported. In cases for which a mean is calculated, the date of the last component result should be reported, and the most applicable analytical method should be reported.

The 'sample taken' could be analysed for the same parameter more than once to perform a counter-analysis or to confirm a positive sample. In these cases, the only result to be reported is the one for which the evaluation is performed and the 'sample analysed portion' should not be used.

| Example                                                                              | XML                                             |
|--------------------------------------------------------------------------------------|-------------------------------------------------|
| Each portion into which the laboratory sample is divided and analysed for aflatoxins | <pre>&lt;anPortSeq&gt;1&lt;/anPortSeq&gt;</pre> |

## J.01 Identification code of the laboratory (labId) (mandatory)

A unique code to identify each laboratory providing laboratory results must be reported here (e.g. the national laboratory code). This code must also be used when providing information on participation in proficiency tests in National Reports. The mapping between code and laboratory name is the reporting country's responsibility (i.e. competent authority/organisation reporting data to EFSA). It should be updated in case of a request by Commission, EURLs or EFSA.

| Example                                 | XML                      |
|-----------------------------------------|--------------------------|
| National Reference Laboratory of Poland | <labId>PolandNRL</labId> |

## J.02 Laboratory accreditation (labAccred) (mandatory)

This element indicates whether the laboratory performing the analysis has been accredited as required by Article 37 of Regulation (EU) No 2017/625.

For pesticide monitoring only two codes from the LABACC catalogue may be used (see table below).

Codes can be selected from the LABACC catalogue.

| Example                                                                                                                                                                       | XML                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Accredited according to ISO/IEC 17025                                                                                                                                         | <labAccred>L001A</labAccred> |
| For results generated by laboratories not or not yet accredited according to ISO/IEC 17025 (e.g. when the laboratory is awaiting the final audit from the accreditation body) | <labAccred>L003A</labAccred> |

## J.03 Laboratory country (labCountry) (mandatory)

Indicate using ISO 3166-1-alpha-2 country codes the country where the laboratory is located. The country code must be unique for each labId provided.

Codes can be selected from the COUNTRY catalogue.

| Example | XML                         |
|---------|-----------------------------|
| Germany | <labCountry>DE</labCountry> |

## K.01 Type of parameter (paramType) (optional)

This data element indicates whether the parameter (paramCode K.02) reported has been analysed in full or partially; it also makes it possible to indicate that the selected paramCode refers to the analysis of a single component of 'Multicomponent' residue definitions (VMPP and pesticide residues) or of an individual parameter of a 'summed' parameter (e.g. dioxins TEQ).

A substance's residue definitions or marker compound used for VMPP and pesticide residue MRLs can be broadly split into two types:

- 'Simple': Compounds that can be analysed using one single calibration substance (in terms of identity: same substance or same substance mix of isomers, etc.).
- 'Multicomponent': Compounds that can be analysed using several different calibration substances (e.g. the parent compound and one or more metabolites).

Starting from the 2021 data collection this data element has been made optional, and the use and meaning of paramType were revised according to definitions reported in Table 5, with the aim of making data reporting easier. EFSA automatically pre-assigns the paramType in those cases in which the paramCode/paramType combination is unambiguous. However, in case of multicomponent paramCode calculated from measurements of individual components or the sum of individual parameters, the paramType should still be reported to indicate if all the expected parts as described by the parameter name have been analysed and summed up or not.

Codes to assign the paramType are selected from the PARAMTYP catalogue; however, in order to harmonise and clarify the scope of the definition, a simplified version will be used as described in Table 5 below.

Table 5: paramType codes and correspondent interpretation

| Code                                      | Code interpretation      | Note                                                                                                                                          |
|-------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P002A</b>                              | Part of a sum            | Result from analysis of a component that is part of a calculated result reported for a paramCode (either P004A or P005A)                      |
| <b>P004A</b>                              | Sum based on subset      | Result for which the full analysis has not been performed, and one or more components are not part of the calculated result for the paramCode |
| <b>P005A</b>                              | paramCode fully analysed | Result for which the full analysis has been performed, and all components are part of the calculated result for the paramCode                 |
| <b>P001A and P003A are not to be used</b> |                          |                                                                                                                                               |

EFSA develops and updates a separate document with the full list of paramCode and pre-assigned paramType, available in Knowledge Junction with other supporting tools, following the criteria explained here:

- Individual components of a multicomponent/sum paramCode: paramType =P002A.
- paramCode measured with one calibration compound (or a mix of compounds), covering the full description of paramCode: paramType =P005A.
- Multicomponent/sum paramCode calculated from measurements of individual components: paramType =P004A/P005A.

In the last case data providers are requested to report either paramType 'P005A' if all the expected parts have been analysed as specified by paramCode, or 'P004A' otherwise.

For pesticide residues, in order to check for MRL compliance it is a legal requirement to analyse the full residue definitions as defined in Regulation (EU) No 396/2005 (i.e. all the components) and reflected in the correspondent paramCode; the current valid residue

definitions and paramCodes are also available in the EFSA Legal Limits database. Additionally, results on those individual components associated with P002A are strongly recommended to be reported too. These will normally not be included in the data analysis presented in the EU Report on pesticide residues. However, for specific data analysis the results labelled as P002A might be used by EFSA in later mandates received by the Commission on specific substances (e.g. acetamiprid<sup>46</sup>).

For veterinary medicinal products, a similar approach is established. However, in case of negative<sup>47</sup> screening results the reporting of the single components associated with the P002A is sufficient and the multicomponent/sum paramCode does not need to be reported. In this case, EFSA will automatically generate the record related to the complex paramCode in the sDWH, which will be then accounted for and totalised in the summary results presented in both the National and EU VMPP Reports.

For contaminants, the single substances or congener-specific occurrence data must be reported, in addition to the sum (when required for assessment against maximum levels) or complex mixtures of occurrence. These data are essential for dietary exposure assessments since animal and human exposure estimates on food/feed required substance/congener-specific occurrence data. The sums of congener/substance groups reported alone are of very limited use for EFSA.

For food additives, the analytical result for the individual substances must be reported expressed as specified in the Regulation N°1333/2008 (e.g., sorbic acid-sorbates expressed as the free acid). Additionally, the sum should be reported if it complements the individual substance, as the maximum permitted levels (MPLs) are regulated at the sum level for the reported legislative category.

Here some examples of paramType assignments are shown.

| Example                                                                                                                                                                                                                        | XML                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Result of analysis of the sum of malachite green and leucomalachite green (paramCode=RF-00004513-PAR) in meat of aquaculture products (Minimum Required Performance Limit (MRPL) defined); both components analysed and summed | <code>&lt;paramType&gt;P005A&lt;/paramType&gt;</code><br>EFSA pre-assignment: P004A/P005A<br>Data providers should report paramType P005A |
| Result of partial analysis of sum of malachite green and leucomalachite green (paramCode=RF-00004513-PAR); only leucomalachite green analysed                                                                                  | <code>&lt;paramType&gt;P004A&lt;/paramType&gt;</code><br>EFSA pre-assignment: P004A/P005A<br>Data providers should report paramType P004A |
| Result of analysis for leucomalachite green (paramCode=RF-00001727-PAR) which is one of the components of the sum reported with paramCode=RF-00004513-PAR (see above)                                                          | <code>&lt;paramType&gt;P002A&lt;/paramType&gt;</code><br>Automatically assigned by EFSA                                                   |
| Result of analysis for permethrin (sum of isomers) (MRL defined) (paramCode=RF-0842-001-PPP)                                                                                                                                   | <code>&lt;paramType&gt;P005A&lt;/paramType&gt;</code><br>Automatically assigned by EFSA                                                   |

<sup>46</sup> <https://open.efsa.europa.eu/questions/EFSA-Q-2022-00589>

<sup>47</sup> In this context, EFSA flags the 'negative' results those reported with the following combinations of codes: RESTYPE='LOD' or 'LOQ' or 'CCB' or 'BIN' or 'CCA' AND evalCode is different from J041A='Detected'



| Example                                                                                                                                                                                                                | XML                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Result reported for aldicarb (sum of aldicarb, its sulfoxide and its sulfone, expressed as aldicarb) (paramCode=RF-0020-001-PPP) in plant commodities where all the components have been analysed                      | <pre>&lt;paramType&gt;P005A&lt;/paramType&gt;</pre> <p>EFSA pre-assignment: P004A/P005A<br/>Data providers should report paramType P005A.</p> <p>Additionally, aldicarb (paramCode=RF-0020-002-PPP) as P002A, aldicarb sulfoxide (paramCode=RF-0020-003-PPP) as P002A and aldicarb sulfone (paramCode=RF-0020-004-PPP) as P002A</p> |
| Result of analysis for Betamethasone in cow meat (paramCode=RF-00000273-VET) (MRL defined)                                                                                                                             | <pre>&lt;paramType&gt;P005A&lt;/paramType&gt;</pre> <p>Automatically assigned by EFSA</p>                                                                                                                                                                                                                                           |
| Result of analysis for Betamethasone in sheep meat (paramCode=RF-00000273-VET) (no MRL defined – not authorised use)                                                                                                   | <pre>&lt;paramType&gt;P005A&lt;/paramType&gt;</pre> <p>Automatically assigned by EFSA</p>                                                                                                                                                                                                                                           |
| Result of analysis for Benzo-a-pyrene in smoked meat (paramCode=RF-00000046-ORG) (ML defined)                                                                                                                          | <pre>&lt;paramType&gt;P005A&lt;/paramType&gt;</pre> <p>Automatically assigned by EFSA</p>                                                                                                                                                                                                                                           |
| Result of analysis for Benzo-a-pyrene in smoked paprika (paramCode=RF-00000046-ORG) (no ML defined)                                                                                                                    | <pre>&lt;paramType&gt;P005A&lt;/paramType&gt;</pre> <p>Automatically assigned by EFSA</p>                                                                                                                                                                                                                                           |
| Result of analysis for the sum of 2 polycyclic aromatic hydrocarbons (PAH) indicators (benzo[a]pyrene and chrysene) in smoked paprika (paramCode=RF-00004427-PAR) (no ML defined); both components analysed and summed | <pre>&lt;paramType&gt;P005A&lt;/paramType&gt;</pre> <p>EFSA pre-assignment: P004A/P005A<br/>Data providers should report paramType P005A</p>                                                                                                                                                                                        |
| Result of analysis for sum of 2 PAH indicators (benzo[a]pyrene and chrysene) in smoked paprika (paramCode=RF-00004427-PAR) (no ML defined); only chrysene analysed                                                     | <pre>&lt;paramType&gt;P004A&lt;/paramType&gt;</pre> <p>EFSA pre-assignment: P004A/P005A<br/>Data providers should report paramType P004A</p>                                                                                                                                                                                        |

## K.02 Coded description of parameter (paramCode) (mandatory) and K.03 Parameter text (paramText) (optional)

The parameter code is used to indicate the substance identified in the laboratory analysis.

It should contain codes linked to the ChemMonRep hierarchy of the PARAM catalogue. This means that if a specific paramCode is in the PARAM catalogue but is not in the ChemMonRep hierarchy, it cannot be transmitted.

| Example                                                                              | XML                                                           |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Results of analysis for aflatoxins sum of B1, B2, G1 and G2                          | <pre>&lt;paramCode&gt;RF-00000435-TOX&lt;/paramCode&gt;</pre> |
| Results of analysis for the aflatoxin B1 alone as component of the sum of aflatoxins | <pre>&lt;paramCode&gt;RF-00000150-TOX&lt;/paramCode&gt;</pre> |
| Results of analysis for cloxacillin                                                  | <pre>&lt;paramCode&gt;RF-00000566-VET&lt;/paramCode&gt;</pre> |
| Results of testing for the sweetener saccharin                                       | <pre>&lt;paramCode&gt;RF-00000013-ADD&lt;/paramCode&gt;</pre> |

| Example                                         | XML                                    |
|-------------------------------------------------|----------------------------------------|
| Results of analysis for terbacil                | <paramCode>RF-0912-001-PPP</paramCode> |
| Results of analysis for the flavouring caffeine | <paramCode>RF-00000038-NTR</paramCode> |

Each code of the ChemMonRep hierarchy is also present in one or more specific “analysis” hierarchies that are used in EFSA sDWH to classify data according to legal limits or legislation, to generate National and Annual Reports and to include results in specific data analysis or exposure assessment.

Consequently, **data providers should make sure that the paramCodes used for reporting results intended for a specific domain are included in the hierarchy of that domain as explained in the table below** (Table 6). This information can be found in the PARAM catalogue, in the EFSA Catalogue Browser, on the right-hand side under ‘Reportability’.

Table 6: List of analysis hierarchies of PARAM catalogue and associated domain

| Domain                     | Analysis hierarchy |
|----------------------------|--------------------|
| <b>VMPPR</b>               | vmpprParam         |
| <b>Pesticides residues</b> | pestParam          |
| <b>Contaminants</b>        | chemAnalysis       |
| <b>Food additives</b>      | addAnalysis        |
| <b>Food flavourings</b>    | flavAnalysis       |

Regarding the paramText, please avoid the use of the ampersand character (&) when preparing the XML files<sup>48</sup>.

For food additives, paramText must be filled in as “Expressed as free acid” or “Expressed as salt” when the food additives are regulated as groups (e.g., sorbic acid and sorbates). This information is mainly required when the reported substance can be expressed as the free acid or converted to the salt. If information is not provided, the analytical value will be considered expressed as free acid.

## L.01 Analytical method identification (anMethRefId) (mandatory)

This element should contain a code not longer than 50 characters to identify an analytical method used within the laboratory that links all results obtained from the same analytical method.

<sup>48</sup> The ampersand character is not allowed as such in XML format as well as in other tagged formats or file types (like HTML or JSON), causing different types of reading errors when these files are parsed by some application. As the paramText field is not a mandatory SSD2 field and it is not used for any validation purpose, you can modify its content without any impact by ‘&’ (also in this case easily doable by a find-replace function of your text editor). You can also consider that all the forbidden characters can be entered in an XML file once “isolated” with a CDATA session (e.g. <![CDATA[&]]> or <![CDATA[Substance Name & name & name]]>).

| Example                                                                                                                 | XML                                                  |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Delvo test for antibacterial substances in bulk milk validated in 2017                                                  | <anMethRefId>BulkMilkAntibioticSCR2017</anMethRefId> |
| Gas chromatography–mass spectrometry analysis for pesticide residues in honey used in the national reference laboratory | <anMethRefId>NRLGC-MSHoney</anMethRefId>             |
| Liquid chromatography–mass spectrometry analysis for aspartame in soft drinks                                           | <anMethRefId>2020-LC-MS-E951</anMethRefId>           |
| Coupling liquid and gas chromatography with subsequent flame ionisation detection to analyse MOH in vegetable oils      | <anMethRefId>2019-LC-GC-FID-Veg-Oil</anMethRefId>    |

The 'Analytical method reference code' (anMethRefCode), the 'Analytical method code' (anMethCode), the 'Analytical method text' (anMethText), and the 'Additional information on the analytical method' (anMethInfo) must be constant (the same) for all results with the same 'Analytical method identification' (anMethRefId). Please see the example below when coding your anMethRefId

| Example |        |            |            |                  | Combination     |
|---------|--------|------------|------------|------------------|-----------------|
| Lab     | Result | AnMethCode | AnMethText | AnmethInfo       | anMethRefId     |
| 1       | 1      | F027A      | LC-MS/MS   | combinationMS_LC | XX12_LC_MS_Lab1 |
|         | 2      | F027A      | LC-MS/MS   | combinationMS_LC | XX12_LC_MS_Lab1 |
|         | 3      | F027A      | LC-MS/MS   | combinationMS_LC | XX12_LC_MS_Lab1 |
| 2       | 4      | F027A      | LC-MS/MS   | combination_lab2 | XX12_LC_MS_Lab2 |
|         | 5      | F027A      | LC-MS/MS   | combination_lab2 | XX12_LC_MS_Lab2 |
|         | 6      | F027A      | LC-MS/MS   | combination_lab2 | XX12_LC_MS_Lab2 |

### L.03 Analytical method type (anMethType), L.04 Analytical method code (anMethCode) (mandatory)

The analytical method type is used to indicate whether the analysis was performed to detect the presence of a substance/class of substances ('screening' AT06A) or to quantify/unequivocally identify the substance ('confirmation' AT08A). 'Screening' code should only be selected when a qualitative method returns a negative result; 'confirmation' should be reported for quantitative/semi-quantitative analytical methods.

The analytical method code describes the type of analysis performed by the laboratory. It is strongly advised to report the specific analytical method, instead of reporting the anMethCode code F001A='Classification not possible'. For chemical contaminants, food additives and food flavourings, the system returns an error message if the anMethCode is reported as code F001A='Classification not possible'; if the generic code F001A is reported, enter also anMethText.

Codes can be selected from the ANLYTYP catalogue and the ANYLMD catalogue.

| Example                                                                                                 | XML                                                              |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Charm II test to screen for tetracyclines in animal tissues                                             | <anMethType>AT06A</anMethType><br><anMethCode>F580A</anMethCode> |
| Multiresidue method using gas chromatography with tandem mass spectrometry                              | <anMethType>AT08A</anMethType><br><anMethCode>F049A</anMethCode> |
| Liquid chromatography–tandem mass spectrometry used to quantify unauthorised residues in animal samples | <anMethType>AT08A</anMethType><br><anMethCode>F027A</anMethCode> |

## M.01 Result identification code (resId) (mandatory)

This element must be provided for every record in the dataset and **must be unique** for an analytical result reported for a sample across all data collections from a data provider. This identifier is also used for communication between EFSA and the data provider during the transmission and validation phases. When validating data, it is essential to be able to detect, which results are actually counted and to be able to identify which results may need to be amended. resId will be displayed in validation reports for non-compliant results (in aggregated tables by drill-down).

It is recommended that certain prefixes or suffixes are included to ensure the resId is unique within the country.

| Example                                                                         | XML                                    |
|---------------------------------------------------------------------------------|----------------------------------------|
| Result reported by an Estonian veterinary laboratory in 2017                    | <resId>EEVetLab2017_0009845634</resId> |
| Result reported by the Italian pesticides national reference laboratory in 2017 | <resId>ITNRL2017_ADE0000456792</resId> |
| Result reported by a Danish food testing laboratory in 2017                     | <resId>DKDTU2017_K0000034597X</resId>  |

## M.02 Accreditation procedure for the analytical method (accredProc) (mandatory)

This code describes the validation/accreditation status of the method linked to anMethRefId.

Codes can be selected from the MDACC catalogue.

| Example                                                                                                                                                                                                                 | XML                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Method accredited according to ISO/IEC17025 and validated according to Commission Implementing Regulation (EU) 2021/808 <sup>49</sup> on the performance of analytical methods for veterinary medicine residue controls | <accredProc>V007A</accredProc> |

<sup>49</sup> Commission Implementing Regulation (EU) 2021/808 of 22 March 2021 on the performance of analytical methods for residues of pharmacologically active substances used in food-producing animals and on the interpretation of results as well as on the methods to be used for sampling and repealing Decisions 2002/657/EC and 98/179/EC. OJ L 180, 21.5.2021, p. 84–109.

## M.03 Result unit (resUnit) (mandatory)

This element indicates the units of measurement for the numerical values resLOD, resLOQ, resVal, resLLWR, resULWR, resValUncert, resValUncertSD, CCalpha, CCbeta, evalLowLimit, or evalHighLimit. Codes can be selected from the chemUnit hierarchy of the UNIT catalogue. This hierarchy reflects the standard International System of Units (SI) for concentrations: grams, milligrams, micrograms, picograms or nanograms per kilogram, gram, or litre (for liquids). Restriction to these units allows the results to be converted by EFSA to a single unit type for consistent presentation in tables of reports or comparison in data analysis. Liquid samples will be converted by EFSA on the assumption that one litre corresponds to one kilogram, irrespective of the density. The table below lists the resUnit codes typically reported in the context of different residue domains.

| Example                                                                                            | XML                      |
|----------------------------------------------------------------------------------------------------|--------------------------|
| Reporting the results of pesticide monitoring in milligram/kilogram                                | <resUnit>G061A</resUnit> |
| Reporting the results of testing for beta-agonists (VMPP) in microgram/kilogram                    | <resUnit>G050A</resUnit> |
| Reporting the results of testing for mycotoxins in microgram/kilogram                              | <resUnit>G050A</resUnit> |
| Reporting the results of testing for metals in milligram/kilogram                                  | <resUnit>G061A</resUnit> |
| Reporting the results of testing for dioxins and PCBs in picogram/gram                             | <resUnit>G080A</resUnit> |
| Reporting the results of testing for polycyclic aromatic hydrocarbons (PAHs) in microgram/kilogram | <resUnit>G050A</resUnit> |

## M.04 Result limit of detection (resLOD) (optional)

The limit of detection (LOD) is the lowest concentration that can be determined to be statistically different from a 'blank' analytical result. Results with the LOD reported may be used by EFSA to assess new scenarios when estimating the consumer's chronic exposure. resLOD must be reported if resType=LOD.

| Example                                                                    | XML                    |
|----------------------------------------------------------------------------|------------------------|
| Result reported as LOD (i.e. resType=LOD) with a limit of detection =0.001 | <resLOD>0.001</resLOD> |

Specifically for food additives and food flavourings, if resType=BIN is reported and resQualValue is NEG, then it is highly recommended to report the LOD. Otherwise, the data cannot be used for dietary exposure estimations.

## M.05 Result limit of quantification (resLOQ) (optional)

The resLOQ, the numerical value of the limit of quantification (LOQ), is the lowest validated residue concentration of the analyte, which can be quantified and reported by routine monitoring with validated methods.

This data element is always required unless:

- unvalidated methods are used (infrequent cases);
- the summed LOQ of the 'Multicomponent' pesticide residue definition/sum of contaminants cannot be calculated (where resInfo.notSummed='Y', see also Section M.20 below);
- the values for resLOD, CCbeta or CCalpha are reported for VMPPR;
- resType = BIN, where resLOD should be reported.

An LOQ is required for the following reasons<sup>50,51,52</sup>:

- 1 Uncertainty: the LOQ is used by EFSA as a substitution method for the calculation of the middle and upper bound for the left-censored results on residue/contaminant concentrations.
- 2 Sensitivity and method evaluation: the LOQ is required to ensure the quality and comparability of analytical results, to ensure that acceptable accuracy is achieved and to ensure that false positives or false negatives are avoided.
- 3 To apply quality criteria: if a cut-off value is applied based on the LOQ, this will affect both quantified and left-censored data.

In cases where neither the LOD nor LOQ is provided, results cannot be used in exposure assessments and scientific reports.

For veterinary monitoring, one of resLOD, resLOQ, CCalpha and CCbeta must be reported for each result.

| Example                      | XML                    |
|------------------------------|------------------------|
| LOQ=0.005 milligram/kilogram | <resLOQ>0.005</resLOQ> |

## M.08 CCalpha (CCalpha) and M.09 CCbeta (CCbeta) (optional)

Decision limit (CCalpha) means the limit at and above which it can be concluded with an error probability of  $\alpha$  that a sample is non-compliant. Since CCalpha accounts for

<sup>50</sup> 'Use of cut-off values on the limits of quantification reported in datasets used to estimate dietary exposure to chemical contaminants.' (EFSA, 2018c).

<sup>51</sup> The guidance in this document is intended for laboratories involved in the official control of pesticide residues in food and feed across the European Union. (European Commission, 2018)

<sup>52</sup> 'In order to decide if the analytical methods used for pesticide monitoring in accordance with Article 28 of Regulation (EC) No 396/2005 is sufficiently sensitive to identify samples containing residues that exceed the legal limits, a sensitivity check needs to be performed by the laboratories.' (European Commission, 2015, p. 3).



measurement uncertainty, this value must be reported for confirmatory results where the result is evaluated for compliance<sup>53,54</sup>□.

Detection capability (CCbeta) means the smallest content of the substance that may be detected, identified and/or quantified in a sample with an error probability of  $\beta$ . For substances for which no permitted limit has been established, the detection capability is the lowest concentration at which a method is able to detect truly contaminated samples with a statistical certainty of  $1 - \beta$ . For substances with an established permitted limit, this means that the detection capability is the concentration at which the method can detect permitted limit concentrations with a statistical certainty of  $1 - \beta$ . The detection capability must be reported for screening methods for veterinary residues.

For multicomponent residue definitions/marker compounds for MRL, where one or more of the components is quantified, it is sufficient to report the CCalpha of the substance used for the evaluation of compliance.

In cases of multicomponent residue definitions/marker compounds for MRL where no component can be quantified, the CCalpha of the usual main component should be reported as 'representative' for confirmatory tests.

For screening tests, it is sufficient to report the CCbeta for the individual components.

For veterinary medicinal residues, when the analytical method is validated according to Commission Implementing Regulation (EU) 2021/808 the reporting of either CCalpha or CCbeta is required. For A3b and B1b substances when the validation is done according to the Pesticides domain CCalpha and CCbeta are not required.

The use of CCalpha and CCbeta is also permitted for certain mycotoxins in food of animal origin (for instance AFM1 in milk or OTA A in pig meats).

| Example                                  | XML                   |
|------------------------------------------|-----------------------|
| CCalpha reported for a confirmatory test | <CCalpha>20</CCalpha> |
| CCbeta reported for a screening test     | <CCbeta>350</CCbeta>  |

## M.10 Result value (resVal) (optional)

The resVal must be used to report the measured concentration of the substance in the product expressed in the unit reported in resUnit. resVal is mandatory if resType=VAL and must be greater than 0. If a sample was analysed using (qualitative) screening methods, the data element resVal must be left blank. See also the paragraph on M.16 Type of results (resType).

For processed products in general, the results must be reported for the sample analysed, i.e. the processed product, without any recalculation of the result to the unprocessed

<sup>53</sup> Commission Implementing Regulation (EU) 2021/808 of 22 March 2021 on the performance of analytical methods for residues of pharmacologically active substances used in food-producing animals and on the interpretation of results as well as on the methods to be used for sampling and repealing Decisions 2002/657/EC and 98/179/EC. OJ L 180, 21.5.2021, p. 84–109

<sup>54</sup> Commission Decision 2002/657/EC.



product. If the pesticide residues domain and results expressed on fruits have stones (e.g. peaches, mangoes, avocados), the analytical result should be recalculated back to the whole fruit when checking for MRL compliance. However, when exposure assessments are undergone the pulp (without the stone) will be mapped with the consumption data. Therefore, when reporting either options the use of facet F20 is recommended to indicate if the analytical result is expressed with stone (A07QJ) or without stone (A07QK).

The MRL will only be validated considering the processed pesticide residue and VMPP samples, as MRL are set for fresh/unprocessed samples in both legal frameworks, i.e. the MRL will not be used to validate the plausibility of resVal, resType and evalCode reported if the sample is 'processed'. Further, MRL will not be validated for 'Wild' game samples analysed for VMPP substances. See also Section 7 describing the Legal Limits database structure and its use.

The distribution and descriptive statistics for quantifiable results for specific residues/markers in different matrices may be presented in the validation and web reports. These values would also be used in exposure assessments.

| Example                                                                    | XML                  |
|----------------------------------------------------------------------------|----------------------|
| Measured concentration of a residue in a sample is 5.6 milligrams/kilogram | <resVal>5.6</resVal> |

## M.11 Result value recovery rate (resValRec) and M.12 Result value corrected for recovery (resValRecCorr) (optional)

The results from analytical methods which do not include an extraction step or analytical methods which use certified reference material at a certified concentration must be reported uncorrected for extraction recovery during the sample preparation.

Calculated recovery corrections are typically used to assess the performance of the method. These should not be applied to adjust the numerical results of pesticide residue analysis.

However, for mycotoxins as described in Commission Regulation (EU) 2023/2782, erucic acid as described in Commission Regulation (EU) 2023/2783, nitrates as described in Commission Regulation (EC) No 1882/2006 and lead, cadmium, mercury, inorganic tin, 3-MCPD and benzo(a)pyrene as described in Commission Regulation No 333/2007<sup>55</sup> the results must be adjusted for recovery and the recovery rate reported (unless specific conditions occur, e.g. for result significantly lower or higher than the maximum level).

The result value recovery rate (resValRec) associated with the concentration measurement is expressed as a percentage (%), i.e. 100 should be reported for a 100% rate.

Unless otherwise specified, the result expressed is considered by EFSA as not corrected for recovery. It is strongly recommended to report whether the analytical result has been or

<sup>55</sup> Commission Regulation (EC) No 333/2007 of 28 March 2007 laying down the methods of sampling and analysis for the official control of the levels of lead, cadmium, mercury, inorganic tin, 3-MCPD and benzo(a)pyrene in foodstuffs. OJ L 88, 29.3.2007, p. 29–38.

not corrected for recovery within the data element `resValRecCorr` ('Yes' ('Y') or 'No' ('N')). In case of an empty field in the data element '`resValRecCorr`', the system will return a Warning message with respect to the chemical contaminants domain. If the data provider reports a result corrected for recovery, the data element `resValRecCorr` must be set to 'Yes' ('Y'). If reported, the results recovery must be greater than '0'.

| Example                                                                    | XML                                                                                           |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| The concentration of aflatoxin M1 was corrected for a recovery rate of 63% | <pre>&lt;resValRec&gt;63&lt;/resValRec&gt; &lt;resValRecCorr&gt;Y&lt;/resValRecCorr&gt;</pre> |

## M.13 Expression of result percentage (`exprResPerc`) and M.14 Expression of result type (`exprResType`) (mandatory for some domains)

These elements are used to indicate when the concentration is expressed as a percentage of a component of the sample, for example, on a dry weight basis. In the cases where `exprResType` should be reported (mandatory), `exprResPerc` should be also reported (dependent mandatory).

For pesticide residues, the MRLs for eggs and milk apply to the whole product. However, the pesticides that are fat soluble tend to concentrate on the fat part of the product. Therefore, when reporting pesticides in eggs or milk it is necessary to report if the result is expressed as 'whole weight', code B001A, (the result is on the whole egg - after removal of the shell- or whole milk - based on a fat content of 4% by weight-) or if the result is expressed on the 'fat weight' code B003A, (the laboratory has separated the fat phase and the result is provided on the fat part. This information will allow EFSA to correctly check the consistency of result evaluation in view of the exposure assessment.

For food additives and food flavourings, `exprResType` should be reported, as it became mandatory in 2025 and 2026 respectively. If not reported, the system returns an error message.

In Regulation (EU) 2023/915, where the maximum levels for certain contaminants in food are established, there are specific remarks on the matrix to which the maximum levels apply. For example, the established maximum levels for aflatoxin B1, ochratoxin A, deoxynivalenol, zearalenon, fumonisins apply to the dry matter of the baby food and processed cereal-based foods for infants and young children.

For product of animal origin concerning meat, it is to be considered that the MRLs apply to the muscle or to the fat. If fat soluble pesticide is analysed in meat sample and `exprResType` is reported as 'fat weight' (code B003A), the results will be checked against the MRL for fat. While if `exprResType` is expressed as 'whole weight' (B001A) the fat percentage under `exprResPerc` should be reported. If it is not reported, EFSA will consider default values whenever possible. If the pesticide analysed is not fat soluble, it is highly recommended to express the result as whole weight so is checked against the MRLs for muscle.

Directive (EC) 2002/32 describes the maximum levels of undesirable substances in mg/kg in the feedstuffs with a moisture content of 12% (88% dry matter); in order to check compliance and to derive concentration levels usable in exposure assessment it is recommended to report feed contaminants data expressed on 88% dry matter (code B004A). The `exprResType` has to be reported for feed; if the analysis has been performed on whole weight, this should be indicated in the `exprResType` with the code 'B001A' and the moisture percentage of the sample in the `exprResPerc.moistPerc` field must be reported. The system will return an Error message if this BR is not followed.

In cases where percentage of moisture is not available, but percentage of dry matter is instead, EFSA recommends calculation of the percentage of moisture by applying the calculation '100 – percentage of dry matter' = percentage of moisture.

Codes can be selected from the EXPRES catalogue.

| Example                                                                                                                           | XML                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Results for a fat-soluble pesticide measured in a butter sample expressed on a fat weight basis                                   | <pre>&lt;exprResPerc&gt;fatPerc=80&lt;/exprResPerc&gt; &lt;exprResType&gt;B003A&lt;/exprResType&gt; OR &lt;exprResPerc.fatPerc&gt;80&lt;/exprResPerc.fatPerc&gt; &lt;exprResType&gt;B003A&lt;/exprResType&gt;</pre>       |
| Results (moisture percentage) for heavy metals as occurred in seaweed samples expressed on a whole weight basis                   | <pre>&lt;exprResPerc&gt;moistPerc=40&lt;/exprResPerc&gt; &lt;exprResType&gt;B001A&lt;/exprResType&gt; OR &lt;exprResPerc.moistPerc&gt;40&lt;/exprResPerc.moistPerc&gt; &lt;exprResType&gt;B001A&lt;/exprResType&gt;</pre> |
| Results (moisture and fat percentage) for heavy metals expressed as occurred in seaweed samples expressed on a whole weight basis | <pre>&lt;exprResPerc&gt;moistPerc=40\$fatPerc=5&lt;/exprResPerc&gt; &lt;exprResType&gt;B001A&lt;/exprResType&gt;</pre>                                                                                                    |

## M.15 Result qualitative value (`resQualValue`) (optional)

When qualitative screening results (for example, biological tests) are reported with `resType=BIN` then `resQualValue` must be reported and the accepted value is NEG. For confirmatory results or quantifiable results `resQualValue` must not be reported.

As mentioned also in section M04, for food additives and food flavourings, if `resType=BIN` and `resQualValue` is NEG, then it is highly recommended to report the LOD. Otherwise, the data cannot be used for dietary exposure estimations.

| Example                                                                                             | XML                                                     |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Negative result for the presence of amoxicillin in a milk sample and using a Delvo test (screening) | <pre>&lt;resQualValue&gt;NEG&lt;/resQualValue&gt;</pre> |

## M.16 Type of results (`resType`) (mandatory)

The `resType` indicates the type of analytical result obtained for a substance in a product.

Codes can be selected from the VALTYP catalogue.

| Example                                                                   | XML                    |
|---------------------------------------------------------------------------|------------------------|
| The result can be quantified at a validated level and resVal is reported  | <resType>VAL</resType> |
| The residue can be quantified but is below the reported value for CCalpha | <resType>CCA</resType> |
| The residue is below the LOQ                                              | <resType>LOQ</resType> |
| The residue is below the LOD and resLOD is reported                       | <resType>LOD</resType> |
| The result of a qualitative screening test                                | <resType>BIN</resType> |
| The residue cannot be detected and CCbeta is reported                     | <resType>CCB</resType> |

In some cases, a chromatographic peak below the limit of quantification (LOQ) may be seen. In order to refine exposure assessments in the most consistent approach among all chemical domains, when resType = LOQ, LOD or BIN, resVal value must be empty. The system will return an Error message.

When resType= 'VAL', then resLOQ must be numerically lower than resVal. In the case of multicomponent paramCodes to overcome this rule, resInfo.notSummed='Y' can be reported (see Section M.20).

For test results in which the residue can be quantified at a validated level then the resType must be reported as VAL and the resVal must be greater than 0; in that case it is expected that the resType='confirmation' (see also paragraph on anMethType).

For VMPPR monitoring, these additional cases apply. If the residue cannot be quantified and CCalpha is reported, then the resType='CCA' must be reported. If, however, the resLOQ is reported and it is below CCalpha, then the resType='LOQ' should be reported if the result is below the resLOQ. For qualitative screening method results, the expected values for resType are BIN (in association with resQualValue=NEG and resVal=blank) with detection capability reported in CCbeta, or alternatively the LOD reported in resLOD. For positive screening results, the quantitative confirmatory test results should be reported and not binary positive screening results. Quantitative screening method results can be reported in a similar way to confirmatory results (see above) with a value for CCbeta reported in place of a value for CCalpha.

In exceptional cases, for the veterinary medicinal products domain also the code AWR='Value above the upper limit of the working range' can be selected, but only when one of the six above codes would not be fit for purpose and when:

- 1 the identity of the residue has been analytically confirmed;
- 2 the laboratory is certain of the 'positive' occurrence of the residue in the tested sample;
- 3 even though quantifiable, the residue was not quantified with a given precision/accuracy as the residue concentration was above the upper level of the validated/calibrated/working analytical method range for residue concentration.

This specific code is expected to be selected only to report 'non-standard' samples, whose results are considered relevant at the national level for risk management reasons; for

example, an 'environmental' sample consisting of a 'contaminated' syringe clearly containing a prohibited substance whose presence and identity could be confirmed, but its 'concentration' was well above the calibrated concentration curve. Thus, results coded with AWR can have sufficient information for risk management but will not be useful for risk assessment. If the resType code=AWR is selected, then a numerical value for the data elements CCalpha (or resLOQ) and resVal must be reported. If the resType code AWR is selected, EFSA would consider it as 'corresponding' to resType=VAL when e.g. counting the number of 'positive' detections.

## M.17 Result value uncertainty (resValUncert) (optional)

The resValUncert indicates the expanded uncertainty value (usually 95% confidence interval) associated with the measurement. This uncertainty is expressed in the same unit as the one reported in the field resUnit.

For pesticide residue monitoring, it is recommended to populate this field especially if single residue methods are used and/or if for a multiresidue method the expanded measurement uncertainty (MU) is different than 50%. The resValUncert is requested only when resType = VAL<sup>56</sup>

According to Regulation (EU) 2023/2782 (on mycotoxins), Regulation (EC) No 705/2015 (on erucic acid), Regulation (EC) No 333/2007, and Regulation (EC) No 644/2017, the condition for acceptance of a lot should also take into consideration measurement uncertainty.

## M.20 resInfo.notSummed (optional)

In line with the European Commission (2015) document on pesticides, the following provisions for reporting resLOQ according to the residue definition type are described:

- For multicomponent pesticide residue definitions, EFSA requests that the individual LOQ for each component quantified is reported separately from that residue definition and a summed LOQ, which is calculated by the reporting country.
- If the reporting country does not report the summed LOQ, then resLOQ can be null for a given paramCode set with paramType equal to P005A or P004A and resInfo.notSummed equal to 'Y'. Thus, the paramCode of the multicomponent and the resInfo.notSummed set to Y, must be reported. In this case, upon submission of the individual component LOQ (for paramType P002A substances) will be calculated by EFSA based on the LOQs reported for the individual components associated with paramType P002A. For this reason, the individual component resLOQ associated to paramType P002A is mandatory. If no individual resLOQ is reported for at least one component for a given paramCode with paramType P005A/P004A, no 'sum of LOQ' calculation will be done by EFSA.

<sup>56</sup> For result below the LOQ value, no MU is to be reported as it is '*a parameter characterising the dispersion of quantify values being attributed to a measurand*' (International vocabulary of metrology, ISO-2007: ISO/IEC guide 99, VIM 3).

For the calculated sum of contaminants, there are currently no standard guidelines on how to sum LOQs of individual substances. The reporting country can decide to report the LOQ by summing up the LOQs of all the individual substances to which the sum refers or to indicate `resInfo.notSummed='Y'` and `resLOQ=null`, if the results for the single substances and correspondent LOQs are reported. This approach is to avoid storing a false value of '99999' in `resLOQ` when the summed LOQ was not possible. Business rules apply to ensure that `resLOQ` can only be NULL where `resInfo.notSummed='Y'`. No value such as '99999' or similar (e.g. '999', '9999', '999999') shall be reported.

## N.01 Limit for the result evaluation (`evalLowLimit`), N.03 Type of limit for the result evaluation (`evalLimitType`) (optional)

These two elements are used to report the numerical value of the legal maximum residue limit or maximum limit enforced when assessing a laboratory result and the type of legal limit.

In case of a non-regulated processed product the maximum limit to be indicated is the one of the regulated unprocessed product (Article 3 of Regulation (EU) 2023/915). The transformation factor (see Art. 3 of Regulation (EU) 2023/915) could be indicated in `evalInfo.com` or in a `resInfo.com`.

EFSA developed a harmonised database (Legal Limit Database) containing MRLs established for pesticides and veterinary medicine residues in Regulation (EC) No 396/2005 and Regulation (EC) No 37/2010. This database is used to validate the plausibility of the reported result evaluation (`evalCode`) with respect to the numerical comparison of result value (`resVal`) and the MRL when the reported sample is 'unprocessed'. Thus, the reporting of these elements would only be required in cases in which limits other than those defined in the above-mentioned regulations or other than the official EU limits are in use, e.g. national limits.

The Legal Limit Database has also been developed for the food additives and flavourings domains, containing the maximum permitted levels (MPLs) established in Regulations (EC) No 1333/2008 and (EC) No 1334/2008.

Limit type codes can be selected from the LMTTYP catalogue.

For pesticide monitoring the typically reported value for the type of limit for the result evaluation (`evalLimitType`) data element is W002A (MRL). If the national or local limit (W990A) is reported, the result will be disregarded from the report analysis.

For VMPPR monitoring the following limit types can be reported in `evalLimitType`: W002A=MRL, W005A=Minimum required performance limit, W006A=Reference point of action, W012A=Presence, W001A=Maximum limit (this applies only to coccidiostats, histomonostats and chemical elements) and W007A=Action level.



| Example                         | XML                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Reporting of an EU MRL in place | <pre>&lt;evalLowLimit&gt;0.003&lt;/evalLowLimit&gt;</pre> <pre>&lt;evalLimitType&gt;W002A&lt;/evalLimitType&gt;</pre> |

## N.04 Evaluation of the result (evalCode) (mandatory)

The evaluation (evalCode) must be applied at the level of each residue or marker within the analytical method. It provides the judgement of the reporting country on whether the result reported was considered to exceed the legal limit applicable to the sample or non-compliant due to, for example, the presence of forbidden/prohibited substances.

It is sufficient to report 'Above the level of concern' (evalCode=J003A) for results that were found to clearly exceed the limit (taking into account the measurement uncertainty) or level of concern, or 'Below or equal the level of concern' (evalCode=J002A) for results that are below the limit or level of concern.

The result evaluation for each of the single components of a multicomponent residue definition/marker compound or of a contaminants group (i.e. associated with the paramType=P002A) should be reported with the code J029A= 'results not evaluated'.

For VMPPR monitoring 'detected' (evalCode=J041A) in conjunction with evalLimitType='Presence' (code W012A) can be reported and this will be counted as a non-compliant result, although it is not recommended since the code J003A could be used.

Evaluation codes can be selected from the RESEVAL catalogue. The table below explains the use of codes that should typically be reported.

The count of non-compliant samples and results in EU reports will be based on the values reported in this data element in conjunction with evalInfo.resAsses as explained in aragraph N.06.3

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | XML                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <p>The residue in the sample is considered to be above the level of concern.</p> <p>For residues this code is selected if the numerical value of the quantified residues is clearly above the legal limit taking into account the analytical measurement uncertainty; thus, the result must be evaluated against the legal limit set under the relevant MRL/ML legislation.</p> <p>The code J003A must not be used to indicate whether the measured residue in samples produced in the EU is not approved at EU level according to Regulation</p> | <pre>&lt;evalCode&gt;J003A&lt;/evalCode&gt;</pre> |



| Example                                                                                                                                                                                                            | XML                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1107/2009 <sup>57</sup> (on the approved uses of plant protection products), nor to assess the presence of a pesticide residue (within a legal limit) in organic products that is not permitted in organic farming |                            |
| The residue in the sample is considered to be below the level of concern                                                                                                                                           | <evalCode>J002A</evalCode> |
| The residue was not evaluated; for example, when the substance is only part of the full residue definition/group (paramType=Part of a sum (P002A))                                                                 | <evalCode>J029A</evalCode> |
| The residue was not evaluated as no legal limit applies to the substance or residue measured in the sample                                                                                                         | <evalCode>J029A</evalCode> |
| The result is above the limit, but the residue in the sample is considered to be compliant, taking into account the analytical measurement uncertainty                                                             | <evalCode>J031A</evalCode> |
| The result indicates the occurrence of an illegal/prohibited VMPP included in Group A (no MRL set for these substances) or the detection of an unauthorised food additive                                          | <evalCode>J041A</evalCode> |

## N.05 Action taken (actTakenCode), N.06.1 Conclusion of follow-up investigation (evalInfo.conclusion) and N.06.2 Comment (evalInfo.com) (optional)

Action taken should be reported when a non-conformity is identified during the control activities or if a measured substance is found above the level of concern. Multiple actions can be reported.

This is important for the pesticide samples if found non-compliant, to understand whether the product has been placed on the EU market and consumed or withdrawn without reaching the consumer or destroyed.

It is mandatory for VMPP and pesticides to report the action taken in case of non-compliant results.

<sup>57</sup> Regulation (EC) No 1107/2009 of the European Parliament and of the Council of 21 October 2009 concerning the placing of plant protection products on the market and repealing Council Directives 79/117/EEC and 91/414/EEC. OJ L 309, 24.11.2009, p. 1–50.

Conclusion is used to classify the findings of follow-up investigations. The comment element allows additional details on non-compliant results or non-conformities. This element may be included in EFSA Annual Reports listing non-compliant results.

EvalInfo.conclusion and evalInfo.com can also be used to indicate when results are above a residue legal limit, but the final evaluation is compliant, e.g. cases of natural occurrence or environmental contamination.

EvalInfo.conclusion should be used for reporting a long shelf-life product (code 'C08A='Long shelf-life product' from CONCLUS catalogue) for which the MRL in place at the time of placing it on the market was different than the MRL applicable on the current data collection for the given pesticide/crop combination. The BRs checking against the MRLs will not be applicable.

For the additives and flavourings domains, evalInfo.conclusion is highly recommended to be used to indicate whether the food additive or flavouring was specified on the label of the analysed sample, or the positive analytical result was due to natural occurrence.

The codes to be used are those from the CONCLUS catalogue selecting the fa\_ff hierarchy. The two main options to be used are C19A ('Yes, present on label/added') and C20A ('No, not present on label/not added'). A third option, 'Natural occurrence' (C05A), can also be used in those cases in which the substance is in the sample as part of the matrix itself. As an example, C05A could be reported in combination with code C20A (C20A\$C05A) if the substance is not listed on the label/added but it occurs naturally. Some examples are shown below:

| Example (additives/flavourings)                                                                                                         | XML                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ascorbic acid added as FA and naturally occurring                                                                                       | <evalInfo.conclusion>C19A\$C05A</evalInfo.conclusion>                                                                   |
| Ascorbic acid added as ingredient and naturally occurring                                                                               | <evalInfo.conclusion>C19A\$C05A</evalInfo.conclusion><br><evalInfo.com>Ascorbic acid added as ingredient</evalInfo.com> |
| XXX added as carry-over, and not as FA                                                                                                  | <evalInfo.conclusion>C19A</evalInfo.conclusion><br><evalInfo.com>Carry-over</evalInfo.com>                              |
| Caffeine added as flavouring and declared on the label.<br>"Flavourings" is declared on the label, without specifying "ethyl vanillate" | <evalInfo.conclusion>C19A</evalInfo.conclusion><br><evalInfo.conclusion>C20A</evalInfo.conclusion>                      |

'Action taken codes' can be selected from the ACTION catalogue.

'Conclusion' codes can be selected from the CONCLUS catalogue.

'evalInfo.com' can be reported with a free text, as appropriate. In the table below some examples are provided.

| Example                                                                                                                          | XML                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Follow-up investigation indicates illegal treatment and additional sampling is planned                                           | <pre>&lt;actTakenCode&gt;I&lt;/actTakenCode&gt; &lt;evalInfo&gt;conclusion=C04A\$com=increased testing for this product in aquaculture is planned&lt;/evalInfo&gt; OR &lt;actTakenCode&gt;I&lt;/actTakenCode&gt; &lt;evalInfo.conclusion&gt;C04A&lt;/evalInfo.conclusion&gt; &lt;evalInfo.com&gt;increased testing for this product in aquaculture is planned &lt;/evalInfo.com&gt;</pre>                                                                                                                                |
| The result was non-compliant and there has been a rapid alert notification in RASFF                                              | <pre>&lt;evalCode&gt;J003A&lt;/evalCode&gt; &lt;actTakenCode&gt;R&lt;/actTakenCode&gt; &lt;evalInfo&gt;com=Reference number 2018.1567&lt;/evalInfo&gt; OR &lt;actTakenCode&gt;R&lt;/actTakenCode&gt; &lt;evalInfo.com&gt;Reference number 2018.1567 &lt;/evalInfo.com&gt;</pre>                                                                                                                                                                                                                                          |
| Result considered to be compliant since investigations indicate exceedance may be due to the natural occurrence of the substance | <pre>&lt;evalCode&gt;J002A&lt;/evalCode&gt; &lt;actTakenCode&gt;I&lt;/actTakenCode&gt; &lt;evalInfo&gt;conclusion=C05A\$com=investigations indicate exceedance may be due to natural occurrence of the substance &lt;/evalInfo&gt; OR &lt;evalCode&gt;J002A&lt;/evalCode&gt; &lt;actTakenCode&gt;I&lt;/actTakenCode&gt; &lt;evalInfo.conclusion&gt;C05A&lt;/evalInfo.conclusion&gt; &lt;evalInfo.com&gt;investigations indicate exceedance may be due to natural occurrence of the substance &lt;/evalInfo.com&gt;</pre> |
| Result considered compliant, for a food additive present on the label and natural occurring in the food                          | <pre>&lt;evalCode&gt;J002A&lt;/evalCode&gt; &lt;evalInfo.conclusion &gt;C19A\$C05A&lt;/evalInfo.conclusion&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                     |
| Result considered non-compliant, for a food additive not present on the label while found in the food                            | <pre>&lt;evalCode&gt;J003A&lt;/evalCode&gt; &lt;evalInfo.conclusion&gt;C20A&lt;/evalInfo.conclusion&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                            |

### N.06.3 Result assessment (evalInfo.resAsses) (optional)

This element can be used to report the single result overall evaluation ('Compliant' or 'Non-compliant'), e.g. when results are numerically above the legal limit (single result numerical evaluation) but deemed compliant by national expert judgement (e.g. by a national risk assessor) due, for example, to natural occurrence. This element is optional and can be different for each result in a sample.

The values reported in evalCode and evalInfo.resAsses will be used by EFSA to determine whether the result in the relevant annual/national reports is counted as compliant or non-compliant.

A sample will consequently be classified as non-compliant for a specific Annual Report if one or more results for substances covered by that regulation are counted as non-compliant.

For example, if evalCode=J003A (Above the level of concern) and evalInfo.resAsses=null then the result is counted as non-compliant; if evalCode=J003A (Above the level of concern) and evalInfo.resAsses=J037A (Compliant) then the result is counted as compliant. If evalCode=J002A (Below or equal the level of concern) and evalInfo.resAsses=J038A (non-compliant) then the result is counted as non-compliant.

However, in the pesticide residues remit, the assessment of compliance or non-compliance is based only on the numerical comparison between the resVal and the MRL. Thus, it will be done based on the evalCode only. Further information can be provided in the evalInfo.resAsses if competent authorities considered it (e.g. if fish or feed is reported, or if organic samples are reported) but this information will not be used to count the number of non-compliance in the EU Annual Report.

Only the codes 'J037A' (Compliant) or 'J038A' (Non-compliant) of the RESEVAL catalogue are valid when reporting evalInfo.resAsses.

The below table shows the example of the element evalInfo.resAsses to be used in combination with evalInfo.com. It is important to note that where evalInfo.resAsses is reported and is different from evalCode, the reason for the difference must be reported in evalInfo.Conclusion. For example, the evalInfo.resAsses could be reported as in the following case: <evalInfo>resAsses=J037A\$conclusion=C05A\$com=Result is within a physiological range for this sample to be a naturally occurring hormone</evalInfo>.

| sampMatCode | paramCode       | evalCode               | evalInfo.conclusion | evalInfo.com                                                               | evalInfo.resAsses     |
|-------------|-----------------|------------------------|---------------------|----------------------------------------------------------------------------|-----------------------|
| Milk, raw   | Chloramphenicol | Above level of concern | Accidental          | Contamination from sampler cannot be excluded                              | Compliant (J037A)     |
| Pig         | Zearalenone     | Below level of concern |                     | Presence of zeranol and absence of mycotoxins indicates illegal occurrence | Non-compliant (J038A) |

The SSD data elements sampEventAsses, sampTkAsses, and sampAnAsses will not be used for EFSA data analysis purposes in counting results/samples non-compliances.

#### N.06.4 Restriction or exception (evalInfo.restrictionException) (optional)

This data element should be used to provide the restrictions/exceptions as described in the Regulation (EC) No 1333/2008 for each authorised food additives across the different additive food categories. This data element can also be used for food flavourings in line with Regulation (EC) No 1334/2008 where the restrictions/exceptions for their use are described.

The codes need to be selected from the ADDFOOD catalogue. The most detailed code that defines the restriction/exception should be chosen and reported in this data element. Few examples are described below:

| Example                                              | XML                                                                      |
|------------------------------------------------------|--------------------------------------------------------------------------|
| Sample of mozzarella cheese analysed for lactic acid | <evalInfo.restrictionException>ADD00197A</evalInfo.restrictionException> |
| Sample of milk chocolate analysed for citric acid    | <evalInfo.restrictionException>ADD00386A</evalInfo.restrictionException> |
| Sample of processed fish analysed for ascorbic acid  | <evalInfo.restrictionException>ADD00622A</evalInfo.restrictionException> |
| No restrictions/exceptions to be reported            | <evalInfo.restrictionException>ADD00881A</evalInfo.restrictionException> |

Despite being optional, the information provided by this data element is relevant since the same substance can sometimes be authorised for the same legislative food category/subcategory at different MPLs depending on restrictions/exceptions. It will also help to identify the correct MPLs for each combination of substance, legislative category and restriction/exception in the EFSA legal limit database for food additives and flavourings. A warning will be raised when the reported combination (substance – category – restriction/exception) is not authorised in the Regulation (EC) 1333/2008 and Regulation (EC) 1334/2008

Table 7: Full list of SSD2 elements, mapping to SSD1 and mandatory SSD2 elements

| Element code | Element name     | Element label                             | Type                | Controlled terminology | Mandatory/optional element for chemical monitoring reporting | SSD1 element name | SSD1 element code |
|--------------|------------------|-------------------------------------------|---------------------|------------------------|--------------------------------------------------------------|-------------------|-------------------|
| <b>A.01</b>  | localOrgId       | Local organisation identification code    | xs:string (100)     |                        | Optional                                                     | localOrgId        | O.1               |
| <b>A.02</b>  | localOrgCountry  | Local organisation country                | xs:string (2)       | COUNTRY                | Optional                                                     | localOrgCountry   | O.2               |
| <b>A.03</b>  | localOrgInfo     | Local organisation additional information | <i>CompoundType</i> |                        | Optional                                                     |                   |                   |
| <b>B.01</b>  | progId           | Sampling programme identification code    | xs:string (100)     |                        | Mandatory                                                    | progCode          | S.31              |
| <b>B.02</b>  | progLegalRef     | Programme legal reference                 | xs:string (100)     | LEGREF                 | Mandatory                                                    | progLegalRef      | S.32              |
| <b>B.03</b>  | sampStrategy     | Sampling strategy                         | xs:string (5)       | SAMPSTR                | Mandatory                                                    | progSampStrategy  | S.33              |
| <b>B.04</b>  | progType         | Programme type                            | xs:string (5)       | PRGTYP                 | Mandatory                                                    | progType          | S.34              |
| <b>B.05</b>  | sampMethod       | Sampling method                           | xs:string (5)       | SAMPMD                 | Optional                                                     | sampMethod        | S.35              |
| <b>B.06</b>  | sampler          | Sampler                                   | xs:string (5)       | SAMPLR                 | Mandatory                                                    |                   |                   |
| <b>B.07</b>  | sampPoint        | Sampling point                            | xs:string (5)       | SAMPNT                 | Mandatory                                                    | sampPoint         | S.39              |
| <b>B.08</b>  | progInfo         | Additional sampling programme information | <i>CompoundType</i> |                        | Optional                                                     |                   |                   |
| <b>C.01</b>  | sampEventId      | Sampling event identification code        | xs:string (100)     |                        | Optional                                                     |                   |                   |
| <b>C.02</b>  | sampUnitType     | Sampling unit type                        | xs:string (5)       | SAMPUNTYP              | Optional                                                     |                   |                   |
| <b>C.03</b>  | sampUnitSize     | Sampling unit size                        | xs:double           |                        | Optional                                                     | lotSize           | S.37              |
| <b>C.04</b>  | sampUnitSizeUnit | Sampling unit size unit                   | xs:string (5)       | UNIT                   | Optional                                                     | lotSizeUnit       | S.38              |
| <b>C.05</b>  | sampUnitIds      | Other sampling unit identifications       | <i>CompoundType</i> |                        | Optional                                                     |                   |                   |
| <b>C.06</b>  | sampEventInfo    | Additional sampling event information     | <i>CompoundType</i> |                        | Optional                                                     |                   |                   |
| <b>D.01</b>  | sampId           | Sample taken identification code          | xs:string (100)     |                        | Mandatory                                                    | labSampCode       | S.01              |
| <b>D.02</b>  | repCountry       | Reporting country                         | xs:string (2)       | COUNTRY                | Optional                                                     |                   |                   |
| <b>D.03</b>  | sampCountry      | Country of sampling                       | xs:string (2)       | COUNTRY                | Mandatory                                                    | sampCountry       | S.04              |
| <b>D.04</b>  | sampArea         | Area of sampling                          | xs:string (5)       | NUTS                   | Optional                                                     | sampArea          | S.05              |
| <b>D.05</b>  | repYear          | Reporting year                            | xs:integer (4)      |                        | Optional                                                     |                   |                   |
| <b>D.06</b>  | sampY            | Year of sampling                          | xs:integer (4)      |                        | Mandatory                                                    | sampY             | S.28              |



| Element code | Element name        | Element label                                                                   | Type            | Controlled terminology | Mandatory/optional element for chemical monitoring reporting | SSD1 element name                                                                 | SSD1 element code                                    |
|--------------|---------------------|---------------------------------------------------------------------------------|-----------------|------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------|
| <b>D.07</b>  | sampM               | Month of sampling                                                               | xs:integer (2)  |                        | Mandatory                                                    | sampM                                                                             | S.29                                                 |
| <b>D.08</b>  | sampD               | Day of sampling                                                                 | xs:integer (2)  |                        | Mandatory                                                    | sampD                                                                             | S.30                                                 |
| <b>D.09</b>  | sampSize            | Sample taken size                                                               | xs:double       |                        | Optional                                                     |                                                                                   |                                                      |
| <b>D.10</b>  | sampSizeUnit        | Sample taken size unit                                                          | xs:string (5)   | UNIT                   | Optional                                                     |                                                                                   |                                                      |
| <b>D.11</b>  | sampInfo.OrigSampId | Additional sample taken information                                             | xs:string (100) |                        | Optional                                                     |                                                                                   |                                                      |
| <b>E.01</b>  | sampMatType         | Type of matrix                                                                  | xs:string (5)   | MTXTYP                 | Optional                                                     |                                                                                   |                                                      |
| <b>E.02</b>  | sampMatCode         | Coded description of the matrix of the sample taken                             | CompoundType    | MTX (FoodEx2)          | Mandatory                                                    | EFSAProdCode, prodProdMeth, prodPack, prodTreat, prodIngred                       | S.12, S.15, S.16, S.17, S.20                         |
| <b>E.03</b>  | sampMatText         | Text description of the matrix of the sample taken                              | xs:string (250) |                        | Optional                                                     | prodText                                                                          | S.14                                                 |
| <b>E.04</b>  | origCountry         | Country of origin of the sample taken                                           | xs:string (2)   | COUNTRY                | Mandatory                                                    | origCountry                                                                       | S.06                                                 |
| <b>E.05</b>  | origArea            | Area of origin of the sample taken                                              | xs:string (5)   | NUTS                   | Optional                                                     | origArea                                                                          | S.07                                                 |
| <b>E.06</b>  | origFishAreaCode    | Area of origin for fisheries or aquaculture activities code of the sample taken | xs:string (10)  | FAREA                  | Optional                                                     | origFishAreaCode                                                                  | S.08                                                 |
| <b>E.07</b>  | origFishAreaText    | Area of origin for fisheries or aquaculture activities text of the sample taken | xs:string (250) |                        | Optional                                                     | origFishAreaText                                                                  | S.09                                                 |
| <b>E.08</b>  | procCountry         | Country of processing of the sample taken                                       | xs:string (2)   | COUNTRY                | Optional                                                     | procCountry                                                                       | S.10                                                 |
| <b>E.09</b>  | procArea            | Area of processing of the sample taken                                          | xs:string (5)   | NUTS                   | Optional                                                     | procArea                                                                          | S.11                                                 |
| <b>E.10</b>  | sampMatInfo         | Additional information on the matrix sampled                                    | CompoundType    |                        | Optional                                                     | prodCom, prodY, prodM, prodD, expiryY, expiryM, expiryD, prodManuf, prodBrandName | S.21, S.22, S.23, S.24, S.25, S.26, S.27, S.19, S.18 |
| <b>F.01</b>  | sampAnId            | Sample analysed identification code                                             | xs:string (100) |                        | Optional                                                     |                                                                                   |                                                      |



| Element code | Element name   | Element label                                         | Type            | Controlled terminology | Mandatory/optional element for chemical monitoring reporting | SSD1 element name | SSD1 element code |
|--------------|----------------|-------------------------------------------------------|-----------------|------------------------|--------------------------------------------------------------|-------------------|-------------------|
| <b>F.02</b>  | sampAnRefTime  | Sample analysis reference time                        | xs:string (5)   | REFTM                  | Optional                                                     |                   |                   |
| <b>F.03</b>  | analysisY      | Year of analysis                                      | xs:integer (4)  |                        | Mandatory                                                    | analysisY         | R.02              |
| <b>F.04</b>  | analysisM      | Month of analysis                                     | xs:integer (2)  |                        | Optional                                                     | analysisM         | R.03              |
| <b>F.05</b>  | analysisD      | Day of analysis                                       | xs:integer (2)  |                        | Optional                                                     | analysisD         | R.04              |
| <b>F.06</b>  | sampAnInfo     | Additional information on the sample analysed         | CompoundType    |                        | Optional                                                     |                   |                   |
| <b>G.01</b>  | anMatCode      | Coded description of the analysed matrix              | CompoundType    | MTX                    | Optional                                                     |                   |                   |
| <b>G.02</b>  | anMatText      | Text description of the matrix analysed               | xs:string (250) |                        | Optional                                                     |                   |                   |
| <b>G.03</b>  | anMatInfo      | Additional information on the analysed matrix         | CompoundType    |                        | Optional                                                     |                   |                   |
| <b>H.01</b>  | anPortSeq      | Sample analysed portion sequence                      | xs:string (100) |                        | Optional                                                     | labSubSampCode    | S.02              |
| <b>H.02</b>  | anPortSize     | Sample analysed portion size                          | xs:double       |                        | Optional                                                     |                   |                   |
| <b>H.03</b>  | anPortSizeUnit | Sample analysed portion size unit                     | xs:string (5)   | UNIT                   | Optional                                                     |                   |                   |
| <b>H.04</b>  | anPortInfo     | Additional information on the sample analysed portion | CompoundType    |                        | Optional                                                     |                   |                   |
| <b>I.01</b>  | isolId         | Isolate identification                                | xs:string (100) |                        | Not applicable                                               |                   |                   |
| <b>I.02</b>  | isolParamCode  | Coded description of the isolate                      | CompoundType    | PARAM                  | Not applicable                                               |                   |                   |
| <b>I.03</b>  | isolParamText  | Text description of the isolate                       | xs:string (250) |                        | Not applicable                                               |                   |                   |
| <b>I.04</b>  | isolInfo       | Additional information on the isolate                 | CompoundType    |                        | Not applicable                                               |                   |                   |
| <b>J.01</b>  | labId          | Laboratory identification code                        | xs:string (50)  |                        | Mandatory                                                    | labCode           | L.1               |
| <b>J.02</b>  | labAccred      | Laboratory accreditation                              | xs:string (1)   | LABACC                 | Mandatory                                                    | labAccred         | L.2               |
| <b>J.03</b>  | labCountry     | Laboratory country                                    | xs:string (2)   | COUNTRY                | Mandatory                                                    | labCountry        | L.3               |
| <b>J.04</b>  | labInfo        | Additional information on the laboratory              | CompoundType    |                        | Optional                                                     |                   |                   |
| <b>K.01</b>  | paramType      | Type of parameter                                     | xs:string (5)   | PARAMTYP               | Optional                                                     | paramType         | R.08              |
| <b>K.02</b>  | paramCode      | Coded description of the parameter                    | CompoundType    | PARAM                  | Mandatory                                                    | paramCode         | R.06              |
| <b>K.03</b>  | paramText      | Parameter text                                        | xs:string (250) |                        | Optional                                                     | paramText         | R.07              |
| <b>L.01</b>  | anMethRefId    | Analytical method identification                      | xs:string (50)  |                        | Mandatory                                                    | anMethRefCode     | R.09              |

| Element code | Element name      | Element label                                                    | Type            | Controlled terminology | Mandatory/optional element for chemical monitoring reporting | SSD1 element name  | SSD1 element code |
|--------------|-------------------|------------------------------------------------------------------|-----------------|------------------------|--------------------------------------------------------------|--------------------|-------------------|
| <b>L.02</b>  | anMethRefCode     | Analytical method reference code                                 | xs:string (5)   | ANLYREFMD              | Optional                                                     |                    |                   |
| <b>L.03</b>  | anMethType        | Analytical method type                                           | xs:string (5)   | ANLYTYP                | Mandatory                                                    |                    |                   |
| <b>L.04</b>  | anMethCode        | Analytical method code                                           | CompoundType    | ANLYMD                 | Mandatory                                                    | anMethCode         | R.10              |
| <b>L.05</b>  | anMethText        | Analytical method text                                           | xs:string (250) |                        | Optional                                                     | anMethText         | R.11              |
| <b>L.06</b>  | anMethInfo        | Additional information on the analytical method                  | CompoundType    |                        | Optional                                                     |                    |                   |
| <b>M.01</b>  | resId             | Result identification code                                       | xs:string (100) |                        | Mandatory                                                    | resultCode         | R.01              |
| <b>M.02</b>  | accredProc        | Accreditation procedure for the analytical method                | xs:string (5)   | MDACC                  | Mandatory                                                    | accredProc         | R.12              |
| <b>M.03</b>  | resUnit           | Result unit                                                      | xs:string (5)   | UNIT                   | Mandatory                                                    | resUnit            | R.13              |
| <b>M.04</b>  | resLOD            | Result LOD                                                       | xs:double       |                        | Optional                                                     | resLOD             | R.14              |
| <b>M.05</b>  | resLOQ            | Result LOQ                                                       | xs:double       |                        | Optional                                                     | resLOQ             | R.15              |
| <b>M.06</b>  | resLLWR           | Result lower limit of the working range                          | xs:double       |                        | Optional                                                     |                    |                   |
| <b>M.07</b>  | resULWR           | Result upper limit of the working range                          | xs:double       |                        | Optional                                                     |                    |                   |
| <b>M.08</b>  | CCalpha           | CC alpha                                                         | xs:double       |                        | Optional                                                     | ccAlpha            | R.16              |
| <b>M.09</b>  | CCbeta            | CC beta                                                          | xs:double       |                        | Optional                                                     | ccBeta             | R.17              |
| <b>M.10</b>  | resVal            | Result value                                                     | xs:double       |                        | Optional                                                     | resVal             | R.18              |
| <b>M.11</b>  | resValRec         | Result value recovery rate                                       | xs:double       |                        | Optional                                                     | resValRec          | R.19              |
| <b>M.12</b>  | resValRecCorr     | Result value corrected for recovery                              | xs:string (1)   | YESNO                  | Optional                                                     | resValRecCorr      | R.20              |
| <b>M.13</b>  | exprResPerc       | Expression of result percentage                                  | CompoundType    |                        | Optional                                                     | moistPerc, fatPerc | R.23 R.24         |
| <b>M.14</b>  | exprResType       | Expression of result type                                        | xs:string (5)   | EXPRRES                | Optional                                                     | exprRes            | R.25              |
| <b>M.15</b>  | resQualValue      | Result qualitative value                                         | xs:string (3)   | POSNEG                 | Optional                                                     | resQualValue       | R.26              |
| <b>M.16</b>  | resType           | Type of result                                                   | xs:string (3)   | VALTYP                 | Mandatory                                                    | resType            | R.27              |
| <b>M.17</b>  | resValUncert      | Result value uncertainty                                         | xs:double       |                        | Optional                                                     | resValUncert       | R.22              |
| <b>M.18</b>  | resValUncertSD    | Result value uncertainty standard deviation                      | xs:double       |                        | Optional                                                     | resValUncertSD     | R.21              |
| <b>M.19</b>  | resRefId          | Result reference identification                                  | xs:string (100) |                        | Optional                                                     |                    |                   |
| <b>M.20</b>  | resInfo.notSummed | Indicates LOQ should be calculated during the submission process | xs:string (1)   | 'Y'                    | Optional                                                     | resComm            | R.32              |

| Element code | Element name                   | Element label                                | Type             | Controlled terminology | Mandatory/optional element for chemical monitoring reporting | SSD1 element name | SSD1 element code |
|--------------|--------------------------------|----------------------------------------------|------------------|------------------------|--------------------------------------------------------------|-------------------|-------------------|
| <b>N.01</b>  | evalLowLimit                   | Limit for the result evaluation              | xs:double        |                        | Optional                                                     | resLegalLimit     | R.28              |
| <b>N.02</b>  | evalHighLimit                  | Limit for the result evaluation (high limit) | xs:double        |                        | Optional                                                     |                   |                   |
| <b>N.03</b>  | evalLimitType                  | Type of limit for the result evaluation      | xs:string (5)    | LMTTYP                 | Optional                                                     | resLegalLimitType | R.29              |
| <b>N.04</b>  | evalCode                       | Evaluation of the result                     | xs:string (5)    | RESEVAL                | Mandatory                                                    | resEvaluation     | R.30              |
| <b>N.05</b>  | actTakenCode                   | Action taken                                 | xs:string(1)     | ACTION                 | Optional                                                     | actTakenCode      | R.31              |
| <b>N.06</b>  | evalInfo.conclusion            | Conclusion of follow-up investigation        | xs:string (5)    | CONCLUS                | Optional                                                     |                   |                   |
| <b>N.06</b>  | evalInfo.com                   | Comment                                      | xs:string (250)  |                        | Optional                                                     |                   |                   |
| <b>N.06</b>  | evalInfo.resAsses              | Assessment of the result                     | xs:string (5)    | RESEVAL                | Optional                                                     |                   |                   |
| <b>N.06</b>  | evalInfo.restriction Exception | Restriction or exception                     | xs: string (250) | ADDFOOD                | Optional                                                     |                   |                   |

### 3 Specific requirements for baby foods' data submission

This section refers to matrix classifications (FoodEx2) which are under the high-level code for 'Food products for the young population' (A03PV). When reporting samples of such foods, data providers must note the following specific conditions which apply.

For records to be included in the pesticides Annual Report:

- the progLegalRef value must be N028A (Commission Directives No 2006/125/EC) or N318A or N027A;
- baby foods are always considered to be processed;
- matrix classification code attributes must be as detailed as possible, e.g. for infant formula it is important to know whether it was powder or liquid, reconstituted or not. Baby food legal limits tend to apply to the ready to eat product, i.e. likely to the reconstituted form of the product in contrast, with other food products where the limits tend to apply to product as placed on the market.

Baby food samples are excluded from the VMPPR Annual Report as defined in Regulation (EU) 2022/1646. A BR prevents that baby foods is reported with progLegalRef (N371A). Please see BR CHEMON55 in Table 9.

Baby food samples are accepted in the ChemMon data collection when they have been analysed for any valid ChemMon parameter as long as their progLegalRef is not N371A (Regulation (EU) 2022/1646).

### 4 Specific requirements for chemical contaminants' data submission

For some chemical results, more detailed and specific information is necessary to fully describe the sample and analytical results. These specific requirements (mandatory or recommended) are described here (Table 8). Specific requirements are updated annually to reflect the development of relevant legislation and in response to recommendations for data in EFSA's scientific opinions. Annual amendments are shared with the EFSA Scientific Network on Chemical Monitoring Data.

Table 8: Specific reporting requirements for certain contaminants, food additives and food flavourings.

| Topic                                                     | SSD2 data element                                                                                              | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                                                                                     | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable) |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
| <b>Brominated flame retardants</b>                        | Area of origin for fisheries or aquaculture activities code and text of the sample taken (E.06 and E.07)       | Recommended            | If data are reported on fish, it is recommended that the area of origin for fisheries is specified. When the specific information on the 'Area of origin for fisheries or aquaculture activities code' (E.06) is not available ('Unknown [M99]') as a minimum the fish's original environment should be given ('from freshwater' or 'from saltwater') in area of origin for fisheries or aquaculture activities text (E.07) | CHEMON20                                                  |                                                   |
|                                                           | Expression of result percentage/percentage of fat                                                              | Recommended            | Strongly recommended that the percentage of fat in the original sample is always reported (regardless of whether the result value is expressed in whole weight, in fat weight or dry matter)                                                                                                                                                                                                                                | CHEMON21                                                  |                                                   |
| <b>Copper</b>                                             | Coded description of the matrix of the sample taken (E.02) \ Part-consumed-analysed (F20) and/or Process (F28) | Highly Recommended     | It is recommended that the Part-consumed-analysed descriptors (facet F20) and/or Process (facet F28) are provided in the coded description of the matrix of the sample taken.<br><br>NOTE: EFSA informs from next year onwards the reporting of facets F20 and/or F28 will become mandatory following BR CHEMON90_a                                                                                                         | CHEMON90_a                                                |                                                   |
|                                                           | Result value uncertainty (M.17)                                                                                | Mandatory              | Information on the measurement uncertainty should be provided (range to be given in the same units as the result value (resVal))                                                                                                                                                                                                                                                                                            | CHEMON90_b                                                |                                                   |
| <b>Dioxins and dioxin-like PCBs, non-dioxin-like PCBs</b> | Area of origin for fisheries or aquaculture activities code and text of the sample                             | Recommended            | If data are reported on fish, it is recommended that the area of origin for fisheries is specified. When the specific information on 'Area of origin for fisheries or aquaculture activities code' (E.06) is not available ('Unknown [M99]') as a minimum the fish original environment should be given ('from freshwater' or                                                                                               | CHEMON20                                                  |                                                   |

<sup>58</sup> A '-' in this column means that no business rule is relevant given the type of requirement.

| Topic | SSD2 data element                                                                     | Status of data element | Comment                                                                                                                                                                                                                                                                                    | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable)                |
|-------|---------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------|
|       | taken (E.06 and E.07)                                                                 |                        | 'from saltwater') in area of origin for fisheries or aquaculture activities text (E.07)                                                                                                                                                                                                    |                                                           |                                                                  |
|       | Coded description of the matrix of the sample taken (E.02)                            | Recommended            | It is particularly important to classify the samples of fish meat and fish products at the most detailed FoodEx2 level available. If data are reported on fish, it is recommended that the analysed fish species be precisely reported, especially fish originating from the Baltic region | -                                                         |                                                                  |
|       | Additional information on the matrix sampled (E.10)\comment                           | Recommended            | If data are reported on fish, it is recommended that the length and/or weight of the fish samples be reported <sup>59, 60</sup>                                                                                                                                                            | -                                                         | Measurement Regulation (EC) 850/98 and Regulation (EC) 2187/2005 |
|       | Additional information on the matrix sampled (E.10)\year, month and day of production | Recommended            | If data are reported on fish, it is recommended that when the fish have been caught be reported                                                                                                                                                                                            |                                                           |                                                                  |

<sup>59</sup> Council Regulation (EC) No 850/98 of 30 March 1998 for the conservation of fishery resources through technical measures for the protection of juveniles of marine organisms. OJ L 125, 27.4.1998, p. 1.

<sup>60</sup> Council Regulation (EC) No 2187/2005 of 21 December 2005 for the conservation of fishery resources through technical measures in the Baltic Sea, the Belts and the Sound, amending Regulation (EC) No 1434/98 and repealing Regulation (EC) No 88/98. OJ L 349, 31.12.2005, p.1-23.



| Topic | SSD2 data element                                        | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable)                                                         |
|-------|----------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|       | Coded description of the parameter (K.02)                | Recommended            | The full set of the 29 dioxins and dioxin-like PCBs (17 PCDD/F, 12 dl-PCB), as listed in the Appendix to Annex I of Commission Regulation (EU) 2023/915, is strongly recommended to be reported when reporting results on dioxins and dioxin-like PCBs. For the non-dioxin-like PCBs, the six indicator congeners (PCB 28, PCB 52, PCB 101, PCB 138, PCB 153 and PCB 180) are strongly recommended to be reported. It is strongly recommended to report analytical results at the congener level in the measured concentration | CHEMON09<br>CHEMON10                                      | Commission Regulation (EU) 2023/915                                                                       |
|       | Expression of result percentage (M.13)\percentage of fat | Recommended            | It is recommended to report the percentage of fat in the original sample for food samples with maximum or action levels expressed on a fat basis: products of terrestrial animal origin, marine oils and vegetable oils and fats (marine oils included).<br>It is recommended to report food samples of fish and offal and foods of plant origin (except oils) on a whole weight basis. Feed should be reported on the basis of 88% dry matter                                                                                 | CHEMON21                                                  | Commission Regulations (EU) 278/2012 <sup>61</sup> and Commission Regulations (EU) 644/2017 <sup>62</sup> |
|       | Expression of result percentage (M.13)\percentage of fat | Recommended            | It is recommended that the percentage of fat is provided for the other samples, when the lipid content has been determined                                                                                                                                                                                                                                                                                                                                                                                                     | CHEMON21                                                  |                                                                                                           |
|       | Additional information on the result (M.20)              | Recommended            | If the sample was reconstituted before analysis, e.g. foods for infants and young children, it would be relevant to provide information on the exact reconstitution protocol (ratio dry product: added fluid (description of the fluid used) Examples: '10:90 (milk)', '20:80 (water)')                                                                                                                                                                                                                                        | -                                                         |                                                                                                           |

<sup>61</sup> Commission Regulation (EU) No 278/2012 of 28 March 2012 amending Regulation (EC) No 152/2009 as regards the determination of the levels of dioxins and polychlorinated biphenyls. OJ L 91, 29.3.2012, p. 8–22.

<sup>62</sup> Commission Regulation (EU) No 644/2017 of 5 April 2017 laying down methods of sampling and analysis for the control of levels of dioxins, dioxin-like PCBs and non-dioxin-like PCBs in certain foodstuffs and repealing Regulation (EC) No 589/2014. OJ L 92, 6.4.2017, p. 9–34.



| Topic               | SSD2 data element                                                                    | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                    | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable)                                                                                                           |
|---------------------|--------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mineral oils</b> | Expression of result percentage (M.13)\percentage of moisture                        | Recommended            | It is recommended that the percentage of moisture in the original sample is always reported (regardless of whether the result value is expressed in whole weight, in fat weight or dry matter)                                                                                                                                             | CHEMON11                                                  |                                                                                                                                                             |
|                     | Additional information on the result (M.20)                                          | Recommended            | The data providers are requested to report information on the molecular mass distribution expressed as the number of carbon atoms (referring to n-alkanes) including the range of carbon atoms and the maximum of the distribution curve                                                                                                   | -                                                         |                                                                                                                                                             |
| <b>Acrylamide</b>   | Coded description of the matrix of the sample taken (E.02)\ingredients (F04)         | Recommended            | It is recommended that the list of ingredients is provided in the 'Ingredients' facet for the following food items: 'Potato crisps', 'Pre-cooked French fries, potato products for home cooking', 'Breakfast cereals (excluding muesli and porridge)', 'Substitute coffee (dry)' and 'Baby foods, other than processed cereal-based foods' |                                                           |                                                                                                                                                             |
|                     | Coded description of the matrix of the sample taken (E.02)\legislative classes (F33) | Mandatory              | It is mandatory to provide additional product classification in the 'Legislative classes' facet. Data providers must consult the list of legislative classes applicable to acrylamide in FoodEx2 Catalogue Facet 33: Legislative classes                                                                                                   | CHEMON12                                                  | Commission Regulation 2017/2158 establishing mitigation measures and benchmark levels for the reduction of the presence of acrylamide in food <sup>63</sup> |
|                     | Country of processing of the                                                         | Recommended            | It is recommended that the country where the food was processed is provided (ISO 3166-1-alpha-2)                                                                                                                                                                                                                                           |                                                           |                                                                                                                                                             |

<sup>63</sup> Commission Regulation (EU) 2017/2158 of 20 November 2017 establishing mitigation measures and benchmark levels for the reduction of the presence of acrylamide in food. C/2017/7658. OJ L 304, 21.11.2017, p. 24–44.

| Topic        | SSD2 data element                                         | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable) |
|--------------|-----------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
|              | sample taken (E.08)                                       |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                           |                                                   |
|              | Text description of the matrix of the sample taken (E.03) | Recommended            | It is recommended that the following additional information is mentioned in the text description: Potato crisps (category 2): indicate whether batch fried or continuously fried; Pre-cooked French fries/home products (category 3): indicate whether the product is purchased fresh/frozen, and whether the starting material is fresh potato or potato dough; Crispbread (category 6): indicate whether the product is/is not fermented; Roasted coffee (category 7): indicates the degree of roasting (light, medium, dark)                                                                                                                          | -                                                         |                                                   |
|              | Result value uncertainty (M.17)                           | Recommended            | If available, information on the measurement uncertainty should be provided (range to be given in the same units as the result value (resVal))                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                                                         |                                                   |
|              | Additional information on the result (M.20)               | Recommended            | For the samples of the food category 3 (Pre-cooked French fries/potato product for home cooking), e.g. pre-cooked French fries, which have been prepared 'as consumed' before analysis, it is relevant to provide information on the exact cooking conditions (time, temperature and handling information). In addition, if the sample has been reconstituted before analysis, e.g. instant coffee powder, or foods for infant and young children, it is relevant to provide information on the exact reconstitution protocol (ratio dry product: added fluid; description of the fluid used) Examples: '2:100 (water)', '10:90 (milk)', '20:80 (water)' | -                                                         | Commission Recommendation (2010/307/EC)           |
| <b>Furan</b> | Coded description of the analysed matrix (G.01)           | Recommended            | If the sample has been analysed 'as consumed', then it should be described in the analysed sample as consumed <sup>64</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -                                                         | Commission Recommendation 2007/196/EC             |

<sup>64</sup> Commission Recommendation No 2007/196/EC of 28 March 2007 on the monitoring of the presence of furan in foodstuff. OJ L 88, 29.3.2007, p.56-57.



| Topic                                   | SSD2 data element                                                                   | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                          | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable)                                                                                                                                                             |
|-----------------------------------------|-------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | Additional information on the result (M.20)                                         | Recommended            | If the product is analysed as consumed, it is relevant to provide information on the exact cooking preparation with time, temperature and handling information (using the \$ character to separate extra information). Example 'Heated in microwave\$60°C\$1.5min'                                                                                               | -                                                         |                                                                                                                                                                                                               |
|                                         | Sampling method (B.05)                                                              | Recommended            | For the monitor of furan, 2-methylfuran and 3-methylfuran in food and to ensure that the samples are representative, it is advised to follow the sampling procedures laid down in part B of the Annex to Commission Regulation (EC) No 333/2007<br><sampMethod>N011A</sampMethod>                                                                                |                                                           | Commission Recommendation (EC) 2022/495 & Commission Regulation (EC) 333/2007 <sup>65</sup>                                                                                                                   |
| <b>Polycyclic aromatic hydrocarbons</b> | Coded description of the matrix of the sample taken (E.02)\packaging material (F19) | Recommended            | It is recommended to describe the container or wrapper that holds the product as marketed. This is important since a source of PAH contamination in food is the packaging in which it is marketed. According to legislation, the sample container must be of inert material to eliminate the possibility of migration contamination during sampling and analysis | CHEMON15                                                  | Commission Regulation (EC) 333/2007 laying down the methods of sampling and analysis for the official control of the levels of lead, cadmium, mercury, inorganic tin, 3-MCPD and benzo(a)pyrene in foodstuffs |

<sup>65</sup> Commission Recommendation (EC) 2022/495 of March 2022 on monitoring the presence of furan and alkylfurans in food  
Commission Regulation (EC) No 333/2007 of 28 March 2007 laying down the methods of sampling and analysis for the official control of the levels of lead, cadmium, mercury, inorganic tin, 3-MCPD and benzo(a)pyrene in foodstuffs. OJ L 88, 29.3.2007, p. 29–38.

| Topic                                       | SSD2 data element                                          | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable)                                                                                                |
|---------------------------------------------|------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | Result value recovery rate (M.11)                          | Recommended            | Recommended to provide the recovery value associated with the concentration measurement (as a percentage) when an extraction step is applied in the analytical method                                                                                                                                                                                                                                                                                                     | -                                                         | Commission Regulation (EC) No 333/2007                                                                                                           |
|                                             | Result value uncertainty (M.17)                            | Recommended            | Recommended to provide the expanded uncertainty (95% confidence interval) associated with the concentration measurement (in units of the result)                                                                                                                                                                                                                                                                                                                          |                                                           | Commission Regulation (EC) No 333/2007                                                                                                           |
| <b>3-MCPDs, 2-MCPDs and glycidyl esters</b> | Analytical method code (L.04)                              | Mandatory              | For the analysis of 3- and 2-MCPD esters and glycidyl esters it should be clarified whether the samples have been analysed with one of the four methods validated by ISO (the International Organization for Standardization). For the four methods, the appropriate code from the catalogue ANALYMD must be reported; otherwise, a reference to the different methods must be provided. The same applies for 3- and 2-MCPD total referring to the ester part of the sum. | -                                                         | EN ISO 18363-1:2021 <sup>66</sup><br>EN ISO 18363-2:2018 <sup>67</sup><br>EN ISO 18363-3:2024 <sup>68</sup><br>EN ISO 18363-4:2021 <sup>69</sup> |
|                                             | Expression of result percentage (M.13)\percentage of fat   | Recommended            | It is recommended that the percentage of fat in the original sample is always reported (regardless of whether the result value is expressed in whole weight, in fat weight or dry matter)                                                                                                                                                                                                                                                                                 | CHEMON21                                                  |                                                                                                                                                  |
| <b>Mycotoxins</b>                           | Coded description of the matrix of the sample taken (E.02) | Mandatory              | It is important to classify the food/feed samples of grains at the most detailed FoodEx2 level available. This is particularly needed for 'Grains and grain-based products' and 'Food for infants and small children'                                                                                                                                                                                                                                                     | -                                                         |                                                                                                                                                  |

<sup>66</sup> <https://www.iso.org/obp/ui/en/#iso:std:iso:18363:-1:ed-1:v1:en>

<sup>67</sup> <https://www.iso.org/obp/ui/en/#iso:std:iso:18363:-2:ed-1:v1:en>

<sup>68</sup> <https://www.iso.org/obp/ui/en/#iso:std:iso:18363:-3:ed-1:v1:en>

<sup>69</sup> <https://www.iso.org/obp/ui/en/#iso:std:iso:18363:-4:ed-1:v1:en>

| Topic                          | SSD2 data element                                                                  | Status of data element | Comment                                                                                                                                                                                                                                                                 | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable) |
|--------------------------------|------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
|                                | Coded description of the matrix of the sample taken (E.02)\Production Method (F21) | Recommended            | It is recommended that it is reported whether the sample was obtained from the produce of traditional (non-organic) or organic farming                                                                                                                                  | CHEMON17                                                  |                                                   |
|                                | Result value recovery rate (M.11)                                                  | Recommended            | It is recommended that the recovery value associated with the concentration measurement is provided (as a percentage)                                                                                                                                                   |                                                           | Commission Regulation (EU) 2023/2782              |
|                                | Expression of result percentage (M.13)\percentage of moisture                      | Recommended            | It is recommended that the percentage of moisture in the original sample of feed is reported and for processed cereal-based foods for infants and young children                                                                                                        | CHEMON11                                                  |                                                   |
|                                | Result value uncertainty (M.17)                                                    | Recommended            | It is recommended that the expanded uncertainty (95% confidence interval) associated with the concentration measurement be provided                                                                                                                                     | -                                                         | Commission Regulation (EU) 2023/2782              |
|                                | Additional information on the result (M.20)                                        | Recommended            | If the sample was reconstituted before analysis, e.g. infant formula and follow-on formula, it would be relevant to provide information on the exact reconstitution protocol (ratio dry product: added fluid (description of the fluid used) Examples: '14:86 (water)') | -                                                         |                                                   |
| <b>Arsenic and derivatives</b> | Coded description of the matrix of the sample taken (E.02)                         | Recommended            | For seaweed, it is strongly recommended that information is provided to identify the presence of the alga Hijiki. For rice, it is strongly recommended that information is provided about the type of rice                                                              | -                                                         |                                                   |

| Topic                          | SSD2 data element                                                            | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                                             | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable) |
|--------------------------------|------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
|                                | Coded description of the matrix of the sample taken (E.02)\Process (F28)     | Recommended            | Reporting data on rice (as grains for human consumption). In order to be able to distinguish raw rice and rice that has undergone any kind of treatment, in the 'Process' facet (F28) it is recommended to provide at least the codes for 'unprocessed' or 'processed'. Additionally, it is recommended that it is clearly indicated if the original sample is a dehydrated product | CHEMON18                                                  |                                                   |
|                                | Coded description of the matrix of the sample taken (E.02)\ingredients (F04) | Recommended            | Reporting of rice-based products. If a product contains rice (e.g. a ready-to-eat meal for children, cereal-based; rice cake) it should be specified in this field. Furthermore, it is recommended that the type of algae included in alga-based foods for special nutritional use is described                                                                                     | -                                                         |                                                   |
| <b>Cadmium and derivatives</b> | Result value recovery rate (M.11)                                            | Recommended            | It is recommended that the recovery value associated with the concentration measurement is provided (as a percentage) when an extraction step is applied in the analytical method                                                                                                                                                                                                   | -                                                         | Commission Regulation (EC) No 333/2007            |
|                                | Result value uncertainty (M.17)                                              | Recommended            | It is recommended that the expanded uncertainty (95% confidence interval) associated with the concentration measurement is provided                                                                                                                                                                                                                                                 |                                                           | Commission Regulation (EC) No 333/2007            |
| <b>Lead and derivatives</b>    | Result value recovery rate (M.11)                                            | Recommended            | It is recommended that the recovery value associated with the concentration measurement (as a percentage) is provided when an extraction step is applied in the analytical method                                                                                                                                                                                                   | -                                                         | Commission Regulation (EC) No 333/2007            |
|                                | Result value uncertainty (M.17)                                              | Recommended            | It is recommended that the expanded uncertainty (95% confidence interval) associated with the concentration measurement is provided                                                                                                                                                                                                                                                 |                                                           | Commission Regulation (EC) No 333/2007            |
| <b>Mercury and derivatives</b> | Coded description of the matrix of the sample taken (E.02)                   | Recommended            | It is particularly important to classify the samples of fish meat, seafood and fish/seafood products at the most detailed FoodEx2 level available                                                                                                                                                                                                                                   | -                                                         |                                                   |

| Topic                                                    | SSD2 data element                                                                                        | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                                                                           | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable)                                               |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|                                                          | Area of origin for fisheries or aquaculture activities code and text of the sample taken (E.06 and E.07) | Recommended            | If data are reported on fish, it is recommended that the area of origin for fisheries is specified. If the specific information on 'Area of origin for fisheries or aquaculture activities code' (E.06) is not available ('Unknown [M99]') at least the fish original environment should be given ('from fresh water' or 'from salt water') in area of origin for fisheries or aquaculture activities text (E.07) | CHEMON20                                                  |                                                                                                 |
| <b>Nitrates in vegetables and other food commodities</b> | Result value recovery rate (M.11)                                                                        | Recommended            | It is recommended that the recovery value associated with the concentration measurement is provided (as a percentage) <sup>70</sup>                                                                                                                                                                                                                                                                               |                                                           | Commission Regulation (EC) No 1882/2006                                                         |
|                                                          | Result value uncertainty (M.17)                                                                          | Recommended            | It is recommended that the expanded uncertainty (95% confidence interval) associated with the concentration measurement is provided (in units of the result)                                                                                                                                                                                                                                                      | -                                                         | Commission Regulation (EC) No 1882/2006                                                         |
| <b>Chlorates and perchlorates</b>                        | Coded description of the matrix of the sample taken (E.02)\process (F28)                                 | Recommended            | Some product treatments, like deep-freezing, blanching or treatments at elevated temperatures have a significant effect on the chlorate level in food. Therefore, it is recommended to report facet F28 (Process) and select at least the code for 'processed' or 'unprocessed'; if possible, specify the most appropriate code describing the process from the SSD2 catalogue                                    | CHEMON19                                                  | Commission Recommendation (EU) 2015/682 <sup>71</sup><br>Regulation (EU) 2020/749 <sup>72</sup> |
| <b>Food additives occurrence</b>                         | Coded description of the matrix sampled                                                                  | Mandatory              | It is required to provide additional product classification (food category code) based on Commission Regulation (EC) No 1333/2008 on food additives, as last amended. Mapping from FoodEx2 base terms to legislative categories is automatically                                                                                                                                                                  | CHEMON39                                                  | Regulation (EC) No 1333/2008                                                                    |

<sup>70</sup> Commission Regulation (EC) No 1882/2006 of 19 December 2006 laying down methods of sampling and analysis for the official control of the levels of nitrates in certain foodstuffs. OJ L 364, 20.12.2006, p.25.

<sup>71</sup> Commission Recommendation (EU) 2015/682 of 29 April 2015 on the monitoring of the presence of perchlorate in food. OJ L 111, 30.4.2015 p.32-33.

<sup>72</sup> Commission Regulation (EU) 2020/749 of 4 June 2020 amending Annex III to Regulation (EC) No 396/2005 of the European Parliament and of the Council as regards maximum residue levels for chlorate in or on certain products. OJ L 178, 08/06/2020, p. 7–20



| Topic                              | SSD2 data element                                                       | Status of data element | Comment                                                                                                                                                                                                                                                                                                                                                                                 | Corresponding business rule (see Chapter 6) <sup>58</sup> | Legislative or other reference (where applicable)             |
|------------------------------------|-------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|
|                                    | (E02)\legislative classes (F33)                                         |                        | applied in the EFSA catalogue as implicit facets in most of the cases; if the F33 facet is not implicitly assigned, it is required to add it.                                                                                                                                                                                                                                           |                                                           |                                                               |
| <b>Food flavourings occurrence</b> | Coded description of the matrix sampled (E02)\legislative classes (F33) | Mandatory              | It is required to provide additional product classification (food category code) based on Regulation (EC) No 1334/2008 on food flavourings, as last amended. Mapping from FoodEx2 base terms to legislative categories is automatically applied in the EFSA catalogue as implicit facets in most of the cases; if the F33 facet is not implicitly assigned, it is recommended to add it | CHEMON39                                                  | Regulation (EC) No 1334/2008 and Regulation (EC) No 1333/2008 |
| <b>Bisphenol</b>                   | Coded description of the matrix sampled (E02)\packaging material (F19)  | Mandatory              | It is mandatory to provide additional product classification (food category code) based on Commission Regulation (EC) No 213/2018 on bisphenol <sup>73</sup> . The list of codes to be used, according to the MTX catalogue is provided in the F19 packaging material facet of the MTX catalogue                                                                                        | CHEMON14                                                  | Commission Regulation (EU) 2018/213                           |
| <b>Mineral oil hydrocarbons</b>    |                                                                         |                        | Additional SSD requirements for the reporting on mineral oil hydrocarbon analytical results are detailed in the JRC guidance referenced <sup>74</sup>                                                                                                                                                                                                                                   | -                                                         | Bratinova S and Hoekstra E, 2019                              |

<sup>73</sup> Commission Regulation (EU) 2018/213 of 12 February 2018 on the use of bisphenol A in varnishes and coatings intended to come into contact with food and amending Regulation (EU) No 10/2011 as regards the use of that substance in plastic food contact materials. C/2018/0685. OJ L 41, 14.2.2018, p. 6–12.

<sup>74</sup> S. Bratinova, E. Hoekstra (Editors) Guidance on sampling, analysis and data reporting for the monitoring of mineral oil hydrocarbons in food and food contact materials, Luxembourg: Publications Office of the European Union, 2019 ISBN 978-92-76-00172-0, doi:10.2760/208879, JRC115694



## 5 Algorithm to classify samples according to legislative matrix groups

### 5.1 Veterinary medicinal product residues

For the specific purpose of the National and Annual Reports of VMPR, EFSA has created an algorithm which interprets the matrix code reported (sampMatCode from the FoodEx2 classification and description system) and groups them according to the reportable groups defined in the EU VMPR legislation.

The algorithm is published on Knowledge Junction (EFSA, 2024) with other data collection supporting tools and documents. This is intended to help data providers ensure that each sample and result, which they expect to appear in their national VMPR report and in the EU VMPR Report, will be correctly classified based on the FoodEx2 codes reported in their data transmissions.

### 5.2 Pesticide residues

For the specific purpose of the preparation of the EFSA EU Annual Reports on pesticide residues, EFSA has created an algorithm which converts the reported sampMatCode FoodEx2 term to a MATRIX code according to the reportable groups defined in the pesticide legislation. For raw primary commodities (RPC) these mappings are explicitly recorded in the FoodEx2 catalogue as attributes of each term.

The algorithm is published on Knowledge Junction with other data collection supporting tools and documents. This is intended to help data providers ensure that each sample and result which they expect to appear in their pesticide residues national and annual reporting will be correctly classified based on the FoodEx2 codes reported in their data transmissions.

## 6 Business Rules

Business rules are provided to ensure the data submitted through the EFSA Data Collection Framework (DCF) are fit for purpose for use in EFSA's scientific assessments.

There are three groups of BR:

- 1 General business rules (GBR), which apply to all data submitted to EFSA through the DCF, including microbiological data as well as chemical.
- 2 Chemical monitoring business rules (CHEMON), which apply to all chemical monitoring data.
- 3 Specific business rules that apply to a specific subset of the chemical monitoring data, e.g. there are a small number of rules which enforce legislative or reporting requirements in relation to pesticides, VMPR, food additives, food flavourings or specific contaminants.

Table 9: Full list of general business rules (GBR) with troubleshooting tips

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Error/Warning Message                                                                                                                                                             | Type of error | Status   | Comment                                                                                                                                                                                                                                                                                                   |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GBR1</b>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                   |               | Inactive | This rule originally required that resId was unique and the database key for each record. This is now done through the DCF schema, so the rule is no longer active                                                                                                                                        |
| <b>GBR2</b>        | For the records with the same 'Sample taken identification code' (sampId) and/or when the 'Sampling event identification code' (sampEventId) is reported, then the descriptors for the sampling event, the local organisation and the sampling programme (sections A, B, C) must be constant (the same) for all records with the same 'Sampling event identification code' (sampEventId). The data elements in sections A, B, and C are: localOrgId, localOrgCountry, localOrgInfo, sampStrategy, progType, sampMethod, sampler, sampPoint, progInfo, sampUnitType, sampUnitSize, sampUnitSizeUnit, sampUnitIds, sampEventInfo | The sampling event or local organisation or sampling programme descriptors (sections A, B, C <sup>75</sup> ) are not constant for all records with the same sampId or sampEventId | Error         | Active   | Note: progId and progLegalRef were removed from the list of elements which must be consistent for the first time in the 2020 data collection<br>Check that all required elements in Section C of the SSD2 data model are consistent for all the samples reported for the same sampId or Sampling Event Id |
| <b>GBR3</b>        | The descriptors for the sample taken and the matrix sampled (sections D, E) and the 'Sampling event identification code' (sampEventId) must be constant (the same) for all records with the same 'Sample taken identification code' (sampId). The data elements in the sections D and E                                                                                                                                                                                                                                                                                                                                        | The sample taken or matrix sampled descriptors (sections D, E) or sampEventId are not constant for all records with the same sampId                                               | Error         | Active   |                                                                                                                                                                                                                                                                                                           |

<sup>75</sup> Sections A, B, C, D, E, F, G, H and I refer to the SSD2 data model sections as defined in the Standard Sample Description ver. 2.0 (EFSA 2013) guidance document.

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                              | Error/Warning Message                                                                                                                | Type of error | Status | Comment                                 |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-----------------------------------------|
|                    | are: sampId, repCountry, sampCountry, sampArea, sampY, sampM, sampD, sampSize, sampUnitSize, sampInfo, sampMatType, sampMatCode, sampMatText, origCountry, origArea, origFishAreaCode, origFishAreaText, procCountry, procArea, sampMatInfo                                                                                                                                                                                                              |                                                                                                                                      |               |        |                                         |
| <b>GBR4</b>        | The descriptors for the sample analysed and the matrix analysed (Sections F, G) and the 'Sample taken identification code' (sampId) must be constant (the same) for all records with the same 'Sample analysed identification code' (sampAnId). The data elements in the sections F and G are: sampAnId, sampAnRefTime, analysisY, analysisM, analysisD, sampAnInfo, anMatCode, anMatText, anMatInfo                                                     | The sample analysed or matrix analysed descriptors (Sections F, G) or sampId are not constant for all records with the same sampAnId | Error         | Active |                                         |
| <b>GBR5</b>        | If a value in 'Sample analysed identification code' (sampAnId) and 'Sample analysed portion sequence' (anPortSeq) is reported, then the descriptors for the sample analysed portion (Section H) must be constant (the same) for all records with the same 'Sample analysed identification code' (sampAnId) and 'Sample analysed portion sequence' (anPortSeq). The data elements in the Section H are: anPortSeq, anPortSize, anPortSizeUnit, anPortInfo | The sample analysed portion descriptors (Section H) are not constant for all records with the same sampAnId and anPortSeq            | Error         | Active |                                         |
| <b>GBR6</b>        | If a value in 'Isolate identification' (isolId) is reported, then the descriptors for the isolate (Section I) and the 'Sample taken identification code' (sampId) must be constant (the                                                                                                                                                                                                                                                                  | The isolate descriptors (Section I) or sampId are not constant for all records with the same isolId                                  | Error         | Active | This GBR is relevant to biological data |

|                                                                                      |     |                                          |
|--------------------------------------------------------------------------------------|-----|------------------------------------------|
| <a href="http://www.efsa.europa.eu/publications">www.efsa.europa.eu/publications</a> | 102 | EFSA Supporting publication 2026:EN-9911 |
|--------------------------------------------------------------------------------------|-----|------------------------------------------|

| Business Rule Code | Description                                                                                                                                                                                                                                                      | Error/Warning Message                                                                               | Type of error | Status | Comment |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------|--------|---------|
| <b>GBR13</b>       | The 'Area of origin of the sample taken' (origArea) must be within the 'Country of origin of the sample taken' (origCountry)                                                                                                                                     | origArea is not within origCountry                                                                  | error         | active |         |
| <b>GBR14</b>       | The 'Area of processing of the sample taken' (procArea) must be within the 'Country of processing of the sample taken' (procCountry)                                                                                                                             | procArea is not within procCountry                                                                  | error         | active |         |
| <b>GBR15</b>       | If in the 'Coded description of the matrix of the sample taken' the generic-term facet (sampMatCode.gen) is reported with the descriptor 'Other' (A07XE), then a text must be reported in the 'Text description of the matrix of the sample taken' (sampMatText) | sampMatText is missing, though mandatory if sampMatCode.gen is 'Other' (A07XE)                      | error         | active |         |
| <b>GBR16</b>       | If in the 'Coded description of the analysed matrix' the generic-term facet (anMatCode.gen) is reported with the descriptor 'Other' (A07XE), then a text must be reported in the 'Text description of the matrix analysed' (anMatText)                           | anMatText is missing, though mandatory if anMatCode.gen is 'Other' (A07XE)                          | error         | active |         |
| <b>GBR18</b>       | If the reported value in the 'Analytical method code' (anMethCode.base) is 'Classification not possible' (F001A), then a text must be reported in the 'Analytical method text' (anMethText)                                                                      | anMethText is missing, though mandatory if anMethCode.base is 'Classification not possible' (F001A) | error         | active |         |
| <b>GBR19</b>       | The value in the data element 'Percentage of fat' (exprResPerc.fatPerc) must be expressed as a percentage and so be between '0' and '100' (e.g. '40' must be reported for 40 %)                                                                                  | exprResPerc.fatPerc is not between '0' and '100'                                                    | error         | active |         |
| <b>GBR20</b>       | The value in the data element 'Percentage of moisture' (exprResPerc.moistPerc) must be                                                                                                                                                                           | exprResPerc.moistPerc is not between '0' and '100'                                                  | Error         | Active |         |

| Business Rule Code | Description                                                                                                                                                                              | Error/Warning Message                                                                                 | Type of error | Status | Comment |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------|--------|---------|
|                    | expressed as a percentage and so be between '0' and '100' (e.g. '40' must be reported for 40 %)                                                                                          |                                                                                                       |               |        |         |
| <b>GBR21</b>       | The value in the data element 'Percentage of alcohol' (exprResPerc.alcoholPerc) must be expressed as a percentage and so be between '0' and '100' (e.g. '40' must be reported for 40 %)  | exprResPerc.alcoholPerc is not between '0' and '100'                                                  | Error         | Active |         |
| <b>GBR22</b>       | If the value in the 'Expression of result type' (exprResType) is 'Fat weight' (B003A), then a value should be reported in the 'Percentage of fat' (exprResPerc.fatPerc)                  | WARNING:<br>exprResPerc.fatPerc is missing, though recommended if exprResType is 'Fat weight' (B003A) | Warning       | Active |         |
| <b>GBR23</b>       | If the value in the 'Expression of result type' (exprResType) is 'Dry matter' (B002A), then a value must be reported in the 'Percentage of moisture' (exprResPerc.moistPerc)             | exprResPerc.moistPerc is missing, though mandatory if exprResType is expressed on 'dry matter' basis  | Error         | Active |         |
| <b>GBR24</b>       | If a 'Sampling unit size' (sampUnitSize) is reported, then a 'Sampling unit size unit' (sampUnitSizeUnit) must be reported                                                               | sampUnitSizeUnit is missing, though sampUnitSize is reported                                          | Error         | Active |         |
| <b>GBR25</b>       | If a 'Sample taken size' (sampSize) is reported, then a 'Sample taken size unit' (sampSizeUnit) must be reported                                                                         | sampSizeUnit is missing, though sampSize is reported                                                  | Error         | Active |         |
| <b>GBR26</b>       | If a 'Sample analysed portion size' (anPortSize) is reported, then a 'Sample analysed portion size unit' (anPortSizeUnit) must be reported                                               | anPortSizeUnit is missing, though anPortSize is reported                                              | Error         | Active |         |
| <b>GBR27</b>       | If the value reported in 'Type of result' (resType) is different from 'Qualitative Value (Binary)' (BIN) (i.e. not a qualitative value), then a 'Result unit' (resUnit) must be reported | resUnit is missing, though resType is not 'Qualitative Value (Binary)' (BIN)                          | Error         | Active |         |



| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Error/Warning Message                                                                                                                                                                           | Type of error | Status | Comment |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
| <b>GBR28</b>       | If the value reported in 'Type of result' (resType) is 'Qualitative Value (Binary)' (BIN) (i.e. a qualitative value), then a 'Result qualitative value' (resQualValue) must be reported                                                                                                                                                                                                                                                                                                                                                                                                    | resQualValue is missing, though resType is 'Qualitative Value (Binary)' (BIN)                                                                                                                   | Error         | Active |         |
| <b>GBR29</b>       | If a value is reported in at least one of the following data elements: 'Result LOD' (resLOD), 'Result LOQ' (resLOQ), 'Result lower limit of the working range' (resLLWR), 'Result upper limit of the working range' (resULWR), 'CC alpha' (CCalpha), 'CC beta' (CCbeta), 'Result value' (resVal), 'Result value uncertainty' (resValUncert), 'Result value uncertainty Standard deviation' (resValUncertSD), 'Limit for the result evaluation (Low limit)' (evalLowLimit), 'Limit for the result evaluation (High limit)' (evalHighLimit), then a 'Result unit' (resUnit) must be reported | resUnit is missing, though at least one numeric data element (resLOD, resLOQ, resLLWR, resULWR, CCalpha, CCbeta, resVal, resValUncert, resValUncertSD, evalLowLimit, evalHighLimit) is reported | Error         | Active |         |
| <b>GBR30</b>       | If a value is reported in 'Limit for the result evaluation ' (evalLowLimit), then a 'Type of limit for the result evaluation' (evalLimitType) must be reported                                                                                                                                                                                                                                                                                                                                                                                                                             | evalLimitType is missing, though evalLowLimit is reported                                                                                                                                       | Error         | Active |         |
| <b>GBR32</b>       | The value reported in 'Limit for the result evaluation (High limit)' (evalHighLimit) must be greater than the value reported in 'Limit for the result evaluation (Low limit)' (evalLowLimit)                                                                                                                                                                                                                                                                                                                                                                                               | evalHighLimit is not greater than evalLowLimit                                                                                                                                                  | Error         | Active |         |
| <b>GBR33</b>       | If 'Result value' (resVal) is greater than 'Limit for the result evaluation' (evalLowLimit), then the value in 'Evaluation of the result' (evalCode)                                                                                                                                                                                                                                                                                                                                                                                                                                       | evalCode is 'below or equal the level of concern' (J002A), though resVal is greater than evalLowLimit                                                                                           | Error         | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                         | Error/Warning Message                                                                                                                                   | Type of error | Status | Comment |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
|                    | must be different from 'below or equal the level of concern' (J002A)                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                         |               |        |         |
| <b>GBR34</b>       | If 'Evaluation of the result' (evalCode) is either 'above the level of concern' (J003A) or 'Compliant due to measurement uncertainty' (J031A), and the value in 'Type of limit for the result evaluation' (evalLimitType) is different from 'Minimum Required Performance Limit (MRPL)' (W005A), then 'Result value' (resVal) must be greater than 'Limit for the result evaluation' (evalLowLimit) | resVal is lower than evalLowLimit, though evalCode is either 'above the level of concern' (J003A) or 'Compliant due to measurement uncertainty' (J031A) | Error         | Active |         |
| <b>GBR35</b>       | If 'Evaluation of the result' (evalCode) is 'below or equal to the level of concern' (J002A), then 'Result value' (resVal) must be less than or equal to 'Limit for the result evaluation' (evalLowLimit)                                                                                                                                                                                           | resVal is greater than evalLowLimit, though evalCode is 'below or equal to maximum permissible quantities' (J002A)                                      | Error         | Active |         |
| <b>GBR36</b>       | If the value in the data element 'Type of result' (resType) is 'Non-Detected Value (below LOD)' (LOD), then a value must be reported in the data element 'Result LOD' (resLOD)                                                                                                                                                                                                                      | resLOD is missing, though resType is 'Non-Detected Value (below LOD)' (LOD)                                                                             | Error         | Active |         |
| <b>GBR37</b>       | The value in 'Result LOD' (resLOD) must be less than or equal to the value in 'Result LOQ' (resLOQ)                                                                                                                                                                                                                                                                                                 | resLOD is not less than or equal to resLOQ                                                                                                              | Error         | Active |         |
| <b>GBR38</b>       | The value in 'Result LOD' (resLOD) must be greater than '0'                                                                                                                                                                                                                                                                                                                                         | resLOD is not greater than '0'                                                                                                                          | Error         | Active |         |
| <b>GBR39</b>       | If the value in the data element 'Type of result' (resType) is 'Non-Quantified Value (below LOQ)' (LOQ), then a value must be reported in the data element 'Result LOQ' (resLOQ), except if resInfo.notSummed equal 'Y'                                                                                                                                                                             | resLOQ is missing, though resType is 'Non-Quantified Value (below LOQ)' (LOQ) and resInfo.notSummed is not 'Y'                                          | Error         | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                                        | Error/Warning Message                                                         | Type of error | Status | Comment                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------|--------|--------------------------------------------------------------------------------------------|
| <b>GBR40</b>       | The value in 'Result LOQ' (resLOQ) must be greater than 0                                                                                                                                                                                                          | resLOQ is not greater than 0                                                  | Error         | Active |                                                                                            |
| <b>GBR41</b>       | If the value in the data element 'Type of result' (resType) is 'Value below CCalpha (below CCα)' (CCA) AND accredProc is not equal to V999A 'Not Validated', then a value must be reported in the data element 'CC alpha' (CCalpha) (unvalidated methods excluded) | CCalpha is missing, though resType is 'Value below CCalpha (below CCα)' (CCA) | Error         | Active | The qualification "AND accredProc is not equal to V999A 'Not Validated'" was added in 2019 |
| <b>GBR42</b>       | The value in 'CC alpha' (CCalpha) must be less than the value in 'CC beta' (CCbeta)                                                                                                                                                                                | WARNING: CCalpha is not less than CCbeta                                      | Warning       | Active |                                                                                            |
| <b>GBR43</b>       | The value in 'CC alpha' (CCalpha) must be greater than '0'                                                                                                                                                                                                         | CCalpha is not greater than '0'                                               | Error         | Active |                                                                                            |
| <b>GBR44</b>       | If the value in the data element 'Type of result' (resType) is 'Value below CCbeta (below CCβ)' (CCB) AND accredProc is not equal to V999A 'Not Validated', then a value must be reported in the data element 'CC beta' (CCbeta) (unvalidated methods excluded)    | CCbeta is missing, though resType is 'Value below CCbeta (below CCβ)' (CCB)   | Error         | Active | The qualification "AND accredProc is not equal to V999A 'Not Validated'" was added in 2019 |
| <b>GBR45</b>       | The value in 'CC beta' (CCbeta) must be greater than '0'                                                                                                                                                                                                           | CCbeta is not greater than '0'                                                | Error         | Active |                                                                                            |
| <b>GBR46</b>       | If the value in the data element 'Type of result' (resType) is 'Numerical Value' (VAL), then a value must be reported in the data element 'Result value' (resVal)                                                                                                  | resVal is missing, though resType is 'Numerical Value' (VAL)                  | Error         | Active |                                                                                            |
| <b>GBR47</b>       | If the value in the data element 'Type of result' (resType) is 'Non-Detected Value (below LOD)' (LOD), then the                                                                                                                                                    | resVal is reported, though resType is 'Non-Detected Value (below LOD)' (LOD)  | Error         | Active |                                                                                            |

| Business Rule Code | Description                                                                                                                                                                                                                 | Error/Warning Message                                                                                                             | Type of error | Status | Comment |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
|                    | data element 'Result value' (resVal) must be empty                                                                                                                                                                          |                                                                                                                                   |               |        |         |
| <b>GBR48</b>       | The value in 'Result value' (resVal) must be greater than '0'                                                                                                                                                               | resVal is not greater than '0'                                                                                                    | Error         | Active |         |
| <b>GBR49</b>       | The value in 'Result value recovery rate' (resValRec) must be greater than '0'                                                                                                                                              | resValRec is not greater than '0'                                                                                                 | Error         | Active |         |
| <b>GBR50</b>       | The value in 'Result value uncertainty Standard deviation' (resValUncertSD) must be greater than '0'                                                                                                                        | resValUncertSD is not greater than '0'                                                                                            | Error         | Active |         |
| <b>GBR51</b>       | The value in 'Result value uncertainty' (resValUncert) must be greater than '0'                                                                                                                                             | resValUncert is not greater than '0'                                                                                              | Error         | Active |         |
| <b>GBR53</b>       | The date of the slaughtering, reported in 'Day of slaughtering' (sampEventInfo.slaughterD), 'Month of slaughtering' (sampEventInfo.slaughterM), and 'Year of slaughtering' (sampEventInfo.slaughterY), must be a valid date | The combination of values in sampEventInfo.slaughterD, sampEventInfo.slaughterM, and sampEventInfo.slaughterY is not a valid date | Error         | Active |         |
| <b>GBR54</b>       | The date of the sampling, reported in 'Day of sampling' (sampD), 'Month of sampling' (sampM), and 'Year of sampling' (sampY), must be a valid date                                                                          | The combination of values in sampD, sampM, and sampY is not a valid date                                                          | Error         | Active |         |
| <b>GBR55</b>       | The date of the arrival in the laboratory, reported in 'Arrival Day' (sampInfo.arrivalD), 'Arrival Month' (sampInfo.arrivalM), and 'Arrival Year' (sampInfo.arrivalY), must be a valid date                                 | The combination of values in sampInfo.arrivalD, sampInfo.arrivalM, and sampInfo.arrivalY is not a valid date                      | Error         | Active |         |
| <b>GBR56</b>       | The date of the production, reported in 'Day of production' (sampMatInfo.prodD), 'Month of                                                                                                                                  | The combination of values in sampMatInfo.prodD, sampMatInfo.prodM, and                                                            | Error         | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                           | Error/Warning Message                                                                                                                                                     | Type of error | Status | Comment                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-------------------------------------------|
|                    | production' (sampMatInfo.prodM), and 'Year of production' (sampMatInfo.prodY), must be a valid date                                                                                                                                                   | sampMatInfo.prodY is not a valid date                                                                                                                                     |               |        |                                           |
| <b>GBR57</b>       | The date of the expiry, reported in 'Day of expiry' (sampMatInfo.expiryD), 'Month of expiry' (sampMatInfo.expiryM), and 'Year of expiry' (sampMatInfo.expiryY), must be a valid date                                                                  | The combination of values in sampMatInfo.expiryD, sampMatInfo.expiryM, and sampMatInfo.expiryY is not a valid date                                                        | Error         | Active |                                           |
| <b>GBR58</b>       | The date of the analysis, reported in 'Day of analysis' (analysisD), 'Month of analysis' (analysisM), and 'Year of analysis' (analysisY), must be a valid date                                                                                        | The combination of values in analysisD, analysisM, and analysisY is not a valid date                                                                                      | Error         | Active |                                           |
| <b>GBR60</b>       | The date of the isolation, reported in 'Isolation day' (isolInfo.isolD), 'Isolation month' (isolInfo.isolM), and 'Isolation year' (isolInfo.isolY), must be a valid date                                                                              | The combination of values in isolInfo.isolD, isolInfo.isolM, and isolInfo.isolY is not a valid date                                                                       | Error         | Active | This GBR is applicable to biological data |
| <b>GBR61</b>       | The reporting year, reported in 'Reporting year' (repYear), must be less than or equal to the current year                                                                                                                                            | The reporting year, reported in repYear, is greater than the current year                                                                                                 | Error         | Active |                                           |
| <b>GBR62</b>       | The date of the slaughtering, reported in 'Day of slaughtering' (sampEventInfo.slaughterD), 'Month of slaughtering' (sampEventInfo.slaughterM), and 'Year of slaughtering' (sampEventInfo.slaughterY), must be less than or equal to the current date | The date of the slaughtering, reported in sampEventInfo.slaughterD, sampEventInfo.slaughterM, and sampEventInfo.slaughterY, is not less than or equal to the current date | Error         | Active |                                           |
| <b>GBR63</b>       | The date of the sampling, reported in 'Day of sampling' (sampD), 'Month of sampling' (sampM), and 'Year of sampling' (sampY), must be less than or equal to the current date                                                                          | The date of the sampling, reported in sampD, sampM, and sampY, is not less than or equal to the current date                                                              | Error         | Active |                                           |

| Business Rule Code | Description                                                                                                                                                                                                              | Error/Warning Message                                                                                                                                             | Type of error | Status | Comment                                   |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-------------------------------------------|
| <b>GBR64</b>       | The date of the arrival in the laboratory, reported in 'Arrival Day' (sampInfo.arrivalD), 'Arrival Month' (sampInfo.arrivalM), and 'Arrival Year' (sampInfo.arrivalY), must be less than or equal to the current date    | The date of the arrival in the laboratory, reported in sampInfo.arrivalD, sampInfo.arrivalM, and sampInfo.arrivalY, is not less than or equal to the current date | Error         | Active |                                           |
| <b>GBR65</b>       | The date of the production, reported in 'Day of production' (sampMatInfo.prodD), 'Month of production' (sampMatInfo.prodM), and 'Year of production' (sampMatInfo.prodY), must be less than or equal to the current date | The date of the production, reported in sampMatInfo.prodD, sampMatInfo.prodM, and sampMatInfo.prodY, is not less than or equal to the current date                | Error         | Active |                                           |
| <b>GBR67</b>       | The date of the analysis, reported in 'Day of analysis' (analysisD), 'Month of analysis' (analysisM), and 'Year of analysis' (analysisY), must be less than or equal to the current date                                 | The date of the analysis, reported in analysisD, analysisM, and analysisY, is not less than or equal to the current date                                          | Error         | Active |                                           |
| <b>GBR69</b>       | The date of the isolation, reported in 'Isolation day' (isolInfo.isolD), 'Isolation month' (isolInfo.isolM), and 'Isolation year' (isolInfo.isolY), must be less than or equal to the current date                       | The date of the isolation, reported in isolInfo.isolD, isolInfo.isolM, and isolInfo.isolY, is not less than or equal to the current date                          | Error         | Active | This GBR is applicable to biological data |
| <b>GBR70</b>       | The 'Day of slaughtering' (sampEventInfo.slaughterD) must be between 1 and 31                                                                                                                                            | sampEventInfo.slaughterD is not between 1 and 31                                                                                                                  | Error         | Active |                                           |
| <b>GBR71</b>       | The 'Day of sampling' (sampD) must be between 1 and 31                                                                                                                                                                   | sampD is not between 1 and 31                                                                                                                                     | Error         | Active |                                           |
| <b>GBR72</b>       | The 'Arrival Day' (sampInfo.arrivalD) must be between 1 and 31                                                                                                                                                           | sampInfo.arrivalD is not between 1 and 31                                                                                                                         | Error         | Active |                                           |
| <b>GBR73</b>       | The 'Day of production' (sampMatInfo.prodD) must be between 1 and 31                                                                                                                                                     | sampMatInfo.prodD is not between 1 and 31                                                                                                                         | Error         | Active |                                           |

| Business Rule Code | Description                                                                                                                                       | Error/Warning Message                                                            | Type of error | Status | Comment                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------|--------|-------------------------------------------|
| <b>GBR74</b>       | The 'Day of expiry' (sampMatInfo.expiryD) must be between 1 and 31                                                                                | sampMatInfo.expiryD is not between 1 and 31                                      | Error         | Active |                                           |
| <b>GBR75</b>       | The 'Day of analysis' (analysisD) must be between 1 and 31                                                                                        | analysisD is not between 1 and 31                                                | Error         | Active |                                           |
| <b>GBR77</b>       | The 'Isolation day' (isolInfo.isolD) must be between 1 and 31                                                                                     | isolInfo.isolD is not between 1 and 31                                           | Error         | Active | This GBR is applicable to biological data |
| <b>GBR78</b>       | The 'Month of slaughtering' (sampEventInfo.slaughterM) must be between 1 and 12                                                                   | sampEventInfo.slaughterM is not between 1 and 12                                 | Error         | Active |                                           |
| <b>GBR79</b>       | The 'Month of sampling' (sampM) must be between 1 and 12                                                                                          | sampM is not between 1 and 12                                                    | Error         | Active |                                           |
| <b>GBR80</b>       | The 'Arrival Month' (sampInfo.arrivalM) must be between 1 and 12                                                                                  | sampInfo.arrivalM is not between 1 and 12                                        | Error         | Active |                                           |
| <b>GBR81</b>       | The 'Month of production' (sampMatInfo.prodM) must be between 1 and 12                                                                            | sampMatInfo.prodM is not between 1 and 12                                        | Error         | Active |                                           |
| <b>GBR82</b>       | The 'Month of expiry' (sampMatInfo.expiryM) must be between 1 and 12                                                                              | sampMatInfo.expiryM is not between 1 and 12                                      | Error         | Active |                                           |
| <b>GBR83</b>       | The 'Month of analysis' (analysisM) must be between 1 and 12                                                                                      | analysisM is not between 1 and 12                                                | Error         | Active |                                           |
| <b>GBR85</b>       | The 'Isolation month' (isolInfo.isolM) must be between 1 and 12                                                                                   | isolInfo.isolM is not between 1 and 12                                           | Error         | Active | This GBR is applicable to biological data |
| <b>GBR86</b>       | If the 'Day of slaughtering' (sampEventInfo.slaughterD) is reported, then the 'Month of slaughtering' (sampEventInfo.slaughterM) must be reported | sampEventInfo.slaughterM is missing, though sampEventInfo.slaughterD is reported | Error         | Active |                                           |



| Business Rule Code | Description                                                                                                                                                                                                             | Error/Warning Message                                                                                                                                                 | Type of error | Status | Comment                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-------------------------------------------|
| <b>GBR87</b>       | If the 'Day of sampling' (sampD) is reported, then the 'Month of sampling' (sampM) must be reported                                                                                                                     | sampM is missing, though sampD is reported                                                                                                                            | Error         | Active |                                           |
| <b>GBR88</b>       | If the 'Arrival Day' (sampInfo.arrivalD) is reported, then the 'Arrival Month' (sampInfo.arrivalM) must be reported                                                                                                     | sampInfo.arrivalM is missing, though sampInfo.arrivalD is reported                                                                                                    | Error         | Active |                                           |
| <b>GBR89</b>       | If the 'Day of production' (sampMatInfo.prodD) is reported, then the 'Month of production' (sampMatInfo.prodM) must be reported                                                                                         | sampMatInfo.prodM is missing, though sampMatInfo.prodD is reported                                                                                                    | Error         | Active |                                           |
| <b>GBR90</b>       | If the 'Day of expiry' (sampMatInfo.expiryD) is reported, then the 'Month of expiry' (sampMatInfo.expiryM) must be reported                                                                                             | sampMatInfo.expiryM is missing, though sampMatInfo.expiryD is reported                                                                                                | Error         | Active |                                           |
| <b>GBR91</b>       | If the 'Day of analysis' (analysisD) is reported, then the 'Month of analysis' (analysisM) must be reported                                                                                                             | analysisM is missing, though analysisD is reported                                                                                                                    | Error         | Active |                                           |
| <b>GBR92</b>       | If the 'Completion day of analysis' (sampAnInfo.compD) is reported, then the 'Completion month of analysis' (sampAnInfo.compM) must be reported                                                                         | sampAnInfo.compM is missing, though sampAnInfo.compD is reported                                                                                                      | Error         | Active |                                           |
| <b>GBR93</b>       | If the 'Isolation day' (isolInfo.isolD) is reported, then the 'Isolation month' (isolInfo.isolM) must be reported                                                                                                       | isolInfo.isolM is missing, though isolInfo.isolD is reported                                                                                                          | Error         | Active | This GBR is applicable to biological data |
| <b>GBR94</b>       | The date of the production, reported in 'Day of production' (sampMatInfo.prodD), 'Month of production' (sampMatInfo.prodM), and 'Year of production' (sampMatInfo.prodY), must be less than or equal to the date of the | The date of the production, reported in sampMatInfo.prodD, sampMatInfo.prodM, and sampMatInfo.prodY, is not less than or equal to the date of the expiry, reported in | Error         | Active |                                           |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                          | Error/Warning Message                                                                                                                                                                           | Type of error | Status | Comment                                   |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-------------------------------------------|
|                    | expiry, reported in 'Day of expiry' (sampMatInfo.expiryD), 'Month of expiry' (sampMatInfo.expiryM), and 'Year of expiry' (sampMatInfo.expiryY)                                                                                                                                                                                       | sampMatInfo.expiryD, sampMatInfo.expiryM, and sampMatInfo.expiryY                                                                                                                               |               |        |                                           |
| <b>GBR95</b>       | The date of the production, reported in 'Day of production' (sampMatInfo.prodD), 'Month of production' (sampMatInfo.prodM), and 'Year of production' (sampMatInfo.prodY), must be less than or equal to the date of the sampling, reported in 'Day of sampling' (sampD), 'Month of sampling' (sampM), and 'Year of sampling' (sampY) | The date of the production, reported in sampMatInfo.prodD, sampMatInfo.prodM, and sampMatInfo.prodY, is not less than or equal to the date of the sampling, reported in sampD, sampM, and sampY | Error         | Active |                                           |
| <b>GBR97</b>       | The date of the sampling, reported in 'Day of sampling' (sampD), 'Month of sampling' (sampM), and 'Year of sampling' (sampY), must be less than or equal to the date of the analysis, reported in 'Day of analysis' (analysisD), 'Month of analysis' (analysisM), and 'Year of analysis' (analysisY)                                 | The date of the sampling, reported in sampD, sampM, and sampY, is not less than or equal to the date of the analysis, reported in analysisD, analysisM, and analysisY                           | Error         | Active |                                           |
| <b>GBR99</b>       | The date of the sampling, reported in 'Day of sampling' (sampD), 'Month of sampling' (sampM), and 'Year of sampling' (sampY), must be less than or equal to the date of the isolation, reported in 'Isolation day' (isolInfo.isolD), 'Isolation month' (isolInfo.isolM), and 'Isolation year' (isolInfo.isolY)                       | The date of the sampling, reported in sampD, sampM, and sampY, is not less than or equal to the date of the isolation, reported in isolInfo.isolD, isolInfo.isolM, and isolInfo.isolY           | Error         | Active | This GBR is applicable to biological data |
| <b>GBR100</b>      | The date of the slaughtering, reported in 'Day of slaughtering' (sampEventInfo.slaughterD), 'Month                                                                                                                                                                                                                                   | The date of the slaughtering, reported in sampEventInfo.slaughterD,                                                                                                                             | Error         | Active |                                           |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Error/Warning Message                                                                                                                                       | Type of error | Status | Comment |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
|                    | of slaughtering' (sampEventInfo.slaughterM), must be less than or equal to the date of the sampling, reported in 'Day of sampling' (sampD), 'Month of sampling' (sampM), and 'Year of sampling' (sampY)                                                                                                                                                                                                                                                                                                        | sampEventInfo.slaughterM, and sampEventInfo.slaughterY, is not less than or equal to the date of the sampling, reported in sampD, sampM, and sampY          |               |        |         |
| <b>GBR101</b>      | If the value in the data element 'Coded description of the parameter' (paramCode.base) is different from 'Not in list' (RF-XXXX-XXX-XXX), then the combination of values in the data elements 'Coded description of the parameter' (paramCode.base), 'Sample taken identification code' (sampId), 'Sample analysed identification code' (sampAnId), and 'Sample analysed portion sequence' (anPortSeq) must be unique, i.e. only one result per parameter for each sample analysed portion per sample analysed | More than one paramCode per sample analysed portion for one sample analysed is reported, though paramCode is different from 'Not in list' (RF-XXXX-XXX-XXX) | Error         | Active |         |
| <b>GBR102</b>      | The value in the data element 'Result value' (resVal) must be greater than or equal to the value in the data element 'Result LOD' (resLOD)                                                                                                                                                                                                                                                                                                                                                                     | resVal is not greater than or equal to resLOD                                                                                                               | Error         | Active |         |
| <b>GBR103</b>      | If the value in 'Coded description of the analysed matrix' (anMatCode) is reported, then the value in sampAnId should be reported                                                                                                                                                                                                                                                                                                                                                                              | sampAnId is missing, though anMatCode is reported                                                                                                           | Error         | Active |         |

Table 10: Full list of chemical monitoring business rules with troubleshooting tips

| Business Rule Code       | Description                                                                                                                                                                                | Error Message                                                                                                                                                       | Type of Error  | Status | Comment                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DOMAIN VALIDATION</b> | The record is flagged according to the 'Programme legal reference' (progLegalRef) and the 'Coded description of the parameter' (paramCode.base)                                            | NOTE: The record is flagged according to the 'Programme legal reference' (progLegalRef) and the 'Coded description of the parameter' (paramCode.base)               | Not Applicable | Active | This rule appears in the downloadable set of business rules but does not generate any messages. It only invokes EFSA processes to flag the incoming data and adaptation for Data Provider use could be requested |
| <b>CHEMON03</b>          | A value in the data element 'Analytical method code' (anMethCode.base) must be reported                                                                                                    | anMethCode.base is missing, though a value should be reported                                                                                                       | Error          | Active |                                                                                                                                                                                                                  |
| <b>CHEMON04</b>          | The value in the data element 'Result value recovery' (resValRec) should be greater than or equal to 1 (for 85%, the value 85 should be reported)                                          | WARNING: resValRec is less than 1. Please check whether the resValRec is correctly reported (e.g. for 85%, the value 85 should be reported)                         | Warning        | Active |                                                                                                                                                                                                                  |
| <b>CHEMON05</b>          | The value in the data element 'Result value recovery' (resValRec) should be between 50 and 150                                                                                             | WARNING: resValRec is not between 50 and 150                                                                                                                        | Warning        | Active |                                                                                                                                                                                                                  |
| <b>CHEMON06</b>          | The value in the data element 'Percentage of fat in the original sample' (exprResPerc.fatPerc) should be greater than or equal to 1 (e.g. for 85%, the value 85 should be reported)        | WARNING: exprResPerc.fatPerc is less than 1. Please check whether the exprResPerc.fatPerc is correctly reported (e.g. for 85%, the value 85 should be reported)     | Warning        | Active |                                                                                                                                                                                                                  |
| <b>CHEMON07</b>          | The value in the data element 'Percentage of moisture in the original sample' (exprResPerc.moistPerc) should be greater than or equal to 1 (e.g. for 85%, the value 85 should be reported) | WARNING: exprResPerc.moistPerc is less than 1. Please check whether the exprResPerc.moistPerc is correctly reported (e.g. for 85%, the value 85 should be reported) | Warning        | Active |                                                                                                                                                                                                                  |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                 | Error Message                                                                                                                                                                                                         | Type of Error | Status | Comment                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-------------------------------------------------------------------------------------------|
| <b>CHEMON08</b>    | If the value in the data element 'Result value' (resVal) is equal to the value in the data element 'Result LOQ' (resLOQ), then the value in the data element 'Type of result' (resType) must be equal to 'Numerical Value' (Val)                                                                                                            | resLOQ is equal to resVal then resType must be equal to VAL                                                                                                                                                           | Error         | Active | Numeric results which are equal to the LOQ reported should have a value in resType of VAL |
| <b>CHEMON09</b>    | If the value in the data element 'Parameter code' (paramCode) is a dioxin or dioxin-like PCB, then the full set of the 29 dioxins and dioxin-like PCBs (17 PCDD/Fs, 12 dl-PCBs) should be reported (as listed in the Appendix to Annex I of the Commission Regulation (EU) 2023/915)                                                        | WARNING: The full set of the 29 dioxins and dioxin-like PCBs (17 PCDD/Fs, 12 dl-PCBs) is not reported, though mandatory, as listed in the Appendix to Annex I of the Commission Regulation (EU) 2023/915              | Warning       | Active |                                                                                           |
| <b>CHEMON10</b>    | If the value in the data element 'Parameter code' (paramCode) is a non-dioxin-like PCB, then the six indicator congeners (PCB 28, PCB 52, PCB 101, PCB 138, PCB 153, and PCB 180) should be reported (as listed in the Appendix to Annex I of the Commission Regulation (EU) 2023/915)                                                      | WARNING: The six indicator congeners (PCB 28, PCB 52, PCB 101, PCB 138, PCB 153, and PCB 180) are not all reported, though mandatory, as listed in the Appendix to Annex I of the Commission Regulation (EU) 2023/915 | Warning       | Active |                                                                                           |
| <b>CHEMON11</b>    | If the value in the data element 'Parameter code' (paramCode.base) has as ancestor 'Mineral oils' (RF-00000396-ORG), or 'Mycotoxins' (RF-00000132-TOX), then a value in the data element 'Percentage of moisture in the original sample' (exprResPerc.moistPerc) should be reported (regardless of whether the result value is expressed in | WARNING: exprResPerc.moistPerc is missing, though recommended when paramCode is a mineral oil or a mycotoxin                                                                                                          | Warning       | Active |                                                                                           |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                   | Error Message                                                                                                                                                                                                                                                               | Type of Error | Status | Comment |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
|                    | whole weight, fat weight or dry matter)                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                             |               |        |         |
| <b>CHEMON12</b>    | If the value in the data element 'Parameter code' (paramCode.base) is 'Acrylamide' (RF-00000410-ORG), then the value in the data element 'Legislative classes' (sampMatCode.legis) must be reported and must contain the additional product classification (categories and sub-categories) based on Commission Regulation (EU) 2017/2158 and Commission Recommendation (EU) 2019/1888 on the monitoring of acrylamide in food | sampMatCode.legis is not reported or does not contain a specific product code, though paramCode is acrylamide (it is mandatory to provide additional product classification based on Commission Recommendation 2019/1888 and Commission Regulation 2017/2158 on acrylamide) | Error         | Active |         |
| <b>CHEMON14</b>    | If the value in the data element 'Parameter code' (paramCode.base) has as ancestor 'Bisphenol compounds' (RF-00001240-PAR), then a value in the data element 'Packaging material' (sampMatCode.packmat) must be reported                                                                                                                                                                                                      | sampMatCode.packmat is missing, though mandatory when paramCode is a bisphenol compound                                                                                                                                                                                     | Error         | Active |         |
| <b>CHEMON15</b>    | If the value in the data element 'Parameter code' (paramCode.base) has as ancestor 'Polycyclic aromatic hydrocarbons' (RF-00000040-ORG), then a value in the data element 'Packaging material' (sampMatCode.packmat) should be reported                                                                                                                                                                                       | WARNING: sampMatCode.packmat is missing, though recommended when paramCode is a PAH                                                                                                                                                                                         | Warning       | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                               | Error Message                                                                                                                                                                                             | Type of Error | Status | Comment                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-----------------------------------------------------------------------------------------------------------------------|
| <b>CHEMON17</b>    | If the value in the data element 'Parameter code' (paramCode.base) has as ancestor 'Mycotoxins' (RF-00000132-TOX), then the value in the data element 'Production method' (sampMatCode.prod) should be reported                                                                                                           | WARNING:<br>sampMatCode.prod is missing though it is recommended to report whether the sample was obtained from the produce of traditional (non-organic) or organic farming when paramCode is a mycotoxin | Warning       | Active |                                                                                                                       |
| <b>CHEMON18</b>    | If the value in the data element 'Parameter code' (paramCode.base) has as ancestor 'Arsenic and derivatives' (RF-00000127-CHE), and data are reported on rice (as grains for human consumption), then the value in the data element 'Process' (sampMatCode.process) should be reported                                    | WARNING:<br>sampMatCode.process is not reported, though it is recommended to specify at least the codes for 'processed' or 'unprocessed' when reporting data on rice and paramCode is arsenic             | Warning       | Active |                                                                                                                       |
| <b>CHEMON19</b>    | If the value in the data element 'Parameter code' (paramCode.base) is equal to 'Chlorates' (RF-00000015-CHE), or 'Perchlorate' (RF-00001336-PAR) or 'RF-1078-001-PPP' Quaternary Ammonium Compounds (QACs) or any children thereof, then the value in the data element 'Process' (sampMatCode.process) should be reported | WARNING:<br>sampMatCode.process is not reported, though it is recommended to specify at least the code for processed or unprocessed when paramCode is chlorate, perchlorate or QACs                       | Warning       | Active | Changed introduced starting from the 2022 ChemMon data collection so that it returns a Warning and no longer an Error |



| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Error Message                                                                                                                                               | Type of Error | Status | Comment |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
| <b>CHEMON20</b>    | If the value in the data element 'Coded description of the matrix sampled' (sampMatCode.base) has as ancestor 'Processed or preserved seafood' (A0BZ4), or 'Fish, other aquatic animals and products derived thereof (feed)' (A0BNJ) and the value in the data element 'Parameter code' (paramCode.base) has as ancestor 'Brominated flame retardants' (RF-00000074-ORG), or 'Dioxins and PCBs' (RF-00000114-ORG), or 'Mercury and derivatives' (RF-00000169-CHE), then a value in 'Area of origin for fisheries or aquaculture activities code' (origFishAreaCode) should be reported | WARNING: origFishAreaCode is missing, though recommended when data are reported on fish, and paramCode is BFR, dioxins and PCBs, or mercury and derivatives | Warning       | Active |         |
| <b>CHEMON21</b>    | If the value in the data element 'Parameter code' (paramCode.base) has as ancestor 'Brominated flame retardants' (RF-00000074-ORG), or 'Dioxins and PCBs' (RF-00000114-ORG), or '3-MCPDs' (RF-00000376-ORG), then a value in the data element 'Percentage of fat in the original sample' (exprResPerc.fatPerc) should be reported (regardless of whether the result value is expressed on whole weight, fat weight or dry matter)                                                                                                                                                      | WARNING: exprResPerc.fatPerc is missing, though recommended when reporting data on BFR, dioxins and PCBs, or 3-MCPDs                                        | Warning       | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                              | Error Message                                                                                                      | Type of Error | Status | Comment |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
| <b>CHEMON22</b>    | If the value in 'Link to Original Sample' (sampInfo.origSampId) is reported, i.e. a follow-up sample, then the value in 'Sampling Strategy' (sampStrategy) should be 'suspect sampling' (ST30A)                                                                                                                          | WARNING: sampStrategy is not suspect sampling, though sampInfo.origSampId is reported                              | Warning       | Active |         |
| <b>CHEMON23</b>    | The value in the data element 'Analytical Method Type' (anMethType) must be equal to 'Screening' (AT06A), or 'Confirmation' (AT08A)                                                                                                                                                                                      | anMethType is not screening or confirmation                                                                        | Error         | Active |         |
| <b>CHEMON24</b>    | The value in the data element 'Result qualitative value' (resQualValue) must be equal to 'negative/absent' (NEG), because neither positive screening results nor qualitative confirmation results should be reported, unless the reported 'Parameter code' (paramCode) is equal to 'MOAHs 3 to 7 ring' (RF-00011484-PAR) | resQualValue is different from negative/absent                                                                     | Error         | Active |         |
| <b>CHEMON26</b>    | If the value in the data element 'Action Taken' (actTakenCode) is equal to 'Follow-up investigation' (I), then a value in the data element 'Conclusion of follow-up investigation' (evalInfo.conclusion) should be reported                                                                                              | WARNING: evalInfo.conclusion is missing, though actTakenCode is follow-up investigation                            | Warning       | Active |         |
| <b>CHEMON27</b>    | If the result is reported for VMPP or pesticides (is_vet=1 or is_pest=1), then the value in 'Coded description of the matrix of the sample taken' (sampMatCode) should be equal to the value in 'Coded description of the analysed matrix' (anMatCode)                                                                   | WARNING: anMatCode is different from sampMatCode, though the matrix sampled should be equal to the analysed matrix | Warning       | Active |         |

[www.efsa.europa.eu/publications](http://www.efsa.europa.eu/publications)

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                | Error Message                                                                                   | Type of Error | Status | Comment |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------|--------|---------|
|                    | method type' (anMethType) is 'Screening' (AT06A), then a value in the data element 'CC beta' (CCbeta) must be reported                                                                                                                                                                                                                                                                                     |                                                                                                 |               |        |         |
| <b>CHEMON33</b>    | If the value in 'Type of results' (resType) is equal to 'Qualitative value (binary)' (BIN), then the value in the data element 'Analytical method type' (anMethType) should be equal to 'Screening' (AT06A)                                                                                                                                                                                                | resType is BIN but anMethType is not screening                                                  | Error         | Active |         |
| <b>CHEMON34</b>    | If the value in the data element 'Analytical method type' (anMethType) is 'Confirmation' (AT08A), then the value in 'Type of results' (resType) should not be equal to 'Qualitative Value (Binary)' (BIN), and the value in 'Result qualitative value' (resQualValue) should not be reported                                                                                                               | WARNING: resType is equal to BIN or resQualValue is reported, though anMethType is confirmation | Warning       | Active |         |
| <b>CHEMON35</b>    | The value in the data element 'Type of limit for the result evaluation' (evalLimitType) should be equal to 'Maximum Residue Level (MRL)' (W002A), or 'Minimum Required Performance Limit (MRPL)' (W005A), or 'Reference point of action (RPA)' (W006A), or 'Presence' (W012A), or 'Maximum Limit' (W001A), or 'Action level' (W007A), or 'Target Level (TL)' (W008A), or 'National or local limit' (W990A) | WARNING: evalLimitType is not in the list of recommended codes                                  | Warning       | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Error Message                                                                                                                                                                                                  | Type of Error | Status | Comment                                                                                                                                                                                                                                                 |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHEMON36</b>    | If the value in the data element 'Type of limit for the result evaluation' (evalLimitType) is equal to 'Maximum Residue Level (MRL)' (W002A), then the value in 'Evaluation of the result' (evalCode) should be equal to 'Below or equal to maximum permissible quantities (below or equal to level of concern)' (J002A), or 'Above the maximum permissible quantities (above level of concern)' (J003A), or 'Compliant due to measurement uncertainty' (J031A), or 'result not evaluated' (J029A) | WARNING: evalCode is not in the recommended list of codes when evalLimitType is MRL                                                                                                                            | Warning       | Active |                                                                                                                                                                                                                                                         |
| <b>CHEMON37</b>    | If the result is reported for contaminants (is_occ=1), food additives (is_add=1), or food flavourings (is_flav=1) and if the value in 'Evaluation of the result' (evalCode) is equal to 'Detected' (J041A), or 'Above the maximum permissible quantities' (J003A), then a value in the data element 'Action taken' (actTakenCode) should be reported                                                                                                                                               | WARNING: actTakenCode is missing, though recommended when evalCode is detected or greater than the maximum permissible quantities                                                                              | Warning       | Active |                                                                                                                                                                                                                                                         |
| <b>CHEMON39</b>    | If the result is reported for food additives (is_add=1), or food flavourings (is_flav=1), then the value in the data element 'Legislative classes' (sampMatCode.legis) must be used to report the additional product classification (food category code) based on                                                                                                                                                                                                                                  | sampMatCode.legis does not contain specific product code (food category code) based on Regulation (EC) No 1333/2008 on food additives and on Regulation (EC) No 1334/2008 on food flavourings, as last amended | Error         | Active | Starting in 2024, the facet [F33] Legislative classes has been implicitly assigned to most of the FoodEx2 base terms.<br>If the F33 is not present, it must be explicitly assigned by the Data Provider for the food additives and flavourings domains. |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                        | Error Message                                                                                                                                                                          | Type of Error | Status | Comment |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
|                    | Regulations (EC) No 1333/2008 and (EC) No 1334/2008 on food additives, and food flavourings, respectively, as last amended.                                                                                                                                                                                                        |                                                                                                                                                                                        |               |        |         |
| <b>CHEMON40</b>    | If the value in the data element 'Type of result' (resType) is 'Qualitative Value (Binary)' (BIN), or 'Non-Quantified Value (below LOQ)' (LOQ), or 'Non-Detected Value (less than LOD)' (LOD), then the data element 'Result value' (resVal) must be empty                                                                         | A numerical value for resVal is reported, though resType is BIN, LOQ or LOD. Please check the record                                                                                   | Error         | Active |         |
| <b>CHEMON41</b>    | If the value in the data element 'Result type' (resType) is equal to 'Numerical Value' (VAL), then the value in the data element 'Result LOQ' (resLOQ) should not be greater than the value in the data element 'Result value' (resVal) (if the result is a positive detection, the result value cannot be below the reported LOQ) | WARNING: the selected resType is VAL for a result that contains a value less than the reported LOQ                                                                                     | Warning       | Active |         |
| <b>CHEMON42</b>    | If the value in the data element 'Result value recovery corrected' (resValRecCorr) is equal to 'Yes' (Y), then a value in the data element 'Result value recovery' (resValRec) should be reported                                                                                                                                  | WARNING: resValRec is missing, though resValRecCorr is reported if the result is corrected for recovery the corrected value should be reported (mean recovery in the range of 70-120%) | Warning       | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                                                             | Error Message                                                                                  | Type of Error | Status | Comment                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHEMON43</b>    | If the result is reported for pesticides (is_pest=1), or VMPP (is_vet=1), then the value in the data element 'Sampling year' (sampY) should be lower than or equal to the submission year minus 1                                                                                       | sampYear should be lower than or equal to the submission year minus 1                          | Error         | Active | <p>The 'sampY' should typically be less than the year in which a sample is transmitted (e.g. samples transmitted in 2022 should typically have been sampled in 2021).</p> <p>This BR only applies to results related to residue domains for which Annual Reports are to be prepared by EFSA: pesticide residues and VMPP.</p> <p>Change introduced starting with the 2022 ChemMon data collection.</p> |
| <b>CHEMON43_b</b>  | If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1) then the value in the data element 'Sampling year' (sampY) should be lower than or equal to the submission year minus 1                                                                         | sampYear should be lower than or equal to the submission year minus 1                          | Warning       | Active | <p>It will become an error in 2027.</p> <p>CHEMON43 and CHEMON43_b will be merged in 2027.</p>                                                                                                                                                                                                                                                                                                         |
| <b>CHEMON44</b>    | A value in the data element 'Result LOQ' (resLOQ) must be reported, unless an unvalidated method is used, or notSummed=Y or the result is reported for VMPP (is_vet=1), or the resType is BIN                                                                                           | resLOQ is missing, though a value should be reported                                           | Error         | Active |                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>CHEMON45</b>    | A value in at least one of the following data elements must be reported: 'Result LOQ' (resLOQ) or 'Result LOD' (resLOD) or 'CC beta' (CCbeta) or 'CC alpha' (CCalpha) if 'Analytical method' is not an unvalidated methods ('Accreditation procedure' not equal to 'Not validated') and | One of resLOQ, resLOD, CCbeta or CCalpha must be reported unless an unvalidated method is used | Error         | Active |                                                                                                                                                                                                                                                                                                                                                                                                        |



| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                       | Error Message                                                                                                                                                                            | Type of Error | Status   | Comment                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|-----------------------------------------------------------------------------|
|                    | resInfo.NotSummed is not equal to 'Y'                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                          |               |          |                                                                             |
| <b>CHEMON46</b>    | If the value in the data element 'Evaluation of the result' (evalCode) is 'above the level of concern' (J003A), or 'Compliant due to measurement uncertainty' (J031A) and evalLimitType is not equal to 'Presence' (W012A), then the value in 'Type of result' (resType) must be equal to 'VAL'                                                                                                                   | The selected resType code is different from VAL, though evalCode is 'greater than maximum permissible quantities' or 'compliant due to measurement uncertainty'. Please check the record | Error         | Active   |                                                                             |
| CHEMON47           | If the value in the data element 'Result type' (resType) is equal to 'Non-Quantified Value (below LOQ)' (LOQ), then the value in the data element 'Result value' (resVal) should be empty.                                                                                                                                                                                                                        | WARNING: when the selected resType is LOQ, then resVal should be empty                                                                                                                   | Warning       | Inactive | Inactivation of business rule due to the BR CHEMON40 which is more complete |
| <b>CHEMON48</b>    | If the value in the data element 'Type of parameter' (paramType) is different from 'Part of a sum' (P002A) and the value in the data element 'Result value' (resVal) is greater than or equal to the value in the data element 'Limit for the result evaluation' (evalLowLimit), then the value in the data element 'Evaluation of the result' (evalCode) should be different from 'Result not evaluated' (J029A) | WARNING: when resVal greater than or equal to evalLowLimit, then the evalCode is expected to be reported                                                                                 | Warning       | Active   |                                                                             |

[www.efsa.europa.eu/publications](http://www.efsa.europa.eu/publications)

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                             | Error Message                                                                                                                                                         | Type of Error | Status   | Comment                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|-------------------------------------------------------------------------------------------------------------|
| <b>CHEMON52</b>    | If the value in 'Programme legal reference' (progLegalRef) contains 'Regulation (EC) No 396/2005 (amended)' (N027A), the value in 'Programme type' (progType) can only be equal to 'Official (National) programme' (K005A), or 'Official (EU) programme' (K009A), or 'Official (National) programme for Third Country Import' (K038A) or 'Official (National and EU) programme' (K018A) | The reported progType is not one of the values allowed for pesticides                                                                                                 | Error         | Active   | Description updated in 2025.                                                                                |
| CHEMON53           | If 'Programme legal reference' (progLegalRef) is 'Regulation (EC) No396/2005 (amended)' (N027A) and the value in 'Programme type' (progType) is equal to 'EU increased control programme on imported food' (K019A), then the value in 'Sampling strategy' (sampStrategy) can only be equal to 'Suspect sampling' (ST30A)                                                                | The reported combination of codes for progType and progLegalRef is not valid. Please check these variables.                                                           | Error         | Inactive | Due to the update of table 2 in 2025, the specific rule is not active any longer                            |
| <b>CHEMON54</b>    | If the value in 'Programme legal reference' (progLegalRef) is 'Commission Implementing Regulation (EU) No 2019/1793' (N317A), then the value in 'Programme type' (progType) is equal to 'EU increased control programme on imported food' (K019A) and the value in 'Sampling strategy' (sampStrategy) can only be equal to 'Suspect sampling' (ST30A).                                  | sampStrategy is not 'suspect', or progType is not 'EU increased control programme on imported food', though only those values should be reported under Reg. 2019/1793 | Error         | Active   | Description updated in 2025.<br><br>Regulation (EU) 2019/1793 repeals the obsolete Regulation (EC) 669/2009 |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                               | Error Message                                                                                                                              | Type of Error | Status | Comment                      |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|------------------------------|
| <b>CHEMON55</b>    | If the value in the data element 'Programme legal reference' (progLegalRef) is 'Samples of food products falling under Directive 2006/125/EC (N028A) or is 'Samples of food products falling under Regulation (EU) 2016/127' (N318A), then the value in the data element 'Coded description of the matrix of the sample taken' (sampMatCode.base) should have as parent term 'Food products for young population' (A03PV) | The reported sampMatCode does not refer to baby food, although the progLegalRef applies to this type of food. Please check these variables | Error         | Active | Description updated in 2025. |
| <b>CHEMON56</b>    | If the result is reported for pesticides (is_pest=1), then the value in the data element 'Expression of result' (exprResType) can only be equal to 'Whole weight' (B001A), or 'Fat basis' (B003A), or 'Reconstituted product' (B007A)                                                                                                                                                                                     | The reported exprResType is not one of the values allowed for pesticides                                                                   | Error         | Active |                              |
| <b>CHEMON57</b>    | The combination of values in the data elements 'Parameter code' (paramCode) and 'Sample taken identification code' (sampId) must be unique, except for analysis on aflatoxins                                                                                                                                                                                                                                             | The combination of values in paramCode and sampId is not unique for the same sample. Please correct this record                            | Error         | Active |                              |
| <b>CHEMON58</b>    | If the result is reported for pesticides (is_pest=1) or for VMPP (is_vet=1), then the 'Country of sampling' (sampCountry) has to be equal to the country of the reporting organisation                                                                                                                                                                                                                                    | sampCountry does not match with the country of the reporting organisation                                                                  | Error         | Active |                              |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                     | Error Message                                                                                                    | Type of Error | Status | Comment |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------|--------|---------|
| <b>CHEMON59</b>    | If the result is reported for pesticides, then the value in 'Type of limit for the result evaluation' (evalLimitType) can only be 'Maximum Residue Level (MRL)' (W002A), or 'National or local limit' (W990A), or missing                                                                                       | the reported evalLimitType is not one of the values allowed for pesticides                                       | Error         | Active |         |
| <b>CHEMON60</b>    | If the result is reported for pesticides (is_pest=1), then the value in evalCode can only be 'less than or equal to maximum permissible quantities' (J002A), or 'greater than maximum permissible quantities' (J003A), or 'Result not evaluated' (J029A), or 'Compliant due to measurement uncertainty' (J031A) | The reported evalCode is not one of the values allowed for pesticides                                            | Error         | Active |         |
| <b>CHEMON61</b>    | If the result is reported for pesticides (is_pest=1), then the value in 'Type of result' (resType) can only be equal to 'Non-Detected Value (less than LOD)' (LOD), 'Non-Quantified Value (less than LOQ)' (LOQ), 'Qualitative Value (Binary)' (BIN), or 'Numerical Value' (VAL)                                | The reported resType is not one of the values allowed for pesticides                                             | Error         | Active |         |
| <b>CHEMON62</b>    | If resInfo.notSummed is equal to 'Yes' (Y), then the result should be reported for multicomponents/sum and at least the LOQ of one of the components, should be reported                                                                                                                                        | resInfo.notSummed is equal to yes, though the LOQ result is not reported for at least one of the multicomponents | Error         | Active |         |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                             | Error Message                                                                                                                                                                         | Type of Error | Status | Comment                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-------------------------------------------------------------------------------------------------------------------------|
| <b>CHEMON63</b>    | If the coded sample matrix sampMatCode is under the high-level code A03PZ (i.e. the sample is a baby food), the progLegalRef cannot be N371A since these samples are excluded from the VMPR Annual Report as defined in Regulation (EU) 2022/1646 and Regulation (EU) 2022/1644                                                                                                                                                                         | progLegalRef is Regulation (EU) 2022/1646, though the reported sampMatCode is a baby food. Please check these variables since samples of baby food are excluded from VMPR legislation | Error         | Active | Regulation (EU) 2022/1646 and Regulation (EU) 2022/1644 repeal the obsolete Council Directive (EC) No 23/1996 (amended) |
| <b>CHEMON64</b>    | If the sample reported is taken under Regulation (EU) No 2019/1793 (N317A), then the values in 'Country of origin' (origCountry) and in 'Coded description of the matrix of the sample taken' (sampMatCode) can only be those listed in the Regulation                                                                                                                                                                                                  | The selected origCountry and sampMatCode is not a valid code when the sample is taken under Regulation No 2019/1793. Please correct this record                                       | Error         | Active | Description updated in 2025.                                                                                            |
| <b>CHEMON65</b>    | The value in 'Result assessment' (evalInfo.resAsses) can only be equal to 'Compliant' (J037A) or 'Non-compliant' (J038A)                                                                                                                                                                                                                                                                                                                                | The value in evalInfo.resAsses is not equal to compliant or non-compliant, though only these values are allowed                                                                       | Error         | Active |                                                                                                                         |
| <b>CHEMON66_a</b>  | If the value in 'Result assessment' (evalInfo.resAsses) is equal to 'Compliant' (J037A) and the value in 'Evaluation of the result' (evalCode) is equal to 'Above the level of concern' (J003A), or the value in 'Evaluation of the result' (evalCode) is equal to 'Detected' (J041A) and the value in (evalLimitType) is equal to 'Presence' (W012A), then a value in 'Conclusion of follow-up investigation' (evalInfo.conclusion) should be reported | WARNING: evalInfo.conclusion is missing, though the result evaluation reported is an exceedance and the value reported in evalInfo.resAsses is compliant                              | Warning       | Active |                                                                                                                         |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Error Message                                                                                                                                                                                                             | Type of Error | Status | Comment                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|----------------------------------------------------------------------------------------------------------------|
| <b>CHEMON66_b</b>  | If the value in 'Result assessment' (evalInfo.resAsses) is equal to 'Non-compliant' (J038A) and the value in 'Evaluation of the result' (evalCode) is not equal to 'Above the level of concern' (J003A), and the value in 'Evaluation of the result' (evalCode) is not equal to 'Detected' (J041A) and the value in (evalLimitType) is not equal to 'Presence' (W012A), then a value in 'Conclusion of follow-up investigation' (evalInfo.conclusion) should be reported | WARNING:<br>evalInfo.conclusion is missing, though the result evaluation reported is a non-exceedance and the value reported in evalInfo.resAsses is non-compliant                                                        | Warning       | Active |                                                                                                                |
| <b>CHEMON67</b>    | The values in 'Sample taken assessment' (evalInfo.sampTkAsses) and 'Sampling event assessment' (evalInfo.sampEventAsses) should not be reported                                                                                                                                                                                                                                                                                                                          | WARNING:<br>evalInfo.sampTkAsses and/or evalInfo.sampEventAsses are reported. Please note that this information will not be used for further data analysis in the frame of the preparation of pesticides and VMPP reports | Warning       | Active |                                                                                                                |
| <b>CHEMON68</b>    | The domain of the progLegalRef should match with the paramCode domain                                                                                                                                                                                                                                                                                                                                                                                                    | The selected paramCode belongs to a domain different than the progLegalRef domain. Please check this record;                                                                                                              | Error         | Active | New rule was introduced in 2021. It became an Error in 2022                                                    |
| <b>CHEMON69</b>    | If the result is reported for contaminants (is_occ=1) in feed, and the 'Expression of result type' (exprResType) is equal to 'Whole weight' (B001A), then the 'Percentage of moisture' (moistPerc) must be reported                                                                                                                                                                                                                                                      | moistPerc is missing, though the result reported for feed is expressed on whole weight                                                                                                                                    | Error         | Active | New rule was introduced in 2021 only for the contaminants domain and feed matrices. It became an Error in 2022 |



| Business Rule Code | Description                                                                                                                                                                                                                                                                                 | Error Message                                                                                                                           | Type of Error | Status | Comment                                                                                              |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|------------------------------------------------------------------------------------------------------|
| <b>CHEMON70</b>    | If the result is reported for pesticides (is_pest=1) in egg or milk, then the 'Expression of result' (exprResType) 'whole weight' (code B001A) must be reported                                                                                                                             | exprResType is missing, though the MRL is compliant with a whole weight commodity                                                       | Error         | Active | New rule was introduced in 2021 only for pesticides. It became an Error in 2025                      |
| <b>CHEMON71</b>    | If the result is reported for contaminants (is_occ=1) in feed, then the 'Expression of result type' (exprResType) must be reported                                                                                                                                                          | exprResType is missing, though the analysis has been reported in feed                                                                   | Error         | Active | New rule was introduced in 2021. It became an Error in 2022. It only applies to contaminants in feed |
| <b>CHEMON72</b>    | If the result (resType=VAL) is reported for pesticides, then the 'Result value uncertainty' (resValUncert) should be reported                                                                                                                                                               | WARNING: resValUncert is missing, though the analysis has been reported for pesticides                                                  | Warning       | Active | New rule was introduced in 2021. It will continue to be warning in 2025.                             |
| <b>CHEMON73</b>    | When reporting VMPR data on feed and water given to animals, the 'Target consumer' facet (F23) should be reported                                                                                                                                                                           | WARNING: facet F23 – target consumer is missing, though data are reported on feed or water given to animals                             | Warning       | Active |                                                                                                      |
| <b>CHEMON76</b>    | When reporting VMPR results for two or more samples (sampId) with the same sampEventId the F01 facet code of the sampMatCode shall be the same for each sample reported                                                                                                                     | The F01 facet code of the sampMatCode is not the same for each of the samples (sampId) reported for the same sample event (sampEventId) | Error         | Active | New rule was introduced in 2022, which returns an Error message. It only applies to VMPR             |
| <b>CHEMON77</b>    | If the analytical results refer to pooled samples, with sampling method 'sampMethod' code N002A or N031A, the data element 'sampUnitSizeUnit' must be reported with code G005A ('Unit') and the element 'sampUnitSize' must be returned with the actual number of the single samples pooled | sampUnitSizeUnit or sampUnitSize is missing, though a pooled sample has been reported                                                   | Error         | Active | New rule was introduced in 2022. It becomes an Error in 2025.                                        |

| Business Rule Code | Description                                                                                                                                                                                                                                                                         | Error Message                                                                                                        | Type of Error | Status   | Comment                                                                                                                                                                                                                                |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHEMON78</b>    | If the value in the data element 'Type of result' (resType) is 'Qualitative Value (Binary)' (BIN), then the data element resLOQ, must be empty                                                                                                                                      | A numerical value for resLOQ is reported, though resType is qualitative value (binary, BIN). Please check the record | Error         | Active   | New rule was introduced in 2022, which returns a Warning message. It became an Error in 2023.                                                                                                                                          |
| <b>CHEMON79</b>    | If the result is reported for contaminants (is_occ=1), food additives (is_add=1), or food flavourings (is_flav=1), then the value in 'Analytical Method' (anMethCode) should be different from 'Unknown' (F500A) or 'Unspecified' (F598A) or 'Classification not possible' (F001A). | anMethCode must be different from 'Classification not possible', 'Unknown', or 'Unspecified'                         | Error         | Active   | New rule was introduced in 2022 for contaminants, in 2023 for the food additives, and in 2025 for the food flavourings. It becomes an Error in 2026 for all domains.<br><br>CHEMON79_a, CHEMON79_b, and CHEMON79_c are merged in 2026. |
| <b>CHEMON80</b>    | If the result is reported for contaminants (is_occ=1), then a value in the data element 'Result value corrected for recovery' (resValRecCorr) should be reported                                                                                                                    | WARNING: resValRecCorr is missing, though a value should be reported                                                 | Warning       | Active   | New rule was introduced in 2022 for contaminants, which returns a Warning message                                                                                                                                                      |
| <b>CHEMON81</b>    | If the value reported in resVal is less than the value reported in resLOQ, then the value in resType should be equal to 'Below or equal to LOQ' (LOQ)                                                                                                                               | resType is not 'LOQ', though resVal is less than resLOQ                                                              | Error         | Inactive | Due to the update of CHEMON40, the BR CHEMON81 is obsolete                                                                                                                                                                             |
| <b>CHEMON82</b>    | The value in resLOQ should be different from 99999 but also 999, 9999, 999999, 9999999                                                                                                                                                                                              | resLOQ cannot be equal to '99999'; please use the data element 'resInfo.notSummed'=Y to avoid storing a false LOQ    | Error         | Active   | New rule was introduced in 2022 to avoid storing a false LOQ, which returns an Error message                                                                                                                                           |
| <b>CHEMON83</b>    | If the code F10.A18PX on the sampMatCode is reported, then the facet F19 and the facet F18 should not be reported                                                                                                                                                                   | WARNING: packaging material or packaging format reported even though the code F10.A18PX is reported                  | Warning       | Active   | New rule was introduced in 2022.                                                                                                                                                                                                       |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                      | Error Message                                                                                                                                                                    | Type of Error | Status | Comment                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHEMON84_a</b>  | If the result is reported for food additives (is_add=1), or food flavourings (is_flav=1), then the 'Expression of result' (exprResType) must be reported                                                                                                                                                                                                                                         | exprResType is missing, though the analysis has been reported either for food additives or food flavourings                                                                      | Error         | Active | It becomes an Error for food flavourings in 2026.                                                                                                                                            |
| <b>CHEMON84_b</b>  | If the result is reported for chemical contaminants (is_occ=1) in food, then the 'Expression of result Type' (exprResType) is highly recommended to be reported                                                                                                                                                                                                                                  | exprResType is missing, though the analysis has been reported for contaminants in food                                                                                           | Warning       | Active | Business rule remains a Warning message for contaminants                                                                                                                                     |
| <b>CHEMON85</b>    | If the result is reported for VMPP (is_vet=1) or pesticides (is_pest=1) and if the value in 'Evaluation of the result' (evalCode) is equal to 'Detected' (J041A), or 'Above the maximum permissible quantities' (J003A) and results assessment is empty, or result assessment is equal to Non-compliant (J038A), then a value in the data element 'Action taken' (actTakenCode) must be reported | actTakenCode is missing, though mandatory for pesticides and VMPP when evalCode is detected or greater than maximum permissible quantities or evalInfo.resAsses is non-compliant | Error         | Active | New rule was introduced in 2024 for VMPP and Pesticides.<br><br>It became an Error for VMPP and Pesticides in 2024.                                                                          |
| <b>CHEMON86</b>    | If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1), then the physical-state facet of the sample (F03) if not implicitly present in the 'description of the matrix of the sample' (sampMatCode) is highly recommended to be reported when the sample refers to the following food categories/sub-categories:                                                 | WARNING: sampMatCode.state is not reported even though highly recommended                                                                                                        | Warning       | Active | New rule was introduced in 2025 and modified in 2026, which returns a Warning message. From warning to error message in 2027 for the following food categories 12.5, 13, 14.1.4, 14.1.5, 17. |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                               | Error Message                                                                         | Type of Error | Status | Comment                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|--------|-------------------------------------------------------------------|
|                    | 1 - Dairy products and analogues<br>6.3 - Breakfast cereals<br>12.5 - Soups and broths<br>12.6 - Sauces<br>13 - Foods intended for specific groups as defined by Regulation (EU) No 609/2013<br>14.1.2 - Fruit juices and vegetable juices<br>14.1.3 - Fruit nectars and vegetable nectars<br>14.1.4 - Flavoured drinks<br>14.1.5 - Coffee, tea, fruit infusions<br>17 - Food supplements |                                                                                       |               |        |                                                                   |
| <b>CHEMON87</b>    | If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1), then 'conclusion of follow-up investigation' (evalInfo.conclusion) is highly recommended to be reported                                                                                                                                                                                          | WARNING: evalInfo.conclusion is not reported even though highly recommended           | Warning       | Active | New rule was introduced in 2025, which returns a Warning message. |
| <b>CHEMON88</b>    | If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1), then 'restriction or exception' (evalInfo.restrictionException) is                                                                                                                                                                                                                               | WARNING: evalInfo.restrictionException is not reported even though highly recommended | Warning       | Active | New rule was introduced in 2025, which returns a Warning message. |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                 | Error Message                                                                                                                 | Type of Error | Status | Comment                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------|--------|------------------------------------------------------------------------------------------------|
|                    | highly recommended to be reported                                                                                                                                                                                                                                                                           |                                                                                                                               |               |        | Based on the outcome of the pilot this business rule might become an error in 2027.            |
| <b>CHEMON89</b>    | If the result is reported for food additives (is_add=1) or food flavourings (is_flav=1), then the target consumer facet (F23) is highly recommended to be reported when the sample refers to food category 13 (Foods intended for particular nutritional uses as defined by Directive 2009/39/EC)           | WARNING: the target consumer facet (F23) is not reported even though highly recommended                                       | Warning       | Active | New rule was introduced in 2025, which returns a Warning message.                              |
| <b>CHEMON90_a</b>  | If the paramCode "RF-0102-001-PPP" 'Copper' is reported, the use of facet 'Part-consumed-analysis' (F20) and/or 'Process' (F28) is needed                                                                                                                                                                   | WARNING: The use of facet F20 and/or F28 is needed                                                                            | Warning       | Active | New rule was introduced in 2025. Business rule remains a Warning. It becomes an error in 2027. |
| <b>CHEMON90_b</b>  | If the paramCode "RF-0102-001-PPP" 'Copper' is reported, then 'Result Value Uncertainty' (resValUncert) must be reported, if resType=VAL                                                                                                                                                                    | Measurement Uncertainty (resValUncert) when reporting "RF-0102-001-PPP" 'Copper' must be reported                             | Error         | Active | New rule was introduced in 2025. It becomes an Error in 2026.                                  |
| <b>CHEMON91</b>    | If the result is reported for VMPPR (is_vet=1), then only one F33 facet under parent term 'Veterinary Medicinal Product Residues Classes' (A1ANN) should be reported for the 'Coded description of the matrix of the sample taken' (sampMatCode) and 'Coded description of the matrix analysed' (anMatCode) | WARNING: only one F33 facet under "Veterinary Medicinal Product Residues classes (Annex III – 2022/1646)" should be reported. | Warning       | Active | New Warning was introduced in 2025 for VMPPR.                                                  |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Error Message                                                                                                                                                                                         | Type of Error | Status | Comment                                                                                                                                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHEMON92</b>    | If the result is reported for VMPR (is_vet=1), then the base term used for the 'Coded description of the matrix of the sample taken' (sampMatCode and 'Coded description of the matrix analysed' (anMatCode) must belong to the VetDrugRes hierarchy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | the base term reported does not belong to the vetDrugRes hierarchy.                                                                                                                                   | Error         | Active | New rule was introduced in 2025. It becomes an Error in 2025.                                                                                                                                                                                                             |
| <b>CHEMON93</b>    | If the result is reported for VMPR (is_vet=1) and programme type (progType) is not equal to 'Official (National) programme for Third Country Import' (K038A) and the country of origin is equal to the country of reporting organisation, and 'Evaluation of the result' (evalCode) is equal to 'Detected' (J041A), or 'Above the maximum permissible quantities' (J003A) and results assessment is empty, or result assessment is equal to 'Non-compliant' (J038A), then a value in the data element 'Area of sampling' (sampArea) or 'Area of origin of the sample taken' (origArea) or 'Area of origin for fisheries or aquaculture activities code of the sample taken' (origFishAreaCode) or 'Area of processing of the sample taken' (procArea) should be reported' | WARNING: one among sampArea, origArea, origFishAreaCode or procArea should be reported when evalCode is detected or greater than maximum permissible quantities or evalInfo.resAsses is non-compliant | Warning       | Active | New Warning was introduced in 2025 for VMPR.<br><br>For non-compliant results of VMPR Plan 1 and Plan 2, the area where the sample was collected should be reported. This only applies if the country of origin is the same of the country of the reporting organisation. |
| <b>CHEMON94</b>    | If the 'programme type' (progType) is 'Official (National) programme for Third Country Import' (K038A), then sampling point (sampPoint) can only be                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | progType is K038A but sampPoint is not equal to 'Border Control Posts' (E010A)                                                                                                                        | Error         | Active | New rule was introduced in 2025. It becomes an Error in 2026.                                                                                                                                                                                                             |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                  | Error Message                                                                                                                                           | Type of Error | Status | Comment                                                                   |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------------------------------------------------------------------------|
|                    | equal to 'Border Control Posts' (E010A)                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                         |               |        |                                                                           |
| <b>CHEMON95</b>    | If the result is reported for PPP (is_pest=1) and 'Evaluation of the result' (evalCode) is equal to 'greater than maximum permissible quantities' (J003A), then a value in the data element 'country of origin', different from 'XX', 'AA', 'EU', 'XC', 'XD', 'XE' must be reported                                                                          | WARNING: Country of origin must be different from 'Unknown'                                                                                             | Warning       | Active | New rule was introduced in 2025.<br><br>It will remain a Warning in 2026. |
| <b>CHEMON96</b>    | If the result is reported for VMPP (is_vet=1), and programme type (progType) is 'Official (National) programme' (K005A), then sampling strategy (sampStrategy) can only be equal to 'Objective sampling' (ST10A), 'Selective sampling' (ST20A), 'Suspect sampling' (ST30A) or 'Other' (ST90A)                                                                | progType is K005A but sampStrategy is not 'Objective sampling' (ST10A) or 'Selective sampling' (ST20A) or 'Suspect sampling' (ST30A) or 'Other' (ST90A) | Error         | Active | New rule was introduced in 2025.<br><br>It becomes an Error in 2026.      |
| <b>CHEMON97</b>    | If the result is reported for pesticides (is_pest=1), contaminants (is_occ=1), additives (is_add=1) or flavourings (is_flav=1) and programme type (progType) is 'Official (National) programme' (K005A), then sampling strategy (sampStrategy) can only be equal to 'Objective sampling' (ST10A), 'Selective sampling' (ST20A) or 'Suspect sampling' (ST30A) | progType is K005A but sampStrategy is not 'Objective sampling' (ST10A) or 'Selective sampling' (ST20A) or 'Suspect sampling' (ST30A)                    | Error         | Active | New rule was introduced in 2025.<br><br>It becomes an Error in 2026.      |
| <b>CHEMON98</b>    | If the result is reported for contaminants (is_occ=1) and related to control plans, therefore,                                                                                                                                                                                                                                                               | progLegRef is N375A but progType is not 'Official (National) programme'                                                                                 | Error         | Active | New rule was introduced in 2025.<br><br>It becomes an Error in 2026.      |



| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                      | Error Message                                                                                                                 | Type of Error | Status | Comment                                                         |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-----------------------------------------------------------------|
|                    | linked to Commission Delegated Regulation (EU) 2022/931 and Implementing Regulation (EU) 2022/932 (progLegRef N375A), then programme type (progType) can be only 'Official (National) programme' (K005A) or 'Official (National and EU) programme' (K018A) or 'Official (National) programme for Third Country Import' (K038A)                                                   | (K005A) or 'Official (National and EU) programme' (K018A) or 'Official (National) programme for Third Country Import' (K038A) |               |        |                                                                 |
| <b>CHEMON99</b>    | If programme type (progType) is 'Official (National) programme for Third Country Import' (K038A) or 'EU increased control programme on imported food' (K019A) or sampling point is 'Border control posts' (E010A), then country of origin (origCountry) cannot be equal to the sampling country (sampCountry).                                                                   | WARNING: origCountry is equal to sampCountry but progType is equal to K038A or K019A or sampPoint is equal to E010A           | Warning       | Active | New rule is introduced in 2026.<br>It becomes an Error in 2027. |
| <b>CHEMON100</b>   | If the result is reported for VMPP (is_vet=1), then evaluation of the result (evalCode) can only be equal to 'less than or equal to maximum permissible quantities' (J002A), or 'greater than maximum permissible quantities' (J003A), or 'Result not evaluated' (J029A), or 'Compliant due to measurement uncertainty' (J031A) or 'detected' (J041A) or 'not detected' (J040A). | WARNING: The reported evalCode is not one of the values allowed for VMPP                                                      | Warning       | Active | New rule is introduced in 2026.<br>It becomes an Error in 2027. |
| <b>CHEMON101</b>   | If the value in 'Programme legal reference' (progLegalRef) is 'Commission Implementing Regulation (EU) No 2019/1873'                                                                                                                                                                                                                                                             | sampStrategy is not 'suspect', or progType is not 'EU increased control programme on imported food', though                   | Error         | Active | New rule is introduced in 2026.<br>It becomes an Error in 2026. |

| Business Rule Code | Description                                                                                                                                                                                                                                    | Error Message                                                                                                                                                                | Type of Error | Status | Comment                                                             |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------------------------------------------------------------------|
|                    | (N422A), then the value in 'Programme type' (progType) is equal to 'EU increased control programme on imported food' (K019A) and the value in 'Sampling strategy' (sampStrategy) can only be equal to 'Suspect sampling' (ST30A)               | only those values should be reported under Reg. 2019/1873                                                                                                                    |               |        |                                                                     |
| <b>CHEMON102</b>   | If the result is reported for VMPP (is_vet=1), for the records with the same sampEventId, the elements repCountry, sampCountry, sampArea, origCountry, origArea, origFishAreaCode, origFishAreaText, procCountry, procArea should be constant. | WARNING: repCountry, sampCountry, sampArea, origCountry, origArea, origFishAreaCode, origFishAreaText, procCountry, procArea are not constant for the same sampEventId       | Warning       | Active | New rule is introduced in 2026.<br><br>It becomes an Error in 2027. |
| <b>CHEMON103</b>   | 'Organic production' (A07SE) and 'Conventional (non-organic) production' (A0C6Y) should not be reported together in 'Production method' (sampMatCode.prod). Please note that in that specific case the sample will be considered as organic.   | WARNING: sampMatCode.prod is both 'A07SE' and 'A0C6Y', though it is recommended only to report sampMatCode.prod 'A07SE' if the sample is intended to be reported as organic. | Warning       | Active | New rule is introduced in 2026.<br><br>It becomes an Error in 2027. |
| <b>CHEMON104</b>   | If the value in 'Programme legal reference' (progLegalRef) is 'Commission Implementing Regulation (EU) No 2019/1873' (N422A), then no other progLegalRef can be concatenated.                                                                  | progLegRef is N422A, thus no other progLegRef can be included                                                                                                                | Error         | Active | New rule is introduced in 2026.<br><br>It becomes an Error in 2026. |
| <b>CHEMON105</b>   | If the value in 'Programme legal reference' (progLegalRef) is 'Commission Implementing Regulation (EU) No 2019/1793' (N317A), then no other progLegalRef can be used.                                                                          | progLegRef is N317A, thus no other progLegRef can be included                                                                                                                | Error         | Active | New rule is introduced in 2026.<br><br>It becomes an Error in 2026. |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                          | Error Message                                                                                                                                                                                                                                                                                                                                                                                                                | Type of Error | Status | Comment                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|----------------------------------------------------------------|
| <b>CHEMON106</b>   | If the paramCode "RF-00000248-ADD" 'Potassium sorbate' is reported, then the paramText should be used to specify if the value is "Expressed as free acid" or "Expressed as salt".                                                                                                                                                                                                                                    | WARNING: The param text "Expressed as free acid" or "Expressed as salt" is not provided for the paramCode "RF-00000248-ADD" 'Potassium sorbate'                                                                                                                                                                                                                                                                              | Warning       | Active | New rule is introduced in 2026.                                |
| <b>CHEMON107</b>   | If one of the paramCode "RF-00000304-ADD" 'Sorbic acid', "RF-00000248-ADD" 'Potassium sorbate', "RF-00000062-ADD" 'Butylated hydroxytoluene (BHT)", "RF-00000370-TOX" 'Coumarin', "RF-00000333-TOX" 'Hydrocyanic acid' or "RF-00000334-TOX" 'Theobromine' is reported, and ResType is reported as "Qualitative Value (Binary), and resQualValue is reported as 'NEG' then it is highly recommended to report the LOD | WARNING: The LOD is not reported even though highly recommended when the paramCode is one of "RF-00000304-ADD" 'Sorbic acid', "RF-00000248-ADD" 'Potassium sorbate', "RF-00000062-ADD" 'Butylated hydroxytoluene (BHT)", "RF-00000370-TOX" 'Coumarin', "RF-00000333-TOX" 'Hydrocyanic acid' or "RF-00000334-TOX" 'Theobromine' and ResType is reported as "Qualitative Value (Binary), and resQualValue is reported as 'NEG' | Warning       | Active | New rule is introduced in 2026.                                |
| <b>CHEMON108</b>   | If the result is reported for additives (is_add=1) or flavourings (is_flav=1), then the following codes "Food colours" 'A047N', "Artificial food colour" 'A047Q', "Food additives other than flavours, colours and artificial sweeteners" 'A047R', "Food flavourings" 'A047A', "Natural food colour" 'A047P' and "Preparations for food flavouring" 'A0F3T' cannot be reported as sampMatCode                        | WARNING: the sampMatCode reported is equal to one of the following "Food colours" 'A047N', "Artificial food colour" 'A047Q', "Food additives other than flavours, colours and artificial sweeteners" 'A047R', "Food flavourings" 'A047A', "Natural food colour" 'A047P' or "Preparations for food flavouring" 'A0F3T' even though the result is additive or flavouring                                                       | Warning       | Active | New rule introduced in 2026. It might become an error in 2027. |

| Business Rule Code              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                | Error Message                                                                                                                                                        | Type of Error               | Status | Comment                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------|----------------------------------------------------------------|
| <b>CHEMON109</b>                | If the reported sampMatCode contains an implicit additive/flavouring food legislative category (F33), then the explicit additive/flavouring F33 is not needed                                                                                                                                                                                                                                                                                              | WARNING: the reported sampMatCode contains an explicit additive/flavouring food legislative category (F33), even if the implicit additive/flavouring F33 is present. | Warning                     | Active | New rule introduced in 2026. It might become an error in 2027. |
| <b>FOODEX2_SAMPMAT</b>          | The value in sampMatCode should be coded according to FoodEx2 classification rules                                                                                                                                                                                                                                                                                                                                                                         | sampMatCode is not coded according to FoodEx2 classification rules                                                                                                   | Error                       | Active |                                                                |
| <b>FOODEX2_ANMAT</b>            | The value in anMatCode should be coded according to classification rules                                                                                                                                                                                                                                                                                                                                                                                   | anMatCode is not coded according to FoodEx2 classification rules                                                                                                     | Error                       | Active |                                                                |
| <b>LL_01_VMPR<br/>LL_01_PPP</b> | If the value in 'Result type' (resType) is 'Numerical value' (VAL), and the coded sample is unprocessed and value in 'Type of limit for the result evaluation' (evalLimitType) is 'Maximum Residue Level (MRL)' (W002A) or is not reported and the value in 'Result value' (resVal) is greater than the MRL, then the value in 'Evaluation of the result' (evalCode) must be different from 'Less than or equal to maximum permissible quantities' (J002A) | The selected result evaluation (evalCode) is incorrect; result value (resVal) exceeds the result legal limit. Please correct this record                             | Warning (VMPR)/Error (PEST) | Active |                                                                |
| <b>LL_02_VMPR<br/>LL_02_PPP</b> | If the value in 'Result type' (resType) is 'Numerical value' (VAL), and the coded sample is unprocessed and the value in 'Type of limit for the result evaluation' (evalLimitType) is 'Maximum Residue Level (MRL)' (W002A) or is not reported and the value in 'Result value' (resVal) is less than or equal to the MRL,                                                                                                                                  | The selected result evaluation (evalCode) is incorrect; the result value (resVal) is equal to or below the result legal limit. Please correct this record            | Warning (VMPR)/Error (PEST) | Active |                                                                |

| Business Rule Code | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                | Error Message                                                                                                                                                                                                   | Type of Error | Status | Comment         |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|-----------------|
|                    | then the value in 'Evaluation of the result' (evalCode) must be different from 'Greater than maximum permissible quantities' (J003A) and 'Compliant due to measurement uncertainty' (J031A)                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                 |               |        |                 |
| <b>LL_01_FA_F</b>  | If the value in 'Result type' (resType) is 'Numerical value' (VAL), and 'Type of limit for the result evaluation' (evalLimitType) is 'Maximum Permitted Level (MPL)' (W016A) or is not reported and the value in 'Result value' (resVal) is greater than the MPL, then the value in 'Evaluation of the result' (evalCode) should be different from 'Less than or equal to maximum permissible quantities' (J002A) or 'Compliant' (J037A)                   | The 'Evaluation of the result' (evalCode) is reported as 'Less than or equal to maximum permissible quantities' (J002A) or 'Compliant' (J037A), even if the 'Result value' (resVal) is greater than the MPL     | Warning       | Active | New BR in 2026. |
| <b>LL_02_FA_F</b>  | If the value in 'Result type' (resType) is 'Numerical value' (VAL), and the value in 'Type of limit for the result evaluation' (evalLimitType) is 'Maximum Permitted Level (MPL)' (W016A) or is not reported and the value in 'Result value' (resVal) is less than or equal to the MPL, then the value in 'Evaluation of the result' (evalCode) should be different from 'Greater than maximum permissible quantities' (J003A) and 'Non-compliant' (J038A) | The 'Evaluation of the result' (evalCode) is reported as 'Greater than maximum permissible quantities' (J003A) or 'Non-compliant' (J038A), even if the 'Result value' (resVal) is less than or equal to the MPL | Warning       | Active | New BR in 2026. |
| <b>LL_03_FA_F</b>  | The substance (paramCode) reported should be authorised in this legislative category with this                                                                                                                                                                                                                                                                                                                                                             | The reported substance is not authorised in this legislative category with this                                                                                                                                 | Warning       | Active | New BR in 2026. |

| Business Rule Code   | Description                                                                                                                                                                                                                                                                                                                                                                  | Error Message                                                                                                                       | Type of Error | Status | Comment                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|----------------------------------------------------------------------------------------|
|                      | restriction/exception according to Regulation 1333/2008 and 1334/2008.                                                                                                                                                                                                                                                                                                       | restriction/exception according to Regulation 1333/2008 and 1334/2008.                                                              |               |        |                                                                                        |
| <b>LL_03</b>         | The value in paramType must be reported for multicomponent/sum to indicate if the full analysis has been performed as indicated by paramCode                                                                                                                                                                                                                                 | paramType is missing, though it should be reported                                                                                  | Error         | Active | Modified for the data collection of 2021 according to a new paramType mapping approach |
| <b>LL_03_b</b>       | The value in paramType should be equal to the pre-assigned paramType                                                                                                                                                                                                                                                                                                         | WARNING: the reported paramType is different from EFSA pre-assigned paramType                                                       | Warning       | Active | Modified for the data collection of 2021 according to a new paramType mapping approach |
| <b>LL_04/LL_04_b</b> | If the result is reported for pesticides (is_pest=1), and the paramCode is part of the legal limits database, then the combination of the matrix sampled and the 'Parameter code' (paramCode.base) should match with the plausible legal combinations as reflected in the legal limits database (PlausibleCombinationPPP=1), unless the result is reported for a part of sum | The selected paramCode does not refer to the correct legal Residue Definition for the sample analysed. Please correct the paramCode | Error         | Active |                                                                                        |



## 7 Controlled terminology catalogues

Controlled terminology catalogues are available for reference and download from EFSA's Knowledge Junction on Zenodo (EFSA, 2018).

Table 11: Full list of catalogues and the hierarchies in use for chemical monitoring reporting

| Controlled terminology (catalogues) | Chemical monitoring reporting hierarchy*                                 | Analysis hierarchies**                                                   |
|-------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>ACTION</b>                       | Master hierarchy                                                         | Not applicable                                                           |
| <b>ADDFOOD</b>                      | Master hierarchy                                                         | FARestExc and FFRestExc                                                  |
| <b>ANLYMD</b>                       | Master hierarchy                                                         | Not applicable                                                           |
| <b>ANLYREFMD</b>                    | Master hierarchy                                                         | Not applicable                                                           |
| <b>ANLYTYP</b>                      | Master hierarchy                                                         | Not applicable                                                           |
| <b>CONCLUS</b>                      | Master hierarchy                                                         | faff (for food additives and food flavourings)                           |
| <b>COUNTRY</b>                      | Master hierarchy                                                         | Master hierarchy                                                         |
| <b>EXPRRES</b>                      | Master hierarchy                                                         | Not applicable                                                           |
| <b>FAREA</b>                        | Master hierarchy                                                         | Not applicable                                                           |
| <b>LABACC</b>                       | Master hierarchy                                                         | Not applicable                                                           |
| <b>LEGREF</b>                       | ChemMonLegref                                                            | See Table 12 below for legislation applicability to the specific domains |
| <b>LMTTYP</b>                       | Master hierarchy                                                         | Not applicable                                                           |
| <b>MDACC</b>                        | Master hierarchy                                                         | Not applicable                                                           |
| <b>MTX</b>                          | Reporting hierarchy<br>Use food, feed or non-food matrices as applicable | MTXCLS – vmprCIs (for VMPR)***<br>MATRIX (for pesticide residues)***     |
| <b>NUTS</b>                         | NUTS2024                                                                 | Not applicable                                                           |
| <b>PARAM</b>                        | ChemMonRep                                                               | addAnalysis<br>flavAnalysis<br>chemAnalysis<br>pestParam<br>vmprParam    |
| <b>PARAMTYP</b>                     | Master hierarchy                                                         | Master hierarchy                                                         |
| <b>POSNEG</b>                       | Master hierarchy                                                         | Not applicable                                                           |





| Controlled terminology (catalogues) | Chemical monitoring reporting hierarchy* | Analysis hierarchies** |
|-------------------------------------|------------------------------------------|------------------------|
| <b>PRGTYP</b>                       | Master hierarchy                         | Not applicable         |
| <b>REFTM</b>                        | Master hierarchy                         | Not applicable         |
| <b>RESEVAL</b>                      | Master hierarchy                         | Not applicable         |
| <b>SAMPLR</b>                       | Master hierarchy                         | Not applicable         |
| <b>SAMPMD</b>                       | Master hierarchy                         | Not applicable         |
| <b>SAMPNT</b>                       | Master hierarchy                         | vmprClasses (for VMPR) |
| <b>SAMPSTR</b>                      | chemSampStr                              | Not applicable         |
| <b>SAMPUNTYP</b>                    | Master hierarchy                         | Not applicable         |
| <b>UNIT</b>                         | chemUnit                                 | Not applicable         |
| <b>VALTYP</b>                       | chemValTyp                               | Not applicable         |
| <b>YESNO</b>                        | Master hierarchy                         | Not applicable         |

\* All chemical monitoring records will be validated to ensure the values reported exist in the reporting hierarchy specified.

\*\* Analysis hierarchies are specified only when there are legislative or reporting needs which require a specific grouping of reported values. All values which exist in one or more of these will also be in the reporting hierarchy for each catalogue.

\*\*\*This is not a hierarchy as such, but a dedicated catalogue (based on the MTX catalogue), which is needed for specific sample matrices mapping according to the sectoral domain legislation.



Table 12: List of LEGREF values for the progLegalRef data element and domain attributes

| <b>progLegalRef code</b> | <b>LEGALREF description</b>                                                                          | <b>Residue domain applicability</b>  |
|--------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>N023A</b>             | Commission Regulation (EC) No 2073/2005 (amended)                                                    | OCC (CONTAMINANTS)                   |
| <b>N027A</b>             | Regulation (EC) No 396/2005 (amended)                                                                | PEST (PESTICIDES)                    |
| <b>N028A</b>             | Directive 2006/125/EC                                                                                | PEST, OCC, ADD                       |
| <b>N101A</b>             | Regulation (EC) No 767/2009 (amended)                                                                | OCC                                  |
| <b>N112A</b>             | Regulation (EC) No 1333/2008 (amended)                                                               | ADD (ADDITIVES)                      |
| <b>N113A</b>             | Regulation (EC) No 1334/2008 (amended)                                                               | FLAV (FLAVOURINGS)                   |
| <b>N129A</b>             | Regulation (EC) No 178/2002 (amended)                                                                | OCC                                  |
| <b>N139A</b>             | Regulation (EC) No 853/2004 (amended)                                                                | OCC                                  |
| <b>N154A</b>             | Commission Directive (EC) No 40/2003                                                                 | OCC                                  |
| <b>N208A</b>             | Commission Recommendation No 196/2007                                                                | OCC                                  |
| <b>N211A</b>             | Commission Recommendation No 161/2010                                                                | OCC                                  |
| <b>N252A</b>             | Directive 2002/32/EC                                                                                 | OCC                                  |
| <b>N302A</b>             | Commission Recommendation 2015/682                                                                   | OCC                                  |
| <b>N304A</b>             | Commission Recommendation (EU) 2017/84                                                               | OCC                                  |
| <b>N310A</b>             | Commission Regulation (EU) 2017/2158                                                                 | OCC                                  |
| <b>N312A</b>             | Commission Recommendation (EU) 2014/118                                                              | OCC                                  |
| <b>N313A</b>             | Commission Recommendation (EU) 2013/165                                                              | OCC                                  |
| <b>N317A</b>             | Commission Implementing Regulation (EU) 2019/1793                                                    | PEST, OCC, ADD                       |
| <b>N318A</b>             | Commission Delegated Regulation (EU) 2016/127 and Regulation (EU) 2016/128                           | PEST, OCC, ADD                       |
| <b>N319A</b>             | Commission Recommendation 2006/576/EC                                                                | OCC                                  |
| <b>N320A</b>             | Commission Recommendation 2013/711/EU amended                                                        | OCC                                  |
| <b>N323A</b>             | Regulation (EU) No 2017/625                                                                          | OCC, ADD                             |
| <b>N324A</b>             | Commission Recommendation (EU) 2019/1888                                                             | OCC                                  |
| <b>N371A</b>             | Commission Implementing Regulation (EU) 2022/1646 and Commission Delegated Regulation (EU) 2022/1644 | VMPR (VETERINARY MEDICINAL RESIDUES) |
| <b>N372A</b>             | Commission Recommendation (EU) 2022/495                                                              | OCC                                  |
| <b>N374A</b>             | Commission Recommendation (EU) 2016/2115                                                             | OCC                                  |
| <b>N375A</b>             | Commission Delegated Regulation (EU) 2022/931 and Implementing Regulation (EU) 2022/932              | OCC                                  |
| <b>N376A</b>             | Commission Recommendation (EU) 2022/553                                                              | OCC                                  |
| <b>N377A</b>             | Commission Recommendation (EU) 2022/561                                                              | OCC                                  |
| <b>N379A</b>             | Commission Regulation (EU) 2023/915                                                                  | OCC                                  |
| <b>N422A</b>             | Commission Implementing Regulation (EU) 2022/1873                                                    | OCC, PEST, VMPR                      |



It is recommended that data providers use as specific as LEGREF term as possible to ensure appropriate inclusion in reports and other data re-uses. Please note that reporting LEGREF='Regulation (EC) No 178/2002 (amended)' (N129A) will flag the record as is\_occ=1 (see also Section 9 below on reporting flags); this will ensure that data are used for exposure assessment or other specific data analysis but not necessarily included in EU Annual Reports. If a specific piece of legislation is not available in the controlled terminology catalogue LEGREF and/or in the specific catalogue hierarchy 'ChemMonLegRef', data providers should contact EFSA with a suggested new term during the major release consultation period in October–November each year or, if there is an urgent unpredicted need, at any other time.

## 8 Legal Limits database

EFSA developed a harmonised database (Legal Limit Database) containing maximum residue limits (MRLs) established for pesticides and veterinary medicine residues in Regulation (EC) No 396/2005, Regulation (EC) No 37/2010 and Directive (EC) No 141/2006 to support data providers in applying EU MRLs to numerical analytical result evaluation. It is intended that this resource will facilitate checking exceedances and sample compliance evaluation at results level.

In cases where an MRL change during the data collection reference period, the database will contain both MRLs with the dates of applicability. The legal limit applied will be determined by the sampling date of the sample and will only cover raw, unprocessed samples. There is a small number of substances for which legal limits are not harmonised across the legislation which applies to chemical monitoring. A common database of MRLs is considered beneficial for EFSA and the Member States, who should collaborate with the Commission in this respect.

An extraction of the database is visible and downloadable through EFSA MicroStrategy, and it will be a useful tool for data providers to facilitate reporting according to the legislation. It will also contain the history of legal limits throughout the years; data providers must still rely on the legislation as the official source of legal limits.

## 9 Reporting flags in the EFSA Scientific Data Warehouse

Since all chemical monitoring data transmissions to EFSA were merged into one data collection, it became necessary to understand and record in the sDWH the domains that are relevant for each result. There are substances that for instance are both pesticides and veterinary medicinal products or both contaminants and food additives. Many combinations are possible for these dual substances. One of the benefits of a single data collection is to avoid double reporting of such records when they need to appear in more than one domain report.

In order to ensure such a result record can be reported once and used in as many different domains and reports as are relevant, EFSA has implemented a flagging algorithm which identifies the relevance of each result stored in the EFSA sDWH. This has been done based on substance (paramCode) and legislation (progLegalRef) to date and the table below indicates the combinations which determine the flag value. EFSA creates and stores five flags for each result record to indicate its applicability to each of the five chemical monitoring domains: VMPPR, pesticides, contaminants, food additives, and food flavourings.

The flags are created at the time of receiving the records into the DCF which enables them to be used to ensure specific BR only applies to records for which they are relevant.

EFSA is aware that the matrix of the sample tested and measurement Unit reported are also elements which affect the flags and adjustments to the algorithm are needed that will include the use of the Legal Limits database to identify parameter/matrix/legislation combinations that should trigger the flag creation in the future.

|                                                                         | Flag value |
|-------------------------------------------------------------------------|------------|
| Records for which NEITHER progLegalRef NOR paramCode are in the domain* | 0          |
| Records for which progLegalRef AND paramCode are in the domain          | 1          |
| Records for which ONLY paramCode is in the domain                       | 2          |
| Records for which ONLY progLegalRef is in the domain                    | 3          |

\* 'In the domain' means that the term is relevant to the domain. progLegalRef being in the domain is checked through the hierarchy ChemMonLegRef in the attributes of each term ('VMPR', 'PEST', 'OCC', 'ADD' and 'FLAV'). paramCode being in the domain is checked through each domain's analysis hierarchy (vmprParam, pestParam, chemAnalysis, addAnalysis & flavAnalysis). See also Table 12.

For example, under this approach, records would be flagged as follows:

| Example                                                                                          | VMPR<br>(is_vet) | Pesticides<br>(is_pest) | Contaminants<br>(is_occ) | Additives<br>(is_add) | Flavourings<br>(is_flav) |
|--------------------------------------------------------------------------------------------------|------------------|-------------------------|--------------------------|-----------------------|--------------------------|
| progLegalRef=VMPR records whose PARAM is only VMPR                                               | 1                | 0                       | 0                        | 0                     | 0                        |
| progLegalRef=VMPR records whose PARAM is both VMPR and OCC (contaminants)                        | 1                | 0                       | 2                        | 0                     | 0                        |
| progLegalRef=VMPR records whose PARAM is both VMPR and pesticides                                | 1                | 2                       | 0                        | 0                     | 0                        |
| progLegalRef=VMPR and OCC (contaminants) records whose PARAM is both VMPR and OCC (contaminants) | 1                | 0                       | 1                        | 0                     | 0                        |
| progLegalRef=ADD records whose PARAM is both ADD and OCC                                         | 0                | 0                       | 1                        | 1                     | 0                        |
| progLegalRef=FLAV records whose PARAM is only FLAV                                               | 0                | 0                       | 0                        | 0                     | 1                        |

Business rules can be applied according to the domain flags, e.g. CHEMON73 will apply if VMPR=1.

As an example, if a record is reported with progLegalRef=N371A (Regulation (EU) 2022/1646 on VMPR) and paramCode=RF-00000411-VET (Testosterone-17-Alpha), which is included only in 'vmprParam' hierarchy of the PARAM catalogue, then this record will have the flag for VMPR equal to 1 (is\_vet=1) and the flag for the other four domains equal to '0' (is\_pest=0 and is\_occ=0 and is\_add=0 and is\_flav=0). If, instead, a record is reported with progLegalRef=N371A (Regulation (EU) 2022/1646 on VMPR) and paramCode=RF-0024-002-PPP (Amitraz), which is included in 'vmprParam' and 'pestParam' hierarchies, then this record will be flagged as is\_vet=1 and is\_pest=2 and is\_occ=0 and is\_add=0 and is\_flav=0.

## 10 Validation, National and Annual Reports

Data submitted through the ChemMon data collection will be used for EFSA scientific assessments. In addition, EFSA will produce a series of reports based on each calendar year of these data to fulfil customer and legislative requirements. The following table (Table 13) is a summary of the reports which will be derived from data submissions and the filters that will be used to select the data.

Data providers should report the appropriate LEGREF code (or combination of codes), which indicates the legislation under which the samples were taken and analysed in order to fulfil the specific legislative control activity requirements. Please refer to reporting flags in EFSA's sDWH based on LEGREF and PARAM codes as described above in Section 8.

Table 13: Summary of reports

| Customer                                                                                  | Purpose                                                                                                                                                                                    | ChemMon domain flag inclusion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Note                                                                                         |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Data providers and validators</b>                                                      | Validation dashboard reports: <ul style="list-style-type: none"> <li>• Pesticides</li> <li>• VMPPR</li> <li>• Chemical contaminants</li> <li>• Additives</li> <li>• Flavourings</li> </ul> | Only those records with domain flag value=1 will appear in each of the four sections of these validation reports (pesticides, VMPPR, chemical contaminants, food additives and food flavourings).<br><br>Please note that these dashboards with summarised and/or aggregated results are generated by EFSA for each single reporting organisation in the given country                                                                                                                                            | Each submitted record may appear in more than one section of the validation dashboard report |
| <b>Data providers and validators, as supporting tool during the validation process</b>    | National Reports: <ul style="list-style-type: none"> <li>• Chemical contaminants</li> <li>• Food additives</li> <li>• Flavourings</li> </ul>                                               | Reports generated for each of the chemical contaminants and food additives results domain:<br><br>All records with flag value is_occ=1 (contaminants).<br><br>All records with flag value is_add=1 (food additives).<br><br>All records with flag value is_flav=1 (food flavourings).<br><br>Please note that these reports are generated by EFSA for each single reporting organisation in the given country.<br><br>The National Reports can be created at any time by the data providers/national organisation | Each submitted record may appear in more than one domain National Report                     |
| <b>Member States/EFTA/pre-accession countries for delivery to the European Commission</b> | National Reports: <ul style="list-style-type: none"> <li>• VMPPR</li> </ul>                                                                                                                | Reports generated only for VMPPR results domain:<br><br>All records with flag value is_vmppr=1<br><br>Please note that these reports are generated by EFSA for each single reporting organisation in the given country.<br><br>The VMPPR National Report can be created any time by the data providers/national organisation                                                                                                                                                                                      | Each submitted record may appear in more than one domain National Report                     |

| Customer            | Purpose                                                                                                                                                                                     | ChemMon domain flag inclusion                                                                                                                                                                                                                                                                                           | Note                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| European Commission | Statutory EU Annual Reports:<br><ul style="list-style-type: none"> <li>• Pesticides</li> <li>• VMPR</li> </ul>                                                                              | Annual Reports generated only for pesticides and VMPR results domain:<br>All records with flag value is_vmpr=1<br>OR<br>All records with flag value is_vmpr=1, excluding the VMPR matrix class = 'Other'<br>All four ChemMon residue domains official control data (pesticides, VMPR, contaminants and food additives). | Each submitted record may appear in more than one domain EU Annual Report  |
| European Commission | Aggregated data for:<br><ul style="list-style-type: none"> <li>• Pre-filling one of the pre-defined tables (Table 1.4) of the EC Annual Report on Official Control (AROC) Report</li> </ul> | Data extraction from EFSA sDWH coupled with the EC AROC platform to facilitate Member States in submitting data in their Annual Report on the implementation of their multiannual national control plans (MANCP) in a standard model form                                                                               | Each submitted record/sample may appear in more than one domain of EC AROC |



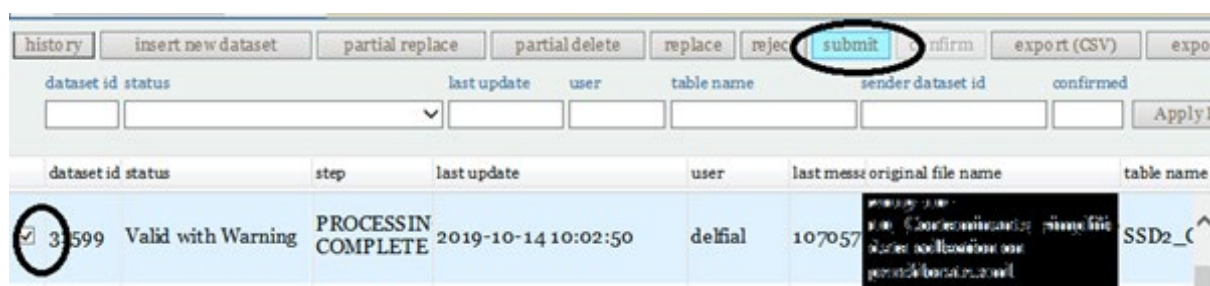
## 11 Data validation and acceptance

Datasets transmitted to the EFSA Data Collection Framework (DCF) will be acknowledged automatically by the system through an acknowledgement (ack) message. The data provider will receive the ack message followed by the results of the BR validation for their dataset. The 'ack details' message can be downloaded as an XML file containing the list of errors found by the system. These process steps are fully described in the DCF user manual for data providers.

To resolve errors and warnings about data quality, data providers are invited to make use of the troubleshooting suggestions in Table 9 and Table 10 to correct and re-transmit their datasets. In addition, the network collaboration site<sup>76</sup> contains information and frequently asked questions that will help to resolve any issues.

If clarification is still required, data providers should contact [data.collection@efsa.europa.eu](mailto:data.collection@efsa.europa.eu). This channel is monitored daily, and each query will be answered as soon as possible by the most appropriate EFSA staff member available.

Once the datasets have reached 'Valid' or 'Valid with Warning' status, the data provider will be able to 'Submit' them to the EFSA sDWH. Datasets can be submitted through DCF by flagging the dataset ID that has to be submitted and by clicking the 'Submit' button as depicted in the image below.



The 'Submitted' data will then be visible through validation reports in MicroStrategy. Both data providers and data validators can view the validation reports in MicroStrategy.

Once the data validator is satisfied that the datasets 'Submitted' are accurate and complete, they will be able to 'Accept' the data through the confirmation document in MicroStrategy. This ensures that data are in the EFSA sDWH and ready for use. Data validators can accept or reject the data, by selecting the final status of the datasets, 'Accepted' or 'Rejected', from the drop-down menu 'Confirm-Reject', in correspondence with the relative dataset ID. At the end of the process, the data validator must submit the final status of the datasets by clicking the 'Submit' button as depicted below.

<sup>76</sup> <https://efsa815.sharepoint.com/sites/chemical-monitoring-data-network/>



| Owner Organization (L1) | Dataset ID | Confirm-Reject  |
|-------------------------|------------|-----------------|
|                         | 29995      | TO BE VALIDATED |
|                         | 31896      | TO BE VALIDATED |
|                         | 31905      | ACCEPTED        |
|                         | 31576      | REJECTED        |

Submit

It is important to note that once data are 'Accepted' in the sDWH, the only way to correct them is through the record 'Update procedure' described in GDE2 Guidance (EFSA, 2014a) (5.2 Amendment Operations p. 18-30). EFSA encourages data providers to ensure a thorough validation of data transmitted before acceptance.



## References

- Bratinova S and Hoekstra E, 2019. Guidance on sampling, analysis and data reporting for the monitoring of mineral oil hydrocarbons in food and food contact materials. Luxembourg, Publications Office of the European Union. Available online: <http://publications.jrc.ec.europa.eu/repository/handle/JRC115694> doi:10.2760/208879
- EFSA (European Food Safety Authority), 2013. Standard Sample Description ver. 2.0. EFSA Journal 2013;11(10):3424, 114 pp. doi:10.2903/j.efsa.2013.3424
- EFSA (European Food Safety Authority), 2014a. Guidance on Data Exchange version 2.0 EFSA Journal 2014;12(12):3945, 173 pp. doi:10.2903/j.efsa.2014.3945
- EFSA (European Food Safety Authority), 2014b. Guidance on the EU Menu methodology. EFSA Journal 2014;12(12):3944, 77 pp. doi:10.2903/j.efsa.2014.3944
- EFSA (European Food Safety Authority), 2015. The food classification and description system FoodEx2 (revision 2). EFSA supporting publication 2015:EN-804, 90 pp. doi: 10.2903/sp.efsa.2015.EN-804
- EFSA (European Food Safety Authority), 2017. Data Collection Framework (DCF) User Manual. Available online in the DCF, accessible by registered DCF users
- EFSA (European Food Safety Authority), 2018. Harmonized terminology for scientific research [Data set]. Zenodo – Knowledge Junction. Available online: <https://doi.org/10.5281/zenodo.779880>
- EFSA (European Food Safety Authority), 2024. H EFSA Legal Limit Data Base for compliance check of chemical monitoring data [Data set]. Zenodo – Knowledge Junction. Available online: <https://zenodo.org/records/11126756>
- EFSA (European Food Safety Authority), Foster D, Locker A, Maldonado A, O'Dea E, Saez Llorente J, Sögel J, Tapanainen H, Thomas S, Tirian A, Richardson J, Marsic I, Pintado C, Czomba B, Ringwald F and Gabbi S, 2019a. Publication of scientific data from EU-coordinated monitoring programmes and surveys. EFSA supporting publication 2019:16(1):EN-1544. 43 pp. doi: 10.2903/sp.efsa.2019.EN -1544
- EFSA (European Food Safety Authority), Ioannidou S, 2019b. EFSA Catalogue Browser User Guide. EFSA supporting publication 2019:EN-1726. 39 pp. doi:10.2903/sp.efsa.2019.EN1726
- EFSA (European Food Safety Authority), Ioannidou S, 2019c. EFSA FoodEx2 Interpreting and Checking Tool user guide. EFSA supporting publication 2019:EN-1727. 20 pp. doi:10.2903/sp.efsa.2019.EN-1727
- EFSA (European Food Safety Authority), 2021. EFSA (European Food Safety Authority) 2021. Chemical Monitoring Reporting Guideline. EFSA supporting publication 2021:EN-4620E. 96 pp. Available online: <https://doi.org/10.2903/sp.efsa.2021.EN-620>
- EFSA (European Food Safety Authority), online-a. Terms of Reference of the EFSA Scientific Network on Chemical Monitoring Data Collection. Available online: [https://www.efsa.europa.eu/sites/default/files/science/Support/Data/Data\\_ToR\\_Chemical\\_Monitoring\\_Network.pdf](https://www.efsa.europa.eu/sites/default/files/science/Support/Data/Data_ToR_Chemical_Monitoring_Network.pdf)
- EFSA (European Food Safety Authority), online-b. Catalogue Browser including FoodEx2 browser. Available online: <https://github.com/openefsa/catalogue-browser>
- European Commission, 2015. Working document on the summing up of LOQs in case of complex residue definitions. SANCO/12574/2014. Available online: [https://food.ec.europa.eu/system/files/2016-10/pesticides\\_mrl\\_guidelines\\_wrkdoc\\_12574.pdf](https://food.ec.europa.eu/system/files/2016-10/pesticides_mrl_guidelines_wrkdoc_12574.pdf)

European Commission, 2024. Guidance document on analytical quality control and method validation procedures for pesticide residues and analysis in food and feed. SANTE/11312/2021 (v2). Available online:

[https://food.ec.europa.eu/system/files/2023-11/pesticides\\_mrl\\_guidelines\\_wrkdoc\\_2021-11312.pdf](https://food.ec.europa.eu/system/files/2023-11/pesticides_mrl_guidelines_wrkdoc_2021-11312.pdf)

## List of abbreviations

|                                         |                                                                |
|-----------------------------------------|----------------------------------------------------------------|
| AOCS                                    | American Oil Chemists' Society                                 |
| AROC                                    | EC Annual Report on Official Controls                          |
| AWR                                     | Above the analytical Working Range                             |
| BR                                      | Business Rules                                                 |
| CC <sub>α</sub> , CC <sub>α</sub> alpha | Decision limit                                                 |
| CC <sub>β</sub> , CC <sub>β</sub> beta  | Detection capability                                           |
| ChemMon                                 | Chemical Monitoring data collection of EFSA                    |
| DCF                                     | Data Collection Framework                                      |
| EEA                                     | European Economic Area                                         |
| EU                                      | European Union                                                 |
| EU MACP                                 | EU Multi-Annual Coordinated Programme                          |
| FoodEx2                                 | Food classification and description system (version 2) of EFSA |
| FCM                                     | Food Contact Materials                                         |
| GDE2                                    | Guidance on Data Exchange (version 2.0)                        |
| GBR                                     | General business rules                                         |
| LOD                                     | Limit of Detection                                             |
| LOQ                                     | Limit of Quantification                                        |
| MANCP                                   | Multi-Annual National Control Plans                            |
| MRL                                     | Maximum Residue Limit                                          |
| MRPL                                    | Minimum Required Performance Limit                             |
| NUTS                                    | Nomenclature of territorial units for statistics               |
| PAH                                     | Polycyclic aromatic hydrocarbons                               |
| RASFF                                   | Rapid Alert System for Food and Feed                           |
| RPC                                     | Raw primary commodities                                        |
| sDWH                                    | Scientific Data Warehouse                                      |
| SSD2                                    | Standard Sample Description (version 2.0)                      |
| TDS                                     | Total diet study                                               |
| TR                                      | Transparency Regulation                                        |
| VMPR                                    | Veterinary Medicinal Product Residues                          |
| XML                                     | extensible mark-up language                                    |